

Dictionary of Networking

You'll never have doubts about it again.

Part 2

Share, don't be selfish, don't think only of yourself.

<https://www.linkedin.com/in/felipe-c-01a8b1280/>

mail-enabled application Any application that includes an e-mail function but that also provides additional services, such as contact-management software, intelligent mail handling, and workflow automation. Also known as message-enabled application.

See also groupware; Lotus Notes; mail-aware application; workflow software.

mailer A program used for sending and receiving e-mail.

mailing list On the Internet, a group of people who share a common interest and who automatically receive all the e-mail posted to the listserver or mailing-list manager program.

Contributions are sent as e-mail to the listserver and then distributed to all subscribers. Mailing lists are private or by invitation only; Usenet newsgroups, by contrast, are open to everyone.

See also LISTSERV; listserver; newsgroup; Usenet.

mail reflector On the Internet, a program that manages a mailing list.

See also LISTSERV; listserver; newsgroup; Usenet.

mailslots In Microsoft Windows NT, a connectionless interprocess communications mechanism that provides one-to-many and many-to-one communications, suitable for broadcasting a message to multiple processes.

In Windows NT, mailslots are implemented via the Mailslot File System, which

manages all mailslot access and connections. Mailslots are used for browse requests and logon authentication.

See also named pipe; pipe; semaphore; shared memory; socket.

mailto An HTML attribute that creates a link to an e-mail address. If a user clicks on the mailto link, the browser opens a window for composing an e-mail message to this address.

See also HyperText Markup Language.

mail transport agent A program that manages the transportation and delivery of e-mail. The mail transport agent accepts the message from a mail user agent or client, performs any translations required, and then routes the message.

See also mail user agent; Simple Mail Transfer Protocol.

mail user agent A client program that provides a user interface for sending and receiving e-mail, as well as all the other features required by any e-mail application, including creating, reading, forwarding, and deleting messages.

mainframe computer A large, fast, multiuser computer system, often utilizing multiple processors, designed to manage huge amounts of data and complex computing tasks.

Mainframes are normally found in large corporations, universities, or military installations and can support thousands of users.

The feature that distinguishes mainframe computers from other types is that they perform all processing at one central location; the terminals accessed by users have no local computing power of their own.

See also dumb terminal; minicomputer; network computer; thin client.

maintenance release A software upgrade that corrects minor bugs or adds a few small features. This type of release is

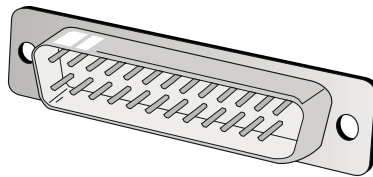
usually distinguished from a major release by an increase in only the decimal portion of the version number; for example, from 3.1 to 3.11 rather than from 3.1 to 4.0.

See also service pack.

male connector Any cable connector with pins designed to engage the sockets on the female connector.

See also female connector.

MALE CONNECTOR



MAN *See* metropolitan-area network.

Management Information Base Abbreviated MIB. A database of network configuration information used by Simple Network Management Protocol (SNMP) and Common Management Information Protocol (CMIP) to monitor or change network settings. MIB provides a logical naming of all resources on the network related to the network's management.

See also Common Management Information Protocol; MIB variables; Simple Network Management Protocol.

Management Information System

Abbreviated MIS. A computer-based information system that integrates data from all the departments that it serves, to provide company management with the information it needs to make timely decisions, track progress, and solve problems.

Manchester encoding In communications, a method used to encode data and timing signals in the same transmitted data stream. The signal state during the first half of the bit period indicates its data value (1 is high; 0 is low). A transition to the opposite state in the middle of the bit period acts as the timing signal.

See also 4B/5B encoding.

mandatory user profile In Microsoft Windows NT, a user profile created and managed by the Administrator and stored on the server and downloaded to the workstation when the user attempts to log on to the network.

The profile is saved with a special extension (.man) so that the user cannot change or alter the profile in any way. Mandatory user profiles can be assigned to a single user or to a group of users.

See also local user profile; roaming user profile.

man pages Short for manual pages. In Unix, the online documentation.

Each man page treats a single topic; some are short, and others are quite long. They are all organized in a standard format using these headings: Name, Synopsis, Description, Files, See Also, Diagnostics, and Bugs.

See also permuted index.

Manufacturing Automation Protocol

Abbreviated MAP. A protocol that was originally developed by General Motors and was designed for use in a manufacturing environment.

See also Technical and Office Protocol.

map 1. To direct a request for a file or a service to an alternative resource. For example, in a virtual memory system, an operating system can translate or map a virtual memory address into a physical address; in a network, drive letters are assigned to specific volumes and directories.

2. An expression of the structure of an object. For example, a memory map describes the use and layout of physical memory.

MAP *See* Manufacturing Automation Protocol.

MAPI *See* Messaging API.

mapped drive Any network hard disk that has been assigned to a drive letter on a client computer.

Marimba, Inc. A company founded by four of the original Sun Microsystems Java team to develop products for the creation and deployment of network-managed applications across the Internet and across corporate intranets.

For more information on Marimba, Inc, see www.marimba.com.

See also Castanet.

mark parity *See* parity.

marquee A scrolling banner on a Web page, usually containing advertising material.

mask A binary number that is used to remove bits from another binary number by use of one of the logical operators (AND, OR, NOT, XOR) to combine the binary number and the mask. Masks are used in IP addresses and file permissions.

See also IP address; subnet mask.

master browser In Microsoft Windows 2000 Server, a computer that maintains a list of computers and services available on the network and makes this information available to other computers on the network.

The master browser provides this list, and the backup browser distributes it to a workgroup or domain.

See also backup browser; browser.

Master Business Continuity

Professional Abbreviated MBCP. A certification from Disaster Recovery Institute International (DRII) that covers advanced information on business continuity planning, disaster recovery, and associated work experience and board certification.

See also Associate Business Continuity Professional; Certified Business Continuity Professional.

Master Certified Novell Engineer Abbreviated Master CNE. A Novell certification program for CNEs who want to focus on the support of enterprise-wide networks. Previously known as Enterprise CNE.

Many consider the Master CNE program the advanced degree of NetWare certification, because candidates must demonstrate proficiency in specialized networking technologies, as well as an in-depth knowledge of the NetWare operating system.

Specializations are available in the general categories of Management, Connectivity, Messaging, and Internet/Intranet Solutions and in the client categories of AS/400 Integration, Unix Integration, and Windows NT Integration.

See also Certified Novell Engineer.

Master Certified Novell Instructor

Abbreviated MCNI. An advanced Novell certification program for trainers who want

to teach Novell courses. An MCNI must have at least two years of teaching experience and be certified as a Master CNE; in addition, an MCNI must also complete an Annual Update Requirement to keep their skills up-to-date.

See also Certified Novell Instructor.

Master CNE *See* Master Certified Novell Engineer.

master file table Abbreviated MFT. In Microsoft Windows 2000, a system file that contains entries for every file and folder on an NTFS volume. The MFT is always the first file on the volume, and for backup purposes, two copies are maintained on each volume.

master replica In Novell NetWare, the main replica for a partition. The master replica must be available during major changes such as partition splitting or merging. Another replica can be assigned as the master replica if the original is lost or damaged.

MAU *See* medium attachment unit; Multi-station Access Unit.

MB *See* megabyte.

Mb *See* megabit.

MBCP *See* Master Business Continuity Professional.

Mbone *See* multicast backbone.

MBps *See* megabytes per second.

Mbps *See* megabits per second.

MCA *See* Microchannel Architecture.

McAfee Associates, Inc. A major provider of antivirus, help desk, security, and encryption software. Merged with Network General Corporation to form Network Associates in 1997.

See also Network Associates, Inc.

MCI *See* Media Control Interface.

MCNE *See* Master Certified Novell Engineer.

MCNI *See* Master Certified Novell Instructor.

MCP *See* Microsoft Certified Professional.

MCR *See* Minimum Cell Rate.

MCSD *See* Microsoft Certified Solutions Developer.

MCSE *See* Microsoft Certified Systems Engineer.

MCT *See* Microsoft Certified Trainer.

MDHCP *See* Multicast DHCP.

MDI *See* multiple document interface.

mean time between failures Abbreviated MTBF. The statistically derived average length of time for which a system component operates before failing. MTBF is expressed in thousands or tens of thousands of hours, also called power-on hours, or POH.

See also mean time to repair.

mean time to repair Abbreviated MTTR. The statistically derived average length of time, expressed in hours, that it takes to make a system-level repair.

See also mean time between failures.

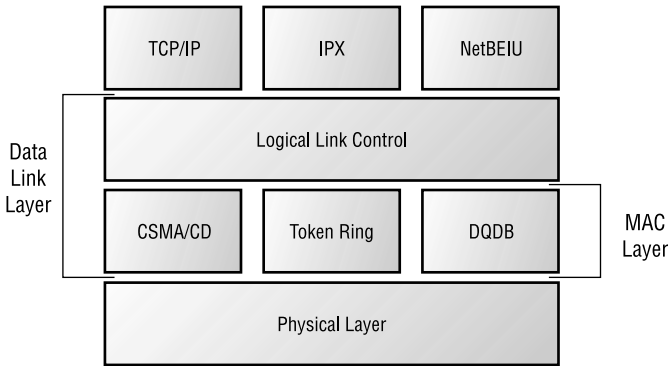
media access control Abbreviated MAC. The lower component of the data-link layer that governs access to the transmission medium. The logical link control layer is the upper component of the data-link layer. MAC is used in CSMA/CD and token-ring LANs as well as in other types of networks.

media access control protocol *See* access control list.

Media Control Interface Abbreviated MCI. A standard interface used for controlling multimedia files and devices. Each device has its own device driver that implements a standard set of MCI functions, such as stop, play, and record.

media filter A device used to convert the output signal from a token-ring adapter board to work with a specific type of wiring. For example, a media filter can link 16Mbps token-ring network interface cards with unshielded twisted-pair (UTP) wiring, thus saving the expense of additional cable runs.

THE MAC SUBLAYER



medium attachment unit Abbreviated MAU. A transceiver that attaches to the AUI (Attachment Unit Interface) port on an Ethernet adapter and provides electrical and mechanical attachments to fiber-optic cable, twisted-pair cable (TP), or other media types.

meg A common abbreviation for megabyte.

mega- Abbreviated M. A prefix meaning one million in the metric system. Because computing is based on powers of 2, mega usually means 2^{20} , or 1,048,576; the power of 2 closest to one million.

megabit Abbreviated Mbit. Usually 1,048,576 binary digits or bits of data. Often used as equivalent to 1 million bits.

See also bit; megabits per second.

megabits per second Abbreviated Mbps. A measurement of the amount of in-

formation moving across a network or communications link in one second, measured in multiples of 1,048,576 bits.

megabyte Abbreviated MB. Usually 1,048,576 bytes. Megabytes are a common unit of measurement for computer memory or hard-disk capacity.

megabytes per second Abbreviated MBps. A measurement of the amount of information moving across a network or communications link in one second, measured in multiples of 1,048,576 bytes.

megahertz Abbreviated MHz. One million cycles per second. A processor's clock speed is often expressed in megahertz.

The original IBM PC operated an Intel 8088 processor running at 4.77 MHz; today's Pentium processor runs at 450 MHz; and IBM has announced an experimental processor capable of speeds of 1000 MHz.

member server In Microsoft Windows NT, a server that participates in a security domain, but that does not act as a domain controller and does not store domain user accounts.

A member server can be used to store shared data, but a user must be authenticated by a primary or a backup domain controller before gaining access to the member server.

memory The primary physical RAM installed in the computer. The operating system copies applications from disk into memory, where all program execution and data processing take place, and then writes the results back to disk. The amount of memory installed in the computer can determine the size and number of programs that it can run, as well as the size of the largest data file.

See also dynamic RAM; static RAM; swapping; virtual memory.

memory address The exact location in memory that stores a particular data item or program instruction.

memory board A printed circuit board containing memory chips. When all the sockets on a memory board are filled and the board contains the maximum amount of memory that it can manage, it is said to be “fully populated.”

memory cache An area of high-speed memory on the processor that stores commonly used code or data obtained from slower memory, eliminating the need to access the system’s main memory to fetch instructions.

The Intel 82385 cache controller chip was used with fast static RAM on some systems to increase performance, but more up-to-date processors include cache-management functions on the main processor.

The Intel Pentium contains two separate 512KB caches, one each for data and instructions.

See also cache; level 2 cache.

memory chip A chip that holds data or program instructions. A memory chip may hold its contents temporarily, as in the case of RAM, or permanently, as in the case of ROM.

memory leak A programming error that causes a program to request new areas of computer memory rather than reusing the memory already assigned to it. This causes the amount of memory in use by the program to increase as time goes on. In a worst case, the application may consume all available memory and stop the computer.

memory management The way in which the operating system handles the use of memory, usually as a combination of physical memory and virtual memory.

When applications are loaded, they are assigned space in which to run and store data. As they are removed, the memory space they occupied is released for use by the next program to run.

See also memory leak.

memory management unit Abbreviated MMU. The part of the processor that manages the mapping of virtual memory addresses to actual physical addresses.

In some systems, such as those based on early Intel or Motorola processors, the MMU was a separate chip; however, in most of today's systems, the MMU is integrated into the processor itself.

memory map The organization and allocation of memory in a computer. A memory map indicates the amount of memory used by the operating system, as well as the amount remaining for use by applications.

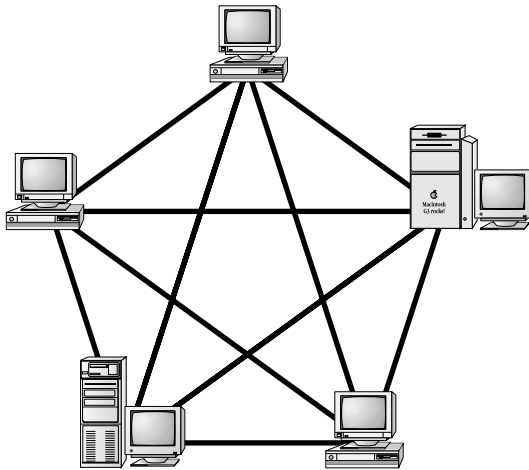
memory-resident Always located in the computer's memory and available for use; not swapped out.

Merge An MS-DOS emulator from Locus Computing that runs MS-DOS and Windows applications under Unix.

Merge provides a complete MS-DOS environment and acts as though your system is running only MS-DOS. Merge is available for UnixWare, SCO, and other systems.

mesh network A network topology in which every device is connected by a cable to every other device on the network. Multiple links to each device are used to provide network link redundancy.

MESH NETWORK



message channel A form of interprocess communication found in multitasking operating systems. Interprocess communications allow two programs running in the same computer to share information.

See also pipe; queue; semaphore.

message-enabled application *See* mail-aware application.

Message Handling Service

Message Handling Service Abbreviated MHS. A protocol for e-mail storage, management, and exchange, originally created by Action Technologies and then licensed by Novell. MHS can collect, route, and deliver e-mail and other files by using store-and-forward technology.

Message Handling System The CCITT X.400 standard protocol for global store-and-forward messaging. The X.400 standard specifies everything from the type of data that a message can contain to rules for converting between different message types, such as from fax to text or vice versa.

See also Consultative Committee for International Telephony and Telegraphy; X.500.

message switching A routing method that uses store-and-forward techniques. Each message contains a destination address and is passed from source to destination through a series of intermediate nodes.

At each intermediate node, the message is stored briefly, reviewed, and then forwarded to the next node. Message switching allows a network operating system to regulate traffic and to use the available communications links effectively.

Messaging API Abbreviated MAPI. An interface used to add messaging capabilities to any Microsoft Windows application. MAPI handles the details of message storage and forwarding and directory services.

Originally developed by Microsoft, MAPI has become a widely supported industry standard.

MAPI Version 3.2 provides for cross-platform messaging independent from the operating system.

See also mail-aware application; Vendor Independent Messaging.

Messenger service In Microsoft Windows NT, a service that sends and receives messages sent by the system administrator or by the Alerter service.

See also Alerter service; service.

metafile A file that contains both data and output control information. For example, a graphics metafile contains not only a graphical image of some kind, but also information on how the image is to be displayed. Use of a metafile allows one single version of the image to be output to a variety of display devices.

metering The process of tracking application software use and availability across a network to ensure that the terms of the license are being met. Metering can also be used to predict when a license will expire and require renewal.

metropolitan-area network Abbreviated MAN. A public, high-speed network, capable of voice and data transmission over a distance of up to 80 kilometers (50 miles). A MAN is smaller than a wide-area network (WAN) but larger than a local-area network (LAN).

MFS *See* Macintosh File System.

MFT *See* master file table.

MHS *See* Message Handling Service.

MHz *See* megahertz.

MIB See Management Information Base.

MIB variables The information stored in a Management Information Base (MIB) database, which can be accessed and managed by network management protocols such as Simple Network Management Protocol (SNMP).

See also Management Information Base; Simple Network Management Protocol.

Microchannel Architecture Abbreviated MCA. A 32-bit, proprietary expansion bus, first introduced by IBM in 1987 for the IBM PS/2 range of computers and also used in the IBM RS/6000 series.

MCA was designed for multiprocessing. It allows expansion boards to identify themselves, thus eliminating many of the conflicts that arose through the use of manual settings in the original bus. The MCA bus can also be driven independently by multiple bus master processors.

MCA is physically and electronically incompatible with expansion boards that follow the earlier 16-bit AT bus standard; the boards are about 50 percent smaller, and the bus depends on proprietary integrated circuits.

See also Extended Industry Standard Architecture; Industry Standard Architecture; local bus; Peripheral Component Interconnect local bus.

microcode Low-level instructions that define how a particular microprocessor works by specifying what the processor does when it executes a machine-language instruction.

See also machine language.

Microcom Networking Protocol Abbreviated MNP. A set of communications protocols from Microcom, Inc., that has become the standard for data compression and error detection and correction, as follows:

- MNP 1 to 4 define hardware error control.
- MNP 5 describes a method of data compression that achieves a 2-to-1 compression ratio.
- MNP 6 describes a communication protocol that begins with V.22 bis modulation and then switches to V.29 when possible.
- MNP 7 describes a method of data compression that achieves a 3-to-1 compression ratio.
- MNP 8 is based on MNP 7 and adds a V.29 technique that lets half-duplex devices operate as full-duplex.
- MNP 9 contains a proprietary technique that provides good performance over a wide variety of link types.
- MNP 10 describes an extremely rigorous error control protocol that is well suited for use on extremely noisy links. MNP 10 has been adopted for use in cellular modems.

These days the CCITT V standards are implemented in modems due to their worldwide acceptance; however, some modem manufacturers offer both.

For more information on Microcom, Inc., see www.microcom.com

See also Consultative Committee for International Telephony and Telegraphy.

microcomputer Any computer based on a single-chip processor. Many of today's microcomputers are as powerful as main-frame models designed just a few years ago; they are also smaller and cheaper.

See also workstation.

microkernel An alternative operating-system kernel design developed by researchers at Carnegie-Mellon University and implemented in the Mach operating system.

Traditionally, the kernel has been a monolithic piece of the operating system, resident in memory at all times. It takes care of operations as varied as virtual memory management, network support, file input/output, and task scheduling.

The microkernel is a stripped-down kernel that is only concerned with loading, running, and scheduling tasks. All other operating system functions (virtual memory management, disk input/output, and so on) are implemented and managed as tasks running on top of the microkernel.

micron A unit of measurement. One millionth of a meter, corresponding to approximately 1/25,000 of an inch. The core diameter of fiber-optic cable for networks is often specified in terms of microns; 62.5 microns is a common size.

Micron Technology, Inc. A major manufacturer of DRAM (dynamic RAM) components, flash memory, synchronous SRAM (static RAM), and graphics DRAM. Micron Technology also owns a majority interest in Micron Electronics, a manufacturer of direct-sales PCs and servers for the

consumer, government, business, and education markets.

For more information on Micron Technology, see www.micronpc.com.

microprocessor A central processor unit on a single chip, often referred to as the processor.

The first microprocessor was developed by Intel in 1969. The microprocessors most often used in Apple Macintosh computers are manufactured by Motorola, and Intel microprocessors are commonly used in the PC.

microsegmentation The division of a network into smaller segments, usually with the aim of increasing bandwidth.

Microsoft Access A popular relational database program from Microsoft.

Microsoft BackOffice A network software suite from Microsoft that runs on Windows NT Server and consists of Microsoft Exchange Server, Microsoft SQL Server, Microsoft Site Server, Microsoft SNA Server, Microsoft Proxy Server, and Microsoft System Management Server.

See also Microsoft Office.

Microsoft Certified Product Specialist *See Microsoft Certified Professional.*

Microsoft Certified Professional Abbreviated MCP. A basic certification from Microsoft designed to establish expertise with at least one Microsoft operating system. Previously known as the Microsoft Certified Product Specialist.

An MCP+Internet certification is also available and covers the use of Internet Explorer, designing and building a Web site, and configuring and troubleshooting Microsoft's implementation of TCP/IP.

Microsoft Certified Solutions

Developer Abbreviated MCSD. An advanced certification from Microsoft for computer professionals who develop custom applications using Microsoft products and computer programming language packages. Three core exams focus on operating system architecture, and two elective exams concentrate on programming and database concepts and skills.

Microsoft Certified Systems

Engineer Abbreviated MCSE. An advanced certification from Microsoft that requires passing four core and two elective exams. The core exams cover the basic concepts and skills involved in installing, using, maintaining, and troubleshooting a Windows NT Server network, and the elective exams cover other server topics and advanced networking concepts.

An MCSE+Internet certification is also available and covers the use of Internet Explorer, designing and building a Web site using Internet Information Server, and configuring and troubleshooting Microsoft's implementation of TCP/IP.

Microsoft Certified Trainer Abbreviated MCT. A certification from Microsoft for technical trainers. An MCT certification is required before you can teach Microsoft Official Curriculum courses at Microsoft Authorized Technical Education Centers.

Microsoft Corporation The world's largest and most successful software company, founded in 1975 by Bill Gates and Paul Allen.

Microsoft's great initial success was in supplying IBM with the PC-DOS operating system for the IBM PC and then providing versions of MS-DOS to the clone manufacturers. Microsoft released Windows 3 in May 1990 and continued to upgrade Windows regularly. The release of Windows NT during 1993 consolidated Microsoft's position as a leading developer of operating systems, and Windows 2000 will continue this trend.

In addition to operating systems and its extensive computer language products, Microsoft markets a wide range of applications, including Microsoft Word, a word processor; Microsoft Excel, a spreadsheet; Microsoft Access, a database program; Microsoft Publisher, a desktop publishing program; and Microsoft Office, an integrated software suite. Microsoft also provides an extensive set of Internet-related products, including Internet Explorer, FrontPage, and Internet Information Server.

The Microsoft consumer division now has more than 50 products, including popular multimedia titles such as Encarta and Cinemania, an interactive movie guide. Microsoft Press publishes and distributes computer-related books and CD-ROM products to bookstores.

For more information on Microsoft, see www.microsoft.com.

Microsoft Disk Operating System

Microsoft Disk Operating System

Abbreviated MS-DOS. A single-user, single-tasking operating system, with either a command-line or shell interface, for use on Intel processors, first introduced in 1981.

Microsoft Excel A popular spreadsheet program from Microsoft.

Microsoft Exchange Server A client/server message-management system from Microsoft. Exchange Server provides enterprise-wide message exchange by integrating e-mail, scheduling, document sharing, and electronic forms and also connects to the Internet and other networks outside the enterprise allowing global messaging. Exchange uses Microsoft Outlook as the client. Exchange Server is a component of Microsoft BackOffice.

Microsoft FrontPage An HTML editor from Microsoft that combines Web-page creation with graphical Web site publication and management.

Microsoft Internet Explorer A popular Web browser from Microsoft, which is also integrated into Windows 98.

Microsoft Internet Information

Server A powerful and capable Web server package from Microsoft that runs on Windows NT Server.

See also Microsoft Site Server.

Microsoft Management Console

Abbreviated MMC. In Microsoft Windows 2000, a framework for Microsoft and third-party management applications that provides a consistent user interface.

Additional components called snap-ins provide the management functions.

See also Computer Management; snap-in.

Microsoft LAN Manager A network operating system, developed by Microsoft and 3Com, based on a version of OS/2; client PCs can run OS/2, MS-DOS, Unix, or Macintosh. Disk mirroring, disk duplexing, and UPS (uninterruptible power supply) monitoring functions are available. The network operating system supports IPX/SPX, TCP/IP, and NetBEUI. LAN Manager interoperates with, and has largely been superseded by, Windows NT Server.

Microsoft NetMeeting An Internet-based audio- and video-conferencing application from Microsoft.

Microsoft NetShow A client/server streaming audio and video application from Microsoft. The NetShow server runs on Windows NT and streams audio, video, and animation to the client, a multimedia player.

Microsoft Office A popular business software suite from Microsoft that runs on Microsoft Windows and the Macintosh and includes Microsoft Word, Microsoft Excel, Microsoft PowerPoint, Microsoft Outlook, Microsoft Access, Microsoft Publisher, Microsoft FrontPage, and Microsoft PhotoDraw.

See also Microsoft BackOffice.

Microsoft Office Expert Abbreviated MOE. An advanced certification from Microsoft that demonstrates complete familiarity with the Microsoft Office suite of

applications, both individually and as a group.

Microsoft Office Expert Specialist

Abbreviated MOES. A middle-level certification from Microsoft that demonstrates familiarity with more complex tasks in the Microsoft Office suite of applications.

Microsoft Office Proficient Specialist

Abbreviated MOPS. A basic certification from Microsoft that demonstrates familiarity with basic tasks in Microsoft Word or Microsoft Excel.

Microsoft Outlook A popular e-mail application from Microsoft, which also includes scheduling and calendar functions, a contact-management module, and a simple project-management tool.

Microsoft Outlook Express An easy-to-use e-mail application, distributed with Internet Explorer; both are available from Microsoft.

Microsoft Services for Macintosh

Software that allows a Macintosh client to share files on a Windows NT server.

Microsoft Services for NetWare Software for NetWare servers that allows NetWare clients to access Windows NT services.

Microsoft Site Server A Microsoft package of tools for creating and managing Internet or intranet Web sites.

Site Server includes Site Analyst, Usage Analyst, Personalization System, Commerce Server, Publishing Solution, Knowledge Management Solution, and the Analysis Solution.

See also Microsoft Internet Information Server.

Microsoft SNA Server A server component from Microsoft that provides Windows, Macintosh, MS-DOS, and OS/2 clients with access to IBM's AS/400 and mainframe systems using System Network Architecture (SNA).

Microsoft SQL Server A Windows NT Server-based relational database management system from Microsoft that also includes development tools, system management tools, data replication processes, and an open development environment.

Microsoft Systems Management Server A set of network management tools from Microsoft, designed to provide a single point for managing network and client hardware and software (SMS hardware and software inventory functions track more than 200 properties for each desktop), software distribution, application metering and licensing, and troubleshooting.

Microsoft Terminal Server A software package that provides clients with access to Windows-based applications running on the server rather than on the local system. The server receives and processes all keystrokes and mouse clicks sent from the client and sends the output back to the appropriate client. The server manages all resources for each connected client and provides each logged-in user with his or her own environment.

Microsoft Transaction Server Often abbreviated MTS. A package from Microsoft for developing and deploying

distributed transaction-processing applications.

See also online transaction processing.

Microsoft Visual Basic Often abbreviated VB. A version of the BASIC programming language that allows developers to create Windows applications quickly and easily.

Microsoft Windows A general name for the family of operating systems available from Microsoft that includes Windows CE, Windows 95/98, and Windows 2000.

Microsoft Windows 2000 A family of powerful operating systems from Microsoft, based on Windows NT, including Windows 2000 Professional for workstation users, Windows 2000 Server, and Windows 2000 Advanced Server. Windows 2000 Datacenter Server is a 64-bit version capable of addressing 64GB of physical RAM and managing 16 processors.

Microsoft Windows 3.1 A 16-bit graphical operating environment that runs on top of MS-DOS on Intel-based PCs, featuring overlapping windowed areas, drop-down menus, and mouse support. The three main elements of Windows 3.1 are the File Manager (used to manage files, directories, and disks), the Program Manager (which manages applications), and the Print Manager (which coordinates printers and printing).

Microsoft Windows 95 A 32-bit, multitasking, multithreaded desktop operating system capable of running MS-DOS, Windows 3.1, and Windows 95 applications. It supports Plug and Play (on the appropriate hardware) and adds an enhanced FAT file

system in the Virtual FAT, which allows filenames of up to 255 characters while also supporting the MS-DOS 8.3 filenames conventions.

Applets include WordPad (a word processor), Paint, and WinPad (a personal information manager), as well as system tools such as Backup, ScanDisk, Disk Defragmenter, and DriveSpace. The Start button and desktop Taskbar make application management easy and straightforward.

Windows 95 supports TCP/IP, IPX/SPX, NetBEUI, NDIS, FTP, SLIP, and PPP communications and networking protocols.

See also protocol.

Microsoft Windows 98 An evolutionary upgrade to Windows 95; includes an integrated Web browser, several other Web-based tools for conferencing, e-mail, Web page creation, and Web publishing.

Windows 98 also includes a complete set of capable and easy-to-use tools for tasks such as defragmenting hard disks, compressing files, testing disks, and monitoring the system and supports new multimedia hardware and entertainment technologies.

You can choose between the classic Windows interface familiar to Windows 95 users and the Active Desktop.

Microsoft Windows CE A small, 32-bit operating system for hand-held portable computers and other specialized devices such as telephones, cable decoder boxes, and television sets.

Microsoft Windows NT Server Microsoft's flagship 32-bit network operating system, which provides high levels of security,

manageability, reliability, and performance. Runs on servers based on Intel, Alpha, and MIPS processors, includes preemptive multitasking and multiple threads of execution, and supports all major networking protocols.

NT Server supports fault tolerance with disk mirroring, disk duplexing, RAID (redundant array of inexpensive disks), and UPS (uninterruptible power supply) monitoring.

The user interface is similar to those available in other members of the Windows family. The latest version of NT Server includes Internet Information Server (a Web server), DNS Server, and multiprotocol router support. All popular interprocess communications protocols for distributed computing are available, including Windows Sockets and Remote Procedure Calls (RPCs).

See also Microsoft Windows 2000.

Microsoft Windows NT Services for

Unix A package from Microsoft Corporation that provides Unix services on a Windows NT system, including several popular command-line utilities, the Perl programming language, and a variety of Unix scripting tools.

Microsoft Windows NT Workstation

Microsoft's high-end 32-bit operating system, which provides high levels of security, manageability, reliability, networking, and performance. Runs on workstations based on Intel, Alpha, and MIPS processors and supports preemptive multitasking and multiple threads of execution. The user interface is similar to those available in other members of the Windows family.

See also Microsoft Windows 2000.

Microsoft Word A popular and fully featured word processor from Microsoft. Wizards help users with tasks such as mail merge and formatting, and you can customize the various toolbars and menu bars to suit your individual needs. Word also supports a powerful macro language

micro-to-mainframe Any form of connection that attaches a PC to a mainframe-based network. Often used to describe software (called terminal-emulation software) that allows the microcomputer to access data and applications on the mainframe system.

See also terminal emulation.

microwave A method of radio transmission that uses high-frequency waves (in the range of 1 to 30 gigahertz) for line-of-sight broadband communications. It requires a repeater station every 20 miles or so because of the curvature of the earth. Microwaves are used for satellite communications, for communications between two buildings in a metropolitan area, and across large open areas such as lakes and rivers where laying a cable may be impractical.

See also broadband network; repeater.

middleware A category of software that shields an application from the underlying mechanics of a network so that the developers of an application do not have to know in advance which network and communications protocols will be used.

Middleware is often implemented in a client/server environment in which it allows systems to exchange information or connect, even though they use different interfaces.

midsplit A special type of broadband cable system that divides the available frequencies into two groups: one for transmission and the other for reception.

See also broadband network.

migration In Novell NetWare, the process of upgrading bindery and other information to a newer version of the network operating system.

See also bindery.

Migration Tool for NetWare A Microsoft Windows NT utility loaded with the Gateway Service for NetWare that duplicates user account and group and security information from a NetWare server onto a Windows NT server, allowing a migration from NetWare to NT.

milli- Abbreviated m. A prefix meaning one-thousandth in the metric system, often expressed as 10^{-3} .

millisecond Abbreviated ms or msec. A unit of measurement equal to one-thousandth of a second. In computing, hard disk and CD-ROM drive access times are often described in milliseconds; the higher the number, the slower the disk system.

millivolt Abbreviated mv. A unit of measurement equal to one-thousandth of a volt.

MIME *See* Multipurpose Internet Mail Extension.

MIME Content Type Information contained in a Multipurpose Internet Mail Extension (MIME) message header indicates the content type of the data contained in the next part of the message. Table M.1 lists the seven major content types as well as the subtypes.

See also Secure MIME.

TABLE M.1 MIME CONTENT TYPES

Content Type	Subtype	Description
Text	Plain	Unformatted text.
	Rich Text	Formatted text.
Multipart	Mixed	Multiple parts of different types.
	Alternative	Multiple parts containing the same data in different formats.
	Digest	Message digest with multiple parts all in the same format.

TABLE M.1 MIME CONTENT TYPES (CONTINUED)

Content Type	Subtype	Description
	Parallel	Multiple parts to be viewed simultaneously, such as audio and video data.
Message	RFC822	E-mail message.
	Partial	Message was fragmented for transmission as the original was larger than 64Kb.
	External-body	External data should be referenced, such as data at an ftp site.
Application	Octet-stream	Binary data associated with no known application.
	PostScript	PostScript document ready for printing.
	SGML	SGML data.
Image	JPEG	JPEG image data.
	GIF	GIF image data.
Audio	Basic	Audio data.
Video	MPEG	MPEG video data.

minicomputer A medium-sized computer running a multitasking operating system capable of managing more than 100 users simultaneously, suitable for use in a small company or a single corporate or government department.

See also mainframe computer; workstation.

mini-hard disk A hard disk mounted on a Type III PC Card.

See also PC Memory Card International Association.

Minimum Cell Rate Abbreviated MCR. In Asynchronous Transfer Mode (ATM), the minimum cell rate that must be provided for a connection to conform to a specific quality of service (QoS) class.

See also Asynchronous Transfer Mode; Available Cell Rate; Peak Cell Rate; Sustainable Cell Rate.

MIPS Acronym for millions of instructions per second. A measure of the processing speed of a computer's CPU.

See also benchmark program.

mirroring The process of duplicating stored information in real time to protect vital data from unexpected hardware failures.

See also clustering; disk duplexing; disk mirroring; fault tolerance; redundant array of inexpensive disks.

mirror site 1. A duplicate Web site. A mirror site contains the same information as the original Web site and reduces traffic on that site by providing a local or regional alternative.

2. A duplicate data center. Large companies running mission-critical applications often mirror their entire data center so that the company can continue to function if the main center is hit by a natural disaster.

MIS *See* Management Information System.

mission-critical application A computer application whose function is vital to the operation of the corporation using it; also called line-of-business application.

mixed mode In Microsoft Windows 2000 Server, a domain that contains at least one Windows NT domain controller. As a safety precaution, all Windows 2000 domains start up in mixed mode and must be converted to native mode manually.

See also native mode.

MJ *See* modular jack.

MLID *See* multiple-link interface driver.

MMC *See* Microsoft Management Console.

MMJ *See* modified modular jack.

MMU *See* memory management unit.

mnemonic Pronounced "nee-monic." A name or abbreviation used to help you remember a long or complex instruction. Programming languages use many mnemonics to represent complex instructions.

MNP *See* Microcom Networking Protocol.

MO *See* Magneto-optical storage.

mobile computing 1. The daily use of a portable or laptop computer as a normal part of the workday.

2. Techniques used to establish links to a network by employees who move from one remote location to another, such as members of a sales staff or telecommuters who work from home. Once the connection is made, users log in and access network resources as easily as if they were working from a computer in the corporate office.

See also docking station; laptop computer; portable computer; port replicator; wireless communications.

mobile IP A mechanism that allows mobility on the Internet by allowing a computer to use the same IP address as it moves from one location to another. Mobile IP also allows a user to change from one medium, such as Ethernet, to another, perhaps a wireless connection.

See also Ethernet; Internet Protocol.

modem Contraction of modulator/demodulator; a device that allows a computer to transmit information over a telephone line.

The modem translates between the digital signals that the computer uses and analog signals suitable for transmission over telephone lines. When transmitting, the modem modulates the digital data onto a carrier signal on the telephone line. When receiving, the modem performs the reverse process to demodulate the data from the carrier signal.

Modems usually operate at speeds up to 56Kbps over standard telephone lines and at higher rates over leased lines.

See also baud rate; external modem; internal modem.

modem eliminator A device that allows two computers to be linked without using modems. In an asynchronous system, in which the serial ports of two PCs are connected, the modem eliminator is a null-modem cable. In synchronous systems, a modem eliminator must also provide timing functions to synchronize communications.

modem server A LAN server that allows a network user to dial out of the network into the Public Switched Telephone System or to access leased lines for asynchronous communications. Also called an asynchronous communications server or a dial-in/dial-out server.

moderated newsgroup On the Internet, a Usenet newsgroup or mailing list that is managed by one or more people in an attempt to maintain standards.

All posts to the newsgroup are reviewed by the moderator to make sure they meet the standards the group has set for subject matter and commercial content before being passed on to the whole group. Moderation is not censorship but an attempt to avoid some of the more extreme antics of those who enjoy flame wars.

See also flame war; listserver; mailing list; moderator.

moderator A person or small group of people who review the contents of all posts to a Usenet newsgroup or Internet mailing list to ensure that the postings meet the standards set by the group. Moderators are almost always volunteers, so be nice to them.

See also moderated newsgroup.

modified modular jack Abbreviated MMJ. A six-pin connector developed by Digital Equipment Corporation used to connect serial lines to terminal devices. MMJ jacks have a side-locking tab and so can be distinguished from RJ-11 jacks, which have a center-mounted tab.

See also modular jack; RJ-11.

modular jack Abbreviated MJ. The jack used to connect telephone cables to a wall-mounted face plate.

See also modified modular jack; RJ-11.

modulation In communications, the process used by a modem to add the digital signal onto the carrier signal so that the signal can be transmitted over a telephone line. The frequency, amplitude, or phase of a signal

may be modulated to represent a digital or an analog signal.

See also carrier signal; demodulation.

module In programming, a self-contained portion of a larger program written, tested, and compiled separately. Normally, a module performs only one specific operation.

See also NetWare Loadable Module; Virtual Loadable Module.

MOE *See* Microsoft Office Expert.

MOES *See* Microsoft Office Expert Specialist.

MONITOR A Novell NetWare server utility that displays information about the server, including the time elapsed since the server was booted, percent utilization of the server's processor, number of disk blocks available, the number of blocks waiting to be written to disk, the number of connections to the server, and the number of files being accessed. It also provides information about server disks, LAN drivers, and other attached mass-storage devices.

See also SERVMAN; SET.

Monitrix for NetWare A package of network management utilities from Cheyenne Software, Inc., that includes hardware and software inventory, server monitoring, traffic monitoring, and virus protection. The package also includes automatic task-scheduling and reporting functions.

Moore's Law States that the number of transistors on a chip of a given size doubles approximately every 18 months. Named

for Intel's Gordon Moore, who first made this statement in 1965.

MOPS *See* Microsoft Office Proficient Specialist.

Mosaic A Web browser, released on the Internet in 1993 by the National Center for Supercomputing Applications (NCSA) at the University of Illinois, and placed in the public domain.

Although Mosaic was one of the first graphical Web browsers and was available free for most computing platforms, it has been overtaken by Netscape Navigator and Microsoft Internet Explorer.

motherboard The main printed circuit board in a computer, containing the central processing unit, appropriate coprocessor and support chips, device controllers, and memory. It may also include expansion slots to give access to the computer's internal bus.

Motorola, Inc. A major manufacturer of microprocessors, including those used in Macintosh computers, founded in 1928 in Chicago. Motorola is also extensively involved in radio and data communications and in automotive and industrial products.

For more information on Motorola, see www.motorola.com.

mount 1. The method by which nodes access network resources in the Network File System (NFS) and other networks.

2. To load a disk volume or tape archive so that users can access files and other resources.

mounted drive In Microsoft Windows 2000, a drive or disk partition attached to a junction point on an NTFS volume.

Mounted drives work just like other drives, but have a label or a name rather than a single drive letter.

See also junction point.

mouse A small input device with one or more buttons used with graphical user interfaces. As the mouse moves, an on-screen mouse cursor follows; all movements are relative. Once the pointer is in the correct position on the screen, you press one of the mouse buttons to initiate an action or operation.

See also trackball.

Moving Pictures Experts Group Abbreviated MPEG. A set of image-compression standards and file formats that defines a compression method for desktop audio, animation, and video.

MPEG is a lossy compression method that results in some data loss when a video clip is compressed. The following standards are available:

- **MPEG-1** The original MPEG standard, designed for CD-ROM use, with a bandwidth of 1.5Mbps, two audio channels, and non-interlaced video.
- **MPEG-2** An extension to MPEG-1, designed for broadcast television, including HDTV (High-Definition Television), with a bandwidth of up to 40Mbps, five audio channels, interlaced video, and a wider range of frame sizes.
- **MPEG-3** A standard designed for HDTV until it was discovered that

MPEG-2 covered HDTV. This standard is no longer used.

- **MPEG-4** A standard designed for video phones and multimedia applications, with a bandwidth of up to 64Kbps.

See also lossy compression.

MPEG *See* Moving Pictures Experts Group.

MPOA *See* Multiprotocol over ATM.

MPPP *See* PPP Multilink.

ms *See* millisecond.

MSAU *See* Multi-station Access Unit.

MS-DOS *See* Microsoft Disk Operating System.

MS-DOS prompt A visual confirmation that MS-DOS is ready to receive input from the keyboard. The default prompt includes the current drive letter followed by a colon and a greater-than symbol, as in C:>. You can customize the MS-DOS prompt by using the PROMPT command.
See also command line; command prompt.

msec *See* millisecond.

MTBF *See* mean time between failures.

MTRR *See* mean time to repair.

multicast 1. A special form of broadcast in which copies of a message are delivered to multiple stations but not to all possible stations.

2. A data stream from a server from which multiple viewers can simultaneously watch a video.

See also broadcast; multicasting; unicast.

multicast backbone Abbreviated Mbone. A method of transmitting digital video over the Internet in real time.

The TCP/IP protocols used for Internet transmissions are unsuitable for real-time audio or video; they were designed to deliver text and other files reliably, but with some delay. MBONE requires the creation of another backbone service with special hardware and software to accommodate video and audio transmissions; the existing Internet hardware cannot manage time-critical transmissions.

See also IP Multicast; multicasting; Real-time Transport Protocol; Resource Reservation Protocol; unicast.

Multicast DHCP Abbreviated MDHCP. In Microsoft Windows 2000 Server, an extension of Dynamic Host Control Protocol (DHCP) that distributes multicast address configurations to network clients automatically.

multicasting An Internet standard that allows a single host to distribute data to multiple recipients.

Multicasting can deliver audio and video content in real time so that the person using the system can interact with the data stream. A multicast group is created, and every member of the group receives every datagram. Membership is dynamic; when you join a group, you start to receive the datastream, and when you leave the group, you no longer receive the datastream.

See also broadcast; IP Multicast; unicast.

multidrop line A circuit connecting several stations or nodes on a single logical link; also called a multipoint line.

A multidrop line is often used in IBM's SNA (Systems Network Architecture). It is controlled by a primary station, and the other nodes are considered secondary.

multihomed computer Any computer system that contains multiple network interface cards and is attached to several physically separate networks; also known as a multihomed host. The term can also be applied to a computer configured with multiple IP addresses for a single network interface card.

multihomed host *See* multihomed computer.

multilayer A printed circuit board that contains several layers of circuitry. The layers are laminated together to make a single board, onto which the other discrete components are added.

multilink In Microsoft Windows NT Server Remote Access Server, the ability to combine several slower data streams into one network connection. For example, you can use two 28.8Kbps modems to form a single 57.6Kbps connection.

Multilink Point-to-Point Protocol *See* PPP Multilink.

Multimaster Replication In Microsoft Active Directory, the process by which Active Directory domains replicate and resolve conflicting updates as peers.

All Active Directory domain controllers maintain a writable copy of the domain database, and updating any record on any domain

controller will ensure that all other domain controllers are eventually updated.

See also Active Directory.

multimedia A computer technology that displays information using a combination of full-motion video, animation, sound, graphics, and text, with a high degree of user interaction.

multimode fiber A fiber-optic cable with a wide core that provides multiple routes for light waves to travel.

Its wider diameter of between 25 to 200 microns prevents multimode fiber from carrying signals as far as single-mode fiber due to modal dispersion.

See also single-mode fiber.

multipart virus A form of virus that infects both the boot sector of a hard disk and executable files. Multipart viruses are difficult to locate because they use stealth and polymorphic techniques to avoid detection.

See also boot sector virus; file-infesting virus; macro virus; polymorphic virus; stealth virus; vaccine; virus.

multiple document interface Abbreviated MDI. In Microsoft Management Console, an interface that allows more than one independently-running snap-in to be loaded into the management window at the same time.

See also Microsoft Management Console; snap-in.

multiple-link interface driver Abbreviated MLID. A Novell Open Datalink Interface (ODI) device driver that manages the

sending and receiving of packets to and from the physical network medium.

See also link-support layer; Open Data-link Interface/Network Driver Interface Specification Support.

multiple provider router In Microsoft Windows 2000, a function that allows two or more Win32 network application program interface (API) providers to exist at the same time.

Multiple Universal Naming Convention Provider Abbreviated MUP. In Microsoft Windows 2000, a function that allows two or more Universal Naming Convention (UNC) providers to exist at the same time.

MUP decides which UNC provider should handle the UNC request and forwards the request to that provider automatically.

Multiple Virtual Storage Abbreviated MVS. IBM's standard operating system for large mainframe computers.

See also virtual machine.

multiplexer Often abbreviated mux. A device that merges several lower-speed transmission channels into one high-speed channel at one end of the link. Another multiplexer reverses this process at the other end of the link to reproduce the low-speed channels.

See also frequency-division multiplexing; inverse multiplexing; statistical multiplexing; time-division multiplexing.

multiplexing A technique that transmits several signals over a single communications channel. Frequency-division multiplexing separates the signals by modulating

multipoint line

the data into different carrier frequencies. Time-division multiplexing divides the available time among the various signals. Statistical multiplexing uses statistical techniques to dynamically allocate transmission space depending on the traffic pattern.

See also inverse multiplexing.

multipoint line *See* multidrop line.

multiprocessing The ability of an operating system to use more than one processor in a single computer.

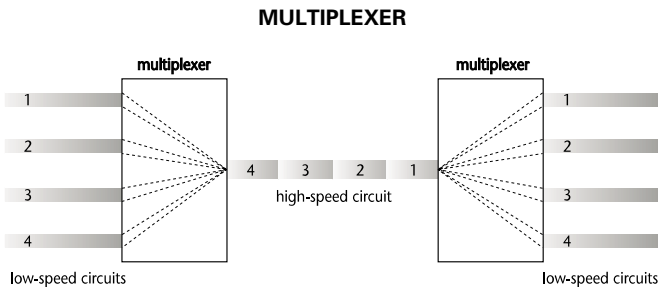
Symmetrical multiprocessing refers to the operating system's ability to assign tasks dynamically to the next available processor. Asymmetrical multiprocessing requires that the original program designer choose the processor to use for a given task when writing the program.

Multiprotocol over ATM Abbreviated MPOA. A proposal to allow layer three network routing protocols over an Asynchronous Transfer Mode Switched network. Multiprotocol over ATM allows corporations to take advantage of ATM's benefits while still maintaining legacy LANs.

See also Integrated-Private Network-to-Network Interface; IP over ATM; LAN Emulation.

Multipurpose Internet Mail Extension Abbreviated MIME. An Internet specification that allows users to send multiple-part and multimedia messages, rather than simple text messages. A MIME-enabled e-mail application can send PostScript images, binary files, audio messages, and digital video over the Internet.

See also MIME Content Type; Secure MIME; uuencode; uudecode.



multiserver network A network that uses two or more file servers.

Multi-station Access Unit Abbreviated MAU; sometimes abbreviated MSAU. A

multiport wiring hub for token-ring networks that can connect as many as eight lobes to a ring network. IBM refers to an MAU that can be managed remotely as a Controlled Access Unit.

multitasking The simultaneous execution of two or more programs in one computer.

multithreading The concurrent processing of several tasks or threads inside the same program. Because several threads can be processed in parallel, one thread does not need to wait for another to finish before starting.

multiuser Describes an operating system that supports more than one simultaneous user. MS-DOS and Windows are single-user operating systems. Unix and its derivatives and networking operating systems are multiuser systems.

multiuser Windows See Terminal services.

MUP See Multiple Universal Naming Convention Provider.

mutual authentication A feature of authentication schemes such as Kerberos that requires both the client and the server to prove their identity to each other before authentication can proceed.

See also Kerberos.

mux See multiplexer.

mv See millivolt.

MVS See Multiple Virtual Storage.

N

n *See* nano-.

NAEC *See* Novell Authorized Education Center.

NAK *See* negative acknowledgment.

named pipe A communications API used by applications running on a network. Named pipes provide connection-oriented messaging between a client and a server using routines similar to those used in normal operations for opening, reading, and writing to files.

See also mailslots; pipe; semaphore; shared memory; socket.

name resolution The process of translating the appropriate numerical IP address, which is required by a computer, into a name that is more easily understood and remembered by a person.

In the TCP/IP (Transmission Control Protocol/Internet Protocol) environment, names such as `www.sybex.com` are translated into their IP equivalents by the Domain Name Service (DNS). In a Microsoft Windows environment, NetBIOS names are resolved into IP addresses by Windows Internet Naming Service (WINS).

namespace In Microsoft Active Directory, a collection of unique domain names. A namespace can be contiguous or disjointed. In a contiguous namespace, a child domain always contains the name of the

parent, for example, `accounting.sybex.com` as a subdomain of `sybex.com`. In a disjointed namespace, a child domain does not share its parent's name, as in `sybex.com` and `sybexnet.com`.

name space In Novell NetWare, the ability of a NetWare volume to support files from non-MS-DOS clients.

Each client sees files on the server in its own format; a Macintosh client sees files as Macintosh files, and a Unix client sees files as Unix files. Name space support is enabled per NetWare volume.

See also name-space NLM.

name-space NLM A specific type of NetWare Loadable Module (NLM) that allows non-MS-DOS filenaming conventions, such as those used in OS/2, Unix, or the Macintosh system, to be stored in NetWare's directory and filenaming systems.

NAMPS *See* Narrowband Advanced Mobile Phone Service.

nano- Abbreviated n. A prefix meaning one-billionth in the American numbering scheme, and one thousand millionth in the British system.

nanosecond Abbreviated ns. One-billionth of a second. The speed of computer memory and logic chips is measured in nanoseconds.

narrowband In communications, a voice-grade transmission channel of 2400 bits per second or less.

Narrowband Advanced Mobile Phone Service Abbreviated NAMPS. A proposed standard from Motorola that combines the AMPS (Advanced Mobile Phone Service) cellular standard with digital signaling information. NAMPS is designed to provide a higher level of performance, reduce the number and incidence of dropped calls, and greatly increase communications capabilities.

See also Advanced Mobile Phone Service; Code Division Multiple Access; Time Division Multiple Access.

Narrow SCSI A Small Computer System Interface (SCSI) or SCSI-2 interface capable of transferring only 8 bits of data at a time

See also Fast SCSI; Fast/Wide SCSI; Wide SCSI.

NAS *See* network attached storage.

NAT *See* network address translation.

National Center for Supercomputing Applications Abbreviated NCSA. At the University of Illinois at Urbana-Champaign, NCSA is credited with the creation of Mosaic, the first ever graphical Web browser.

National Computer Security Center

Abbreviated NCSC. A branch of the U.S. National Security Agency that defines security for computer products. The Department of Defense Standard 5200.28, also known as the Orange Book, specifies

several levels of increasingly complex security measures.

See also C2; Orange Book.

National Information Infrastructure

Abbreviated NII. A planned, high-speed, public-access information service, designed to reach millions of schools, homes, and businesses throughout the United States.

National Semiconductor Corp. A major manufacturer of semiconductor products based in Santa Clara, California. In 1997, National spun off its Fairchild Semiconductor division, and in 1998 acquired Mediamatics, a manufacturer of audio/video decoders, as well as chip maker Cyrix.

For more information on National Semiconductor, see www.national.com.

native mode In Microsoft Windows 2000 Server, a domain that contains only Windows 2000 domain controllers.

See also mixed mode.

nbtstat A utility program used to show active TCP/IP (Transmission Control Protocol/Internet Protocol) connections and statistics for NetBIOS over TCP/IP.

See also ipconfig; netstat; Ping; subnet mask; tracert.

NC *See* network computer.

NCB *See* Network Control Block.

NCIP *See* Novell Certified Internet Professional.

NCP *See* NetWare Core Protocol; network control program.

NCP See Network Control Protocol.

NCP packet signature In Novell NetWare, a security feature that allows each workstation to add a special signature to each NetWare Core Protocol (NCP) packet going to the server. This signature changes dynamically for each packet, protecting both the server and the workstation against unauthorized access or any attempt to use unauthorized network privileges.

See also NetWare Core Protocol.

NCSA See National Center for Supercomputing Applications.

NCSC See National Computer Security Center.

NDD See NetWare Directory Database.

NDIS See Network Driver Interface Specification.

NDPS See Novell Distributed Print Services.

NDS See Novell Directory Services.

NDS for NT An add-on product from Novell that allows the management and integration of Windows NT domains from within NetWare Administrator.

NDS tree In Novell Directory Services (NDS), the container objects and all the leaf objects that make up the hierarchical structure of the NDS database. Also known as the Directory Tree.

See also container object; leaf object; Novell Directory Services.

NEAP See Novell Education Academic Partner.

near-end crosstalk Abbreviated NEXT. Any interference that occurs close to a connector at either end of a cable. NEXT is usually measured near the source of the test signal.

See also crosstalk; far-end crosstalk.

NEC Corporation Major manufacturer of a wide range of electronic products, including semiconductors, communications equipment, the MultiSynch line of monitors, LCD flat-panel monitors, plasma displays, and PCs.

For more information on NEC Corporation, see www.nec.com.

See also Packard Bell NEC.

negative acknowledgment Abbreviated NAK. In communications, a control code, ASCII 21, sent by the receiving computer to indicate that the data was not properly received and should be sent again.

See also acknowledgment.

nested groups In Microsoft Windows 2000 Server, a group placed inside another group.

NETADMIN A Novell NetWare 4 utility used to manage NetWare Directory Services (NDS) objects. This utility has many of the same features and controls many of the same functions as the Windows-based program NWADMIN.

NETADMIN replaces the functions found in several NetWare 3.x utilities, including

DSPACE, SECURITY, SYSCON, and USERDEF, and has itself been replaced in NetWare 5 by NetWare Administrator.

See also NetWare Administrator.

NetBEUI *See* NetBIOS Extended User Interface.

NETBIOS A Novell NetWare 3.x workstation utility used to determine if the client software NETBIOS.EXE is loaded and, if so, which interrupts it is using.

NetBIOS *See* network basic input/output system.

NetBIOS Extended User Interface.

Abbreviated NetBEUI, pronounced “net-boo-ee.” A network device driver for the transport layer supplied with Microsoft’s LAN Manager, Windows for Workgroups, and Windows NT. NetBEUI communicates with the network interface card via the NDIS (Network Driver Interface Specification).

NetBEUI is a small protocol with no networking layer and therefore no routing capability. It is suitable only for small networks; you cannot build internetworks using NetBEUI, and so it is often replaced with TCP/IP.

Microsoft has added extensions to NetBEUI in Windows NT to remove the limitation of 254 sessions per node and calls this extended NetBEUI the NetBIOS Frame (NBF).

See also network basic input/output system; Transmission Control Protocol/Internet Protocol.

NetBSD An implementation of Unix derived from the BSD series of releases;

designed to run on Intel processors. The distribution is usually free, although there may be a small charge to cover the distribution media and packaging.

NetBSD emphasizes multiple platform support and so has been ported to several non-Intel systems.

See also FreeBSD; Linux.

netcasting A method used to deliver Web content automatically to the desktop. Netcasting is referred to as push technology because content is pushed from a Web site to those users who requested receipt of the content. Content can include weather forecasts, stock market quotes, or software updates.

See also Castanet; Marimba, Inc.; Point-Cast, Inc.; server push.

NET.CFG A Novell NetWare workstation configuration file that contains information used to configure the client software on the workstation. The file is read once, during startup of the network files.

For normal network use, the default values established in this file usually work. In some cases, the NET.CFG values must be adjusted to work with particular applications or in certain configurations.

NetDDE *See* Network Dynamic Data Exchange.

netiquette A contraction of *network etiquette*. The set of unwritten rules governing the use of e-mail and other computer and network services.

Like any culture, the online world has its own rules and conventions, and if you understand and observe these conventions, you

can take your place in the online community without problems. Here are a few tips:

- Remember that the people reading your post are human; if you wouldn't say it to their face, don't post it in your e-mail.
- Lurk before you leap. Spend a few days reading the posts in a newsgroup or mailing list before you post anything of your own.
- If you use a signature file to close your e-mail, remember to keep it short; people don't want to read lots of cute stuff every time you post.
- Don't post messages in uppercase as it is the e-mail equivalent of YELLING; to add emphasis, place an asterisk before and after a word.
- Don't flame or mount personal attacks on other users.
- Check your grammar and spelling before you post.
- Don't be shy; if you are an expert, share your knowledge with others.

NetPC An initiative from Compaq, Dell, Hewlett-Packard, Intel, and Microsoft describing a networked PC designed to give system administrators greater control and security than a traditional PC.

NetPC is based on a minimum configuration of a 133MHz Pentium with at least 16MB of memory, running a Windows operating system, with a hard disk and the capability to run applications locally. The computer case is sealed and lockable, and floppy disks and CD-ROM drives are optional. All software distribution is centralized.

See also network computer; thin client; total cost of ownership; Zero Administration for Windows.

Netscape Catalog Server A server for Microsoft Windows NT and Unix systems that allows the publication of complex, hierarchically organized documents stored on the Internet or on a corporate intranet; based on the popular freeware application Harvest.

Netscape Certificate Management System A server for Microsoft Windows NT and Unix systems that manages digital certificates and allows the server to become a certificate authority.

See also certificate authority.

Netscape Collabra Server A Network News Transfer Protocol (NNTP) server for Microsoft Windows NT and Unix systems that allows the creation and management of private and secure newsgroups.

Netscape Commerce Server A server for Microsoft Windows NT and Unix systems that manages electronic commerce on the Internet.

Netscape Communications Corporation A major publisher of software development tools, Web browsers, and Web-server software; based in Menlo Park, California, and founded in 1994 by Jim Clark and Marc Andreessen. Its Web browser, Netscape Navigator, was initially very popular, but later lost market share to Microsoft's Internet Explorer.

In 1998, America Online bought Netscape Communications and spun off the server-based products to Sun Microsystems.

For more information on Netscape Communications, see www.netscape.com.

Netscape Communicator A package that includes the Netscape Navigator Web browser, e-mail support, a newsreader, and Netscape Page composer, an HTML editor. Netscape Communicator is available for Microsoft Windows, the Macintosh, and Unix.

Netscape Directory Server A general-purpose Lightweight Directory Access Protocol (LDAP) directory server that stores, publishes, and manages user, group, and other information in one location on a network and makes this information available to the Netscape SuiteSpot servers.

See also Netscape SuiteSpot.

Netscape Enterprise Server A high-performance and very popular Web server for Microsoft Windows NT, Novell NetWare, and Unix systems that is specifically designed to support large Web sites on the Internet or on a corporate intranet.

Enterprise Server supports standard CGI (Common Gateway Interface), and for application creation, also offers Netscape Server API (NSAPI) for in-process server applications. The LiveWire runtime environment supports server-side JavaScript applications, and for Java developers, the Sun Java Servlet API is also available.

Enterprise Server is administered using a Web-based application called Server Manager, giving access to server settings such as

security, users, and groups, as well as Web site content management.

Netscape FastTrack Server A Web server for Microsoft Windows NT and Unix systems. Designed for non-programmers, it lets users establish a Web presence or an intranet quickly and easily.

Netscape Messaging Server An e-mail server for Microsoft Windows NT and Unix systems that supports Lightweight Directory Access Protocol (LDAP), Simple Mail Transfer Protocol (SMTP), as well as Post Office Protocol 3 (POP3) and Internet Mail Access Protocol (IMAP).

Netscape Navigator A popular Web browser that runs under Windows and Unix and on the Macintosh. Netscape is distributed free by Netscape Communications Corporation, and in an unusual move for a commercial software developer, Netscape has made the source code to Navigator available on the Internet.

Netscape SuiteSpot A popular package of integrated server components designed to provide groupware services, document publishing and management, messaging and e-mail, directory services, and security for corporate intranets. SuiteSpot is available in a Standard Edition and a Professional Edition. The Standard Edition includes Communicator, Enterprise Server, Messaging Server, Calendar Server, Collabra Server, and Directory Server. The Professional Edition includes all the components found in the Standard Edition and adds Compass Server, Certificate Server, Proxy Server, and Netscape Mission Control Desktop for centralized network management.

netstat A utility program used to show server connections running over TCP/IP (Transmission Control Protocol/Internet Protocol) and statistics, including current connections, failed connection attempts, reset connections, segments received, segments sent, and segments retransmitted.

See also ipconfig.

NETSYNC In Novell NetWare, two utility programs for NetWare 3 (NETSYNC3) and NetWare 4 (NETSYNC4) servers that allow NetWare 3 servers to be managed by Novell Directory Services (NDS).

See also Novell Directory Services.

NETUSER In Novell NetWare 4.x, a workstation utility that offers a menu system for performing simple user tasks, including managing print jobs, sending messages to other users, mapping network drives, capturing ports to printers or print queues, and changing the login script and password.

NETUSER replaces the NetWare 3.x utility SESSION and can be invoked from either the workstation or the server.

This utility has many of the same features and controls many of the same functions as the Windows-based program NWUSER.

NetVIEW IBM SNA (Systems Network Architecture) management software that provides monitoring and control functions for SNA and non-SNA devices. This system relies heavily on mainframe data-collection programs, but it also incorporates Token Ring networks, Rolm CBXs, non-IBM modems, and PC-level products.

NetWare *See* Novell NetWare.

NetWare Administrator In Novell NetWare, the main utility used for performing NetWare 4 and higher supervisory and administrative tasks. A graphical application, NetWare Administrator can be run from within Windows or OS/2.

NetWare Client32 A set of 32-bit client software packages released with Novell Netware 5. Client32 software is available for Windows 3.x, Windows 95, Windows 98, and Windows NT.

See also NetWare DOS Requester; NETx.COM.

NetWare Client for DOS and MS

Windows Client software that allows MS-DOS and Windows workstations to connect to a Novell NetWare server and access network resources.

See also NetWare Client32; NetWare DOS Requester; NETx.COM.

NetWare Client for OS/2 Client software that allows OS/2 workstations to connect to a Novell NetWare server and access network resources.

NetWare command files Text files created by the network administrator containing a series of Novell NetWare commands and the appropriate modifying parameters.

NetWare command files execute just as though you typed the commands at the NetWare console, and they have the filename extension .NCF. For example, AUTOEXEC.NCF contains server

configuration information used by SERVER.EXE when starting the server.

NetWare Connect *See* NetWare Internet Access Server.

NetWare Core Protocol Abbreviated NCP. In Novell NetWare, a presentation-layer procedure used by a server when responding to workstation requests. It includes routines for manipulating directories and files, opening semaphores, printing, and creating and destroying service connections.

NetWare Directory Database Abbreviated NDD. In Novell NetWare, a system database that holds the information on all the objects in a Novell Directory Services (NDS) tree. Often referred to as the Directory or the Directory tree.

NetWare Directory Services Abbreviated NDS. In Novell NetWare, a global naming service that maintains information on, and provides access to, every resource on the network, including users, groups, printers, volumes, and servers.

NDS manages all network resources as objects in the NetWare Directory Database (NDD), independent of their actual physical location, and presents them in a hierarchical tree structure. NDS is global to the network, and information is replicated so that a local failure cannot bring down the whole system.

NDD replaces the bindery, the system database for earlier releases of NetWare. The bindery managed the operation of a single NetWare server; NDS supports the whole network, including multiserver networks.

Bindery emulation provides compatibility with previous versions of NetWare.

NDS is based on the 1988 CCITT X.500 standard.

See also Consultative Committee for International Telephony and Telegraphy.

NetWare DOS Requester A group of Virtual Loadable Modules (VLMs) that provide NetWare 3.x and 4.x support for MS-DOS and Windows workstations. The Requester replaces the NetWare shell used by earlier versions of NetWare.

NetWare Hub Services A software package from Novell that supports the management of any hub card that complies with the NetWare Hub Management Interface (HMI) standard.

NetWare Internet Access Server Abbreviated NIAS. In Novell NetWare, the server software that translates data requests from a client using Internetwork Packet eXchange (IPX) into TCP/IP (Transmission Control Protocol/Internet Protocol) for connection to the Internet or other TCP/IP host. Previously known as NetWare Connect.

NetWare/IP A set of NetWare Loadable Modules (NLMs) that provide IP support for NetWare 3.x and 4.x servers by encapsulating the IPX information inside an IP datagram. Native TCP/IP support is available in Novell NetWare 5.x, so this capability is not needed in a pure NetWare 5 environment. NetWare/IP allows a Novell NetWare server to act as a gateway between NetWare and a TCP/IP network.

See also Internet Protocol; Internetwork Packet eXchange; Transmission Control Protocol/Internet Protocol.

NetWare Licensing Services Abbreviated NLS. In Novell NetWare, a service that allows system administrators to monitor and manage the execution of licensed application software.

NetWare Link Services Protocol Abbreviated NLSP. The Novell NetWare IPX (Internetwork Packet eXchange) link-state protocol used by IPX routers to share information about their routes with other devices on the network.

Once the network map is built, information is transferred between routers only when the network changes. NLSP allows large or small internetworks to be connected without causing routing inefficiencies.

See also Routing Information Protocol.

NetWare Loadable Module Abbreviated NLM. Server management programs and LAN drivers that run on a server under Novell NetWare's network operating system. NLMs can be loaded and unloaded dynamically, without interrupting the server, and provide better service than applications that run outside the core operating system. Several kinds of NLMs are available:

- Utilities and application modules allow you to look at or change various configuration options. These NLMs have the .NLM filename extension.
- Name-space modules allow non-DOS filenaming conventions to be used when storing files. These NLMs have the .NAM filename extension.

- Disk drivers give access to hard disks; they have the .DSK filename extension.
- LAN drivers control communications between the network operating system and the network interface cards; they have a .LAN filename extension.

See also NetWare Peripheral Architecture; Virtual Loadable Module.

NetWare for Macintosh A set of NetWare Loadable Modules (NLMs) used to provide file handling, printing, and AppleTalk routing for Macintosh clients on a NetWare network.

See also AppleTalk.

NetWare Management System A network management system for Novell NetWare servers that has been replaced by ManageWise.

See also Novell ManageWise.

NetWare Mobile Services In Novell NetWare, a dial-in remote access client for Windows 3.x and Windows 95.

NetWare Multiprotocol Router PC-based software from Novell that allows network administrators to connect LANs using IPX, TCP/IP, or AppleTalk Filing Protocol (AFP) over a wide range of LANs and WANs. The NetWare Multiprotocol Router supports Ethernet, Fast Ethernet, token ring, FDDI, and ARCnet network architectures.

See also Fiber Distributed Data Interface; Internetwork Packet eXchange; PC-based router; Transmission Control Protocol/Internet Protocol.

NetWare Multiprotocol Router Plus

PC-based software from Novell that provides wide-area connectivity for dispersed heterogeneous networks over T1, fractional T1, X.25, and low-speed synchronous leased lines. NetWare Multiprotocol Router Plus replaces three earlier products: NetWare Link/64, NetWare Link/T1, and NetWare Link/X.25.

See also PC-based router.

NetWare NFS A Novell NetWare Loadable Module (NLM) that adds NFS (Network File System) server capability to an existing NetWare file server. Once loaded, Unix NFS clients see the NetWare server as another NFS server.

See also Network File System.

NetWare NFS Gateway A Novell NetWare Loadable Module (NLM) that lets a NetWare server mount a Unix file system as a NetWare volume. Complete NetWare security is maintained, and access to the Unix system is based on the NetWare client's privileges.

NetWare Peripheral Architecture Abbreviated NPA or NWPA. A feature of NetWare that allows developers to add and support new storage devices and their associated controllers. NPA consists of the following components:

- **Host Adapter Module (HAM)** The adapter-specific driver that controls the interaction between the operating system and the hardware; provided by Novell or by the manufacturer of the adapter, with the filename extension .HAM. There will

be one HAM for each adapter on the server.

- **Host Adapter Interface (HAI)** An API within NPA for communication between the HAM and the NetWare database that tracks the storage devices and media available on the server.
- **Custom Device Module (CDM)** The software that controls the devices attached to the adapter, usually provided by the manufacturer, with the filename extension .CDM. There may be multiple CDMs for a single adapter as each device connected to the adapter needs its own CDM.
- **Custom Device Interface (CDI)** An API within NPA for communication between the CDM and the NetWare database that tracks the storage devices and media available on the server.

The NPA replaces many of the functions found in the NetWare Loadable Module drivers used to control hard disks; the .DSK drivers controlled the hard-disk controller and all the devices attached to that controller.

See also NetWare Loadable Module; Virtual Loadable Module.

NetWare Requester for OS/2 A group of Virtual Loadable Modules (VLMs) that provide NetWare 4.x support for OS/2-based workstations.

NetWare Runtime A single-user version of Novell NetWare, often used as a communications server or applications server that provide its own authentication.

NetWare shell The Novell NetWare program loaded into each workstation's memory that allows the workstation to access the network.

The shell captures the workstation's network requests and forwards them to a NetWare server. In earlier versions of NetWare, the shell program was specific to the version of MS-DOS or Windows in use. The term *shell* is not used in recent versions of NetWare.

See also NetWare Client32; NetWare DOS Requester; NetWare Requester for OS/2; NETx.COM.

NetWare for SNA A set of NetWare Loadable Modules (NLMs) used to provide connectivity to an IBM Systems Network Architecture (SNA) network. With the right access permissions, MS-DOS, Macintosh, Unix, and Windows clients can run applications on the IBM mainframe.

NetWare System Fault Tolerance

See System Fault Tolerance.

NetWare Tools Novell NetWare utilities that allow users to perform a variety of network tasks, such as accessing network resources, mapping drives, managing printing, and sending messages to other network users.

NetWare Tools programs are installed separately from the server installation program, and they are available for MS-DOS, Windows, and OS/2 workstations.

NetWare for Unix A software package that allows Unix clients to access a Novell NetWare server.

NetWare Users International Abbreviated NUI. An organization created to support distributed NetWare user groups, first formed in the mid-1980s.

NUI now has 150 regional chapters and more than 140,000 members worldwide. Users can present a united voice to Novell, giving feedback on new products and how to support and improve existing products. Although supported by Novell, NUI is completely independent from the company.

NetWire Novell's online information service accessed through the commercial service CompuServe. It provides product information, press releases, technical support, downloadable patches, upgrades, and utilities. All the material available through NetWire on CompuServe is also available on the Novell ftp site at [ftp.novell.com](ftp://ftp.novell.com) and on the Web at www.novell.com.

network A group of computers and associated peripheral devices connected by a communications channel capable of sharing files and other resources among several users.

A network can range from a peer-to-peer network connecting a small number of users in an office or department, to a LAN connecting many users over permanently installed cables and dial-up lines, to a MAN or WAN connecting users on several networks spread over a wide geographic area.

network adapter *See* network interface card.

network address translation Abbreviated NAT. A term used to describe the process of converting between IP addresses on

an intranet or other private network and Internet IP addresses.

See also name resolution.

network administrator The person responsible for the day-to-day operation and management of a network; also known as a system administrator. Duties of the network administrator can include the following:

- Planning for future expansion
- Installing new workstations and network peripheral devices
- Adding and removing authorized users
- Backing up the system and archiving important files
- Assigning and changing passwords
- Troubleshooting network problems
- Monitoring system performance
- Evaluating new products
- Installing hardware and software updates
- Training users

See also configuration management.

network analyzer Any device that decodes and analyzes data transmitted over the network. A network analyzer may be hardware, software, or a combination of the two. Some analyzers troubleshoot network problems by decoding packets; others create and transmit their own packets.

See also protocol analyzer; sniffer.

Network and Dial-up Connections In Microsoft Windows 2000, a special folder containing icons used to configure and manage local area network and other connections. Network and Dial-Up Connections

replaces the Windows NT 4 Dial-up Connections and the Network applet in Control Panel.

See also Control Panel.

network architecture The design of a network, including the hardware, software, access methods, and the protocols in use.

Several well-accepted network architectures have been defined by standards committees and major vendors. For example, the International Organization for Standardization (ISO) developed the seven-layer OSI Reference Model for computer-to-computer communications, and IBM designed SNA (Systems Network Architecture). Both architectures organize network functions into layers of hardware and software, with each layer building on the functions provided by the previous layer.

The ultimate goal is to allow different computers to exchange information freely in as transparent a fashion as possible.

Network Associates, Inc. Formed in 1997 by the merger of Network General and McAfee Associates, Network Associates provides virus detection and protection software, encryption software, help desk systems, and network management products. For more information on Network Associates, see www.networkassociates.com.

network attached storage Abbreviated NAS. A collection of mass-storage devices contained in a single chassis with a built-in operating system.

Ethernet connectors allow the NAS to connect directly to the network. A NAS might contain a large hard disk as well as a set of

SCSI connectors to attach additional disks, CD-ROM drives, tape drives, or Iomega Zip and Jaz drives. Because the NAS is managed by its own embedded operating system, adding one to a network adds little in the way of overhead.

See also storage area network.

network backbone *See* backbone.

network basic input/output system

Abbreviated NetBIOS, pronounced “net-by-e-os.” A session-layer network protocol, originally developed in 1984 by IBM and Sytek to manage data exchange and network access.

NetBIOS provides an API (application programming interface) with a consistent set of commands for requesting lower-level network services to transmit information from node to node, thus separating applications from the underlying network operating system. Many vendors provide either their own version of NetBIOS or an emulation of its communications services in their own products.

See also WinSock.

network board *See* network interface card.

network-centric An imprecise term often used to describe an approach to software design and development that includes a strong client/server component.

Network+ certification A certification program from the CompTIA (Computer Technology Industry Association) designed to measure competence in basic networking concepts; aimed at the computer technician.

network computer Abbreviated NC.

An initiative from Apple, IBM, Netscape, Oracle, and Sun Microsystems describing a networked computer designed to give system administrators greater control and security than a traditional PC provides.

The network computer is based on a minimum configuration of a 25MHz Intel i960 with at least 32MB of memory, running a Java Virtual Machine, and downloads applications from the server and runs them locally. Hard disk, floppy disk, and CD-ROM drives and expansion slots are not required.

The initiative fizzled out, in part at least because conventional PCs can do the same job, but also due to Compaq (and other manufacturers) creating the sub-\$1,000 PC.

See also NetPC; thin client; total cost of ownership; Zero Administration for Windows.

Network Control Block Abbreviated NCB. The packet structure used by the NetBIOS transport protocol.

network control program Abbreviated NCP. In an IBM Systems Network Architecture (SNA) environment, performs the routing, error control, testing, and addressing of SNA devices.

Network Control Protocol Abbreviated NCP. A protocol that mediates between network-layer protocols and the underlying data link protocols.

network device driver Software that controls the physical function of a network interface card, coordinating between the card and the other workstation hardware and software.

See also Network Driver Interface Specification.

network directory A directory located on a computer other than the one currently being used. Depending on access privileges, the rest of the disk may or may not be available to the user. On the Macintosh, a network directory is often referred to as a shared folder.

See also network drive; shared folder.

network drive A drive located on a computer other than the one currently being used and that is available to users on the network.

See also local disk; network directory.

Network Driver Interface Specification Abbreviated NDIS. A device driver specification, originally developed by Microsoft and 3Com in 1990, that is independent of both the underlying network interface card hardware and the protocol being used. NDIS also allows multiple protocol stacks to be used at the same time in the same computer.

Windows NT includes the latest version, NDIS 3, which is backward-compatible with the original NDIS and NDIS 2.

See also Open Data-link Interface; Open Data-link Interface/Network Driver Interface Specification Support.

Network Dynamic Data Exchange

Abbreviated NetDDE. A version of Microsoft's DDE that uses NetBIOS to extend DDE features over a network.

Using NetDDE, two or more applications running on networked workstations can dynamically share data.

See also Component Object Model; Distributed Component Object Model; Dynamic Data Exchange; Object Linking and Embedding.

Network File System Abbreviated NFS. A distributed file-sharing system developed well over a decade ago by Sun Microsystems, Inc.

NFS allows a computer on a network to use the files and peripheral devices of another networked computer as if they were local, subject to certain security restrictions. Using NFS, you can share files on your system with other computers running MS-DOS, MacOS, Unix, Novell NetWare, VMS, and many other operating systems, in both local and global environments.

NFS is platform-independent and runs on mainframes, minicomputers, RISC-based workstations, diskless workstations, and personal computers. NFS has been licensed and implemented by more than 300 vendors.

See also Andrews File System; Common Internet File System; WebNFS.

Network Information Service Abbreviated NIS. A recent name for the security and file-access databases on Unix systems, previously known as the Yellow Pages. The NIS for most Unix systems comprises the Unix host files `/etc/hosts`, `/etc/passwd`, and `/etc/group`.

network interface card Abbreviated NIC. In networking, the PC expansion board that plugs into a personal computer or server and works with the network operating system and the appropriate device

drivers to control the flow of information over the network.

The network interface card is connected to the network media (twisted pair, coaxial, or fiber-optic cable) and is designed for a specific type of network such as Ethernet, token ring, FDDI (Fiber Distributed Data Interface), or ARCnet.

Novell NetWare documentation uses the term *network board* rather than the more common term *network interface card*.

Network support is built in to Macintosh computers and does not require an additional expansion board.

network layer The third of seven layers of the OSI Reference Model for computer-to-computer communications. The network layer defines protocols for data routing to ensure that the information arrives at the correct destination node and manages communications errors.

See also OSI Reference Model.

Network Management Protocol Abbreviated NMP. A set of protocols developed by AT&T, designed to control certain network devices, such as modems and T1 multiplexers.

Network Management Vector Transport Abbreviated NMVT. A network management protocol from IBM that provides problem-determination statistics and other network management data within Systems Network Architecture (SNA) services.

Network Monitor In Microsoft Windows NT Server, a graphical utility program used to monitor and troubleshoot network-related problems.

Network Monitor tracks information up to the network layer, filters packets according to the protocol or the source or the destination machine, and performs packet analysis.

A more capable version of Network Monitor is available as part of Microsoft's Systems Management Server package.

Network News Transfer Protocol Abbreviated NNTP. An Internet protocol used for posting, retrieving, and managing posts to newsgroups.

network operating system Abbreviated NOS. In typical client/server architecture LANs, the NOS consists of two parts. The largest and most complex part is the system software running on the file server. This system software coordinates many functions, including user accounts and network access information, security, resource sharing, administration, UPS and power monitoring, data protection, and error detection and control. A much smaller component of the NOS runs on each of the networked PCs or workstations attached to the network.

Network operating systems are available from Banyan (VINES), IBM (OS/2 Warp Server), Microsoft (Windows NT Server and Windows 2000), Novell (IntraNetWare and NetWare), and Sun Microsystems (Solaris), and, of course, many versions of Unix are available.

In peer-to-peer networks, a part of the NOS is installed on each PC or workstation attached to the network and runs on top of the PC operating system. In some cases, the NOS may be installed on one PC designated as a file server, but this PC is not dedicated to the file-server function; it is also available to run applications.

network printer A printer attached to and accessible from the network. A network printer may be attached to a file server or a printer server, or it may have its own direct connection to the network.

See also local printer.

Network Service Access Point Abbreviated NSAP. A 20-octet addressing scheme used in Asynchronous Transfer Mode (ATM) networks for private network addresses.

network topology *See* topology.

NETx.COM The workstation shell software used in Novell NetWare 2.x and 3.x networks. This program is loaded into memory on the workstation and begins transmission when the workstation requests network resources.

In earlier versions of NetWare, NETx.COM was specific to the version of MS-DOS running on the workstation. The only way to unload NETx.COM from memory is to reboot the workstation.

See also NetWare Client32; NetWare DOS Requester.

newsgroup A Usenet e-mail discussion group devoted to a single topic. Subscribers to a newsgroup post articles that can be read by all the other subscribers.

Newsgroup names fit into a formal structure in which each component of the name is separated from the next by a period. The leftmost portion of the name represents the category of the newsgroup, and the name gets more specific from left to right.

The major top-level newsgroup categories are:

- **alt** Newsgroups outside the main structure outlined below
- **comp** Computer science and related topics, including information about operating systems and hardware, as well as more advanced topics such as graphics and robotics
- **misc** Anything that does not fit into any of the other categories
- **news** Information on Usenet and newsgroups
- **rec** Recreational activities, such as hobbies, the arts, movies, and books
- **sci** Discussion groups on scientific topics, including math, physics, and biology
- **soc** Groups that address social and cultural issues
- **talk** Groups that concentrate on controversial subjects

Private newsgroups are often available on corporate intranets, where organization,

structure, and subject matter are decided by the system administrator.

See also newsreader; Usenet.

newsreader An application used to read articles posted to Usenet newsgroups. Newsreaders are of two kinds:

- Threaded newsreaders group the posts into threads of related articles.
- Unthreaded newsreaders present articles in their original order of posting without regard for the subject.

Of the two, threaded newsreaders are much easier to use.

See also newsgroup; thread; Usenet.

New Technology File System Abbreviated NTFS; sometimes NT File System. The native file system used by Microsoft Windows NT, which supports long file-names, reduced file fragmentation, improved fault tolerance, increased system security, and much better recovery after a system crash.

NEXT *See* near-end crosstalk.

Next Hop Resolution Protocol Abbreviated NHRP. An Internet name resolution protocol designed to route IP datagrams across nonbroadcast multiple access networks such as Asynchronous Transfer Mode (ATM), frame relay, Switched Multimegabit Data Service (SMDS), and X.25.

See also Asynchronous Transfer Mode ; IP over ATM; IP switching.

NFS *See* Network File System.

NHRP *See* Next Hop Resolution Protocol.

NIAS *See* NetWare Internet Access Server.

NIC *See* network interface card.

NII *See* National Information Infrastructure.

NIS *See* Network Information Service.

NLM *See* NetWare Loadable Module.

NLS *See* NetWare Licensing Services.

NLSP *See* NetWare Link Services Protocol.

NMP *See* Network Management Protocol.

NMVT *See* Network Management Vector Transport.

NNTP *See* Network News Transfer Protocol.

node Any device attached to the network capable of communicating with other network devices. In Novell NetWare documentation, a workstation is often called a node.

node number The number that uniquely identifies a network interface card and distinguishes it from all others.

Node numbers can be assigned in different ways. Ethernet node numbers are factory-set, so no two Ethernet boards have the same number. On other network interface cards, node numbers are set by jumpers or switches.

noise In communications, extraneous signals on a transmission channel that degrade the quality or performance of the channel. Noise is often caused by interference from nearby power lines, electrical equipment, or spikes in the AC line voltage.

See also crosstalk; far-end crosstalk; near-end crosstalk.

nominal velocity of propagation The speed at which a signal moves through a cable, expressed as a percentage or fraction of the speed of light in a vacuum. Some cable testers use this speed, along with the time it takes for a signal to return to the testing device, to calculate cable lengths.

nondedicated server A server upon which applications are available, while network management software runs in the background. Nondedicated servers are common in peer-to-peer networks.

non-preemptive multitasking Any form of multitasking in which the operating system cannot preempt a running task and move to the next task in the queue.

Programs are easy to write for this environment; however, a single badly written program can take over the whole system. By refusing to relinquish the processor, such a program can cause serious problems for other programs running at the same time. Poorly written non-preemptive multitasking can produce a kind of stuttering effect on running applications, depending on how the programs behave.

See also preemptive multitasking; time-slice multitasking.

nonvolatile memory Any form of memory that holds its contents when the power is removed. ROM (read-only memory), EPROM (erasable programmable read-only memory), and EEPROM (electrically erasable programmable read-only memory) are all forms of nonvolatile memory.

no parity *See* parity.

NOS *See* network operating system.

notebook computer A small, portable computer, about the size of a computer book, with a flat screen and a keyboard that fold together.

A notebook computer is lighter and smaller than a laptop computer. Recent advances in battery technology allow them to run for many hours between charges. Some models use flash memory rather than conventional hard disks for program and data storage; other models offer a range of business applications in ROM. Many offer PCMCIA expansion connections for additional peripheral devices, such as modems, fax modems, and network connections.

See also laptop computer; PC Memory Card International Association.

Novell Application Launcher In Novell NetWare, a utility used with applications for NetWare clients.

The NetWare administrator decides which applications are appropriate and creates an icon group for the users; additional startup and shutdown scripts can be used to add more application control. Novell Application Launcher uses the Universal Naming Convention (UNC) for ease and clarity.

See also Universal Naming Convention.

Novell Authorized Education Center

Abbreviated NAEC. A private organization that provides Novell-approved training courses.

See also NEAP.

Novell BorderManager A network protection package from Novell that provides a firewall, circuit-level gateways, proxy services, and Internet access.

Novell Certified Internet Professional

Abbreviated NCIP. A certification system from Novell structured for professionals who design, build, and maintain Web sites. The following specializations are available: Internet Business Strategist, Web Designer, Intranet Manager, Internet Architect, and Web Developer.

Novell Connect An add-on product from Novell that allows remote users to dial in and access network resources.

See also Novell Connect Services,

Novell Connect Services A collection of network products from Novell that are designed to simplify WAN administration; includes Novell NetWare, Novell Directory Services, ManageWise, Novell Connect, and NetWare Multiprotocol Router.

Novell Directory Services Abbreviated NDS. A distributed directory system from Novell, similar in scope and concept to the X.500 directory services specification.

Originally designed for use with Novell NetWare, NDS is now available for several

other platforms, including Microsoft Windows NT.

NDS uses a distributed database known as the NetWare Directory Database (NDD) to keep track of all network objects, including servers, users, groups, printers, and other networked resources. This information is presented to the user as a collection of containers, arranged in a hierarchical structure according to the organization or corporation structure, and is known as the Directory Tree or simply as the Directory.

Users no longer log in to a single server; they are authenticated by the network through NDS.

See also container object; leaf object; Root object.

Novell Distributed Print Services Abbreviated NDPS. A set of printer services available for Novell NetWare that standardizes and centralizes use and management of printers and printing on a network.

NDPS was developed jointly by Novell, Xerox, and Hewlett-Packard and allows bidirectional communications with newer printers, supports drag-and-drop printing for text and PostScript files without opening the creating application, and supports ISO 10175 Document Printing Architecture, Simple Network Management Protocol (SNMP), Management Information Base (MIB), and automatically downloadable printer drivers.

Networked printers are administered using the NetWare Administrator program.

Novell Education Academic Partner

Abbreviated NEAP. A college or university that provides Novell-approved training courses as a part of its standard curriculum.

See also Novell Authorized Education Center.

Novell GroupWise A groupware software package from Novell that includes e-mail, group scheduling, task management, personal calendaring, document management, workflow routing, and support for threaded discussions.

GroupWise also includes Internet Mail Access Protocol (IMAP), Post Office Protocol 3 (POP3), and Lightweight Directory Access Protocol (LDAP) support.

Novell, Inc. A leading developer of network operating system software, groupware, network management tools, Internet and intranet tools, and security products, based in Orem, Utah.

Best known for its Novell NetWare network operating systems, Novell is moving toward more open systems, the Internet, and corporate intranets and sees Novell Directory Services (NDS) as a significant advance in this direction.

For more information on Novell, see www.novell.com.

Novell IntraNetWare An intranet software package from Novell that includes the NetWare operating system, the Netscape Navigator Web browser, the NetWare Web server, the Multiprotocol Router (MPR), Novell Directory Services (NDS), and the

IPX/IP Gateway—also known as Novell Internet Access Server (NAIS).

Novell LANalyzer A network monitoring and analysis package from Novell used for troubleshooting Ethernet and token-ring networks.

Novell ManageWise A software package from Novell for managing NetWare and Microsoft Windows NT servers.

ManageWise includes server and desktop management, network analysis, software management, virus protection, and automatic network inventory. ManageWise replaces NetWare Management System on Novell servers.

See also Novell Z.E.N.works.

Novell NDS for NT An add-on product for Microsoft Windows NT Server from Novell that allows the integration of Windows NT domains with Novell Directory Services (NDS) and their joint management from within NetWare Administrator.

Novell NetWare A general term for the family of network operating systems available from Novell.

Novell NetWare 3.x A 32-bit network operating system that is designed to take advantage of the features of the Intel 80386 (and later) processors and is suitable for larger, multisegment networks, with up to 250 nodes per server.

NetWare 3.x provides enhanced security, performance, and flexibility and can access up to 4GB of RAM and up to 32TB of storage. A maximum of 100,000 files can be open concurrently on the file server, and

the maximum file size is 4GB. Disk mirroring, disk duplexing, support for optical disks, and UPS-monitoring functions are all available.

Novell NetWare 4.x A 32-bit network operating system from Novell announced in 1993 that builds on the successes of Novell NetWare 3.x.

Its most significant feature was the inclusion of Novell Directory Services (NDS), which allowed system administrators to organize users and network resources such as printers and servers the way people naturally access them. NDS is based on the CCITT X.500 directory standard and replaces the bindery database in earlier versions of NetWare.

NetWare 4.x also adds support for optical disks, CD-ROMs, data compression, and improved login security mechanisms. NetWare 4.1 is the first version of NetWare to integrate Message Handling Service with the directory, and it also adds System Fault Tolerance. NetWare 4.x is suitable for larger, multisegment internetworks, supporting up to 1000 nodes per server. It includes a set of user and administrator utilities featuring a graphical user interface, and it is available in several languages.

Novell NetWare 4.x SFT III A network software package from Novell that provides fault tolerance and protects against downtime by integrating two physically separate servers; if one of the servers fails for whatever reason, the other takes over and continues to support users.

Novell NetWare 5 A 32-bit network operating system from Novell. NetWare 5 continues Novell's transition into open systems, with native TCP/IP support, a Java Virtual Machine that runs on the server, and an enhanced file system.

Improvements have been made to Novell Directory Services (NDS), including the addition of Lightweight Directory Access Protocol (LDAP), as well as to the Remote Access Services for dial-in users. NetWare 5 also includes WAN Traffic Manager, Secure Sockets Layer (SSL), Novell's Public Key Infrastructure Services, Storage Management Services, Netscape's FastTrack Web server, DNS/DHCP services, and a Z.E.N.works starter pack.

Novell NetWare Connect See NetWare Internet Access Server.

Novell NetWare Navigator An automated software distribution package from Novell.

Novell Support Connection A technical information service available by subscription from Novell.

Two update CDs in a searchable format are issued periodically, containing all the latest files, patches, device drivers, and technical information, including product documentation, Novell Appnotes, Novell Developer Notes, and Novell Labs Certification Bulletins.

Novell Z.E.N.works A software management package from Novell that provides desktop management, including hardware

inventory, and adds flexibility to the desktop policy capabilities built into Microsoft Windows. Z.E.N.works can also control the applications a client can run and can operate the client remotely.

Z.E.N.works maintains a hardware inventory for each client, which is updated each time the client logs on to the network, and includes the desktop operating system, the amount of RAM in the workstation, the capacity of the local hard disk, and any connected peripherals.

Z.E.N.works also removes the need to log on twice—once to the operating system and once to the network—under Windows.

See also Novell ManageWise.

NPA *See* NetWare Peripheral Architecture.

ns *See* nanosecond.

NSAP *See* Network Service Access Point.

NTFS *See* New Technology File System.

NT File System *See* New Technology File System.

NUI *See* NetWare Users International.

null A character that has all the binary digits set to zero (ASCII 0) and therefore has no value.

In programming, a null character is used for several special purposes, including padding fields or serving as a delimiter character. In the C programming language, for example,

a null character indicates the end of a character string.

null modem A short serial cable that connects two personal computers so that they can communicate without the use of modems.

The cable connects the two computers' serial ports, and certain lines in the cable are crossed over so that the wires used for sending by one computer are used for receiving data by the other computer.

See also modem eliminator.

NWADMIN A Novell NetWare workstation utility that provides most of the functions needed to administer a Novell NetWare network.

Referred to as the Network Administrator in the Novell documentation, NWADMIN allows you to create users and groups, create, move, delete, and rename Novell Directory Services (NDS) objects, and manage NDS partitions and volumes.

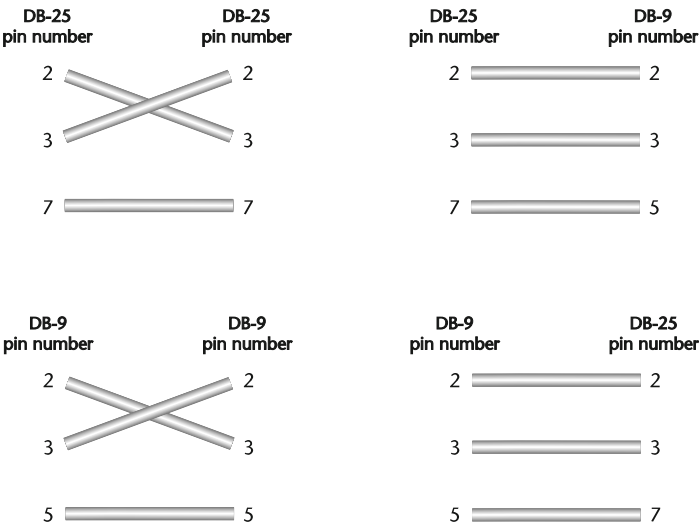
NWLink IPX/SPX In Microsoft Windows NT, a protocol that implements Novell's IPX/SPX protocol.

See also Internetwork Packet eXchange; Sequenced Packet Exchange.

NWLink NetBIOS In Microsoft Windows NT, a protocol that enables Novell NetBIOS packets to be sent between a Novell server running NetBIOS and a Windows NT computer.

See also network basic input/output system.

NULL MODEM



NWPA See NetWare Peripheral Architecture.

NWUSER A Novell NetWare workstation utility that lets users display or modify

their workstation's drive mappings, server attachments, and print queues using a simple graphical user interface.

O

Object 1. In Novell Directory Services (NDS), a representation of a logical or physical network resource, including users, computers, and printers. An Object has both properties and values and may be a Root, leaf, or container object.

2. In Microsoft's Active Directory, a representation of a network resource, including users, computers, and printers. Objects contain properties for definition and configuration.

See also Active Directory; container object; leaf object; Organizational Unit; Root object; Schema.

object 1. Any distinct entity. Program objects can represent applications such as word processors, spreadsheets, and so on. Folder objects can represent a directory and contain a group of files, a group of programs, or a group of other folders. Data-file objects can include information such as text, memos, letters, spreadsheets, video, and sound. Device objects can be printers, fax modems, plotters, servers, and CD-ROMS.

2. In Object Linking and Embedding (OLE), an object can be any user-selected group of data, such as a block of text, a set of spreadsheet cells, a chart, a sound, or a graphical image. This data can be embedded in or linked to another document created by a different OLE application.

3. In object-oriented programming, a program consists of a set of related but self-contained objects that can contain both code and data.

See also container object; leaf object; NetWare Directory Services.

object counters In Microsoft Windows 2000, a container built into each service object that counts the number of times an object performs its designated task.

Object Linking and Embedding Abbreviated OLE, pronounced "oh-lay." A Microsoft protocol for application-to-application exchange and communications using data objects. From a user standpoint, the main benefit to OLE is that it allows any OLE-compliant application to display information created in a different application.

Data objects can be either embedded or linked. If the source data in its original form is actually stored inside the other application program's data file, the data is embedded. In this case, there are two separate copies of the data: the original data and the embedded copy. Any changes made to the original document will not be made in the compound document unless the embedded object is updated.

If the data still exists in a separate file and a set of pointers to this data is stored in the other application program's data file, the data is linked. In this case, only one copy of

the data exists; any changes made in the original document will be made automatically in the compound document.

To determine whether an application supports OLE, check the Edit menu for commands such as Paste Link, Paste Special, and Insert Object. If these commands are present, the application supports OLE.

See also OpenDoc.

object-oriented An over-used term that can be applied to any computer system, operating system, programming language, application, or graphical user interface that supports the use of objects.

See also object-oriented programming.

object-oriented graphics Graphics that are constructed from individual components, such as lines, arcs, circles, and squares.

The image is defined mathematically rather than as a set of dots, as in a bitmapped graphic. Object-oriented graphics are used in illustration, drawing, and CAD (computer-aided design) programs and are also known as vector graphics or structured graphics.

Object-oriented graphics allow the user to manipulate a part of an image without redrawing. Unlike bitmapped graphics, all or parts of object-oriented graphics can be resized or rotated without introducing any distortion.

object-oriented programming Abbreviated OOP. A programming model that views a program as a set of self-contained objects.

These objects interact with other objects by passing messages. Object-oriented programming also lets you create procedures that work with objects whose exact type may not be known until the program actually runs.

In object-oriented programming, each object contains both data and code and is completely self-contained. The program incorporates an object by making it part of a layered hierarchy. Object-oriented programming is the result of many years of theoretical development, and many consider it the current extension of the theory behind modular programming, in which code is combined into reusable modules.

Object Request Broker Abbreviated ORB. A communications mechanism used in an object-oriented distributed computing environment in which program modules can be written in any programming language and still provide services to other applications.

An object makes a request and sends it to the ORB. The ORB locates the requested object and establishes communications between client and server. The receiving object then responds to the request and sends a response to the ORB, which, in turn, sends the response to the original requester.

The physical location of the object that provides the response is unimportant, and to the user, the application appears seamless, even though the various services may be coming from several different parts of the network.

The ORB standard is defined by the Object Management Group, an industry consortium

developing middleware standards based on distributed-object architecture.

See also Common Object Request Broker Architecture; Distributed Component Object Model; middleware.

object rights In Novell Directory Services (NDS), rights granted to a trustee over an object. For example, the Create object right for a container object allows a trustee to create new objects in that container.

See also Object.

OC *See* Optical Carrier.

occupant In Novell Directory Services (NDS), a user who has been assigned to an Organizational Role object. Each Organizational Role object can have multiple occupants.

OCR *See* optical character recognition.

octet The Internet's own term for a unit of data containing exactly eight bits. Some of the computer systems attached to the Internet use bytes with more than eight bits; hence, the need for this term.

See also byte.

ODBC *See* Open Database Connectivity.

odd parity *See* parity.

ODI *See* Open Data-link Interface.

ODI/NDIS Support *See* Open Data-link Interface/Network Driver Interface Specification Support.

ODINSUP *See* Open Data-link Interface/Network Driver Interface Specification Support.

ODSI *See* Open Directory Services Interface.

OE *See* Outlook Express.

OEM *See* original equipment manufacturer.

offline Describes any device that is not in ready mode and is therefore unavailable for use.

See also online.

offline files In Microsoft Windows 2000, a mechanism that caches frequently used network files on the client; also known as client-side caching.

Offline files makes the network seem to respond more quickly to file requests, allows users to ignore minor network outages, and provides a way for mobile users to synchronize laptop and server versions of the same file.

offline reader An application that lets you read postings to Usenet newsgroups without having to stay connected to the Internet.

The program downloads all the newsgroup postings you have not read and disconnects from your Internet Service Provider. You can then read the postings at your convenience without incurring online charges or tying up your telephone line. If you reply to any of these postings, the program will automatically upload them to the correct

newsgroup the next time you connect to your service provider.

See also newsgroup; newsreader; Usenet.

OLAP *See* online analytical processing.

OLE *See* Object Linking and Embedding.

OLTP *See* online transaction processing.

online 1. Most broadly, any work done on a computer instead of by more traditional manual means.

2. Any function available directly on a computer, such as an application's help system.

3. Describes a peripheral device, such as a printer or a modem, when it is directly connected to a computer and ready to operate.

4. In communications, describes a computer connected to a remote computer over a network or a modem link.

See also offline.

online analytical processing Abbreviated OLAP. A category of software used to analyze historical business data to find previously hidden patterns.

Analysts use OLAP software to view data in a multidimensional form, rather than in the more usual two-dimensional row and column format. In a multidimensional format, the intersection of important data is much more obvious, and data is easier to group and categorize.

See also data mining; data warehousing.

online service A service that provides an online connection via modem for access to

various services. Online services fall into these main groups:

- **Commercial services** Services such as America Online charge a monthly membership fee for access to online forums, e-mail services, software libraries, and on-line conferences.
- **Internet** The Internet is a worldwide network of computer systems and is not always easy to use, but the wealth of information available is staggering. The main problem for casual users is that there is no central listing of everything that is available.
- **Specialist databases** Specific databases aimed at researchers can be accessed through online services such as Dow Jones News/Retrieval for business news and Lexis and Nexis for legal information and news archives.

online transaction processing Abbreviated OLTP. A business system that operates in real time, collecting and posting transaction-related data and making changes to shared databases.

A transaction is a single, discrete unit of work, which is normally part of a business process. Examples of OLTP systems include airline and hotel reservation systems, inventory control systems, and banking systems.

See also transaction processing.

OOP *See* object-oriented programming.

open architecture A vendor-independent design that is publicly available and well understood within the industry.

See also closed architecture.

Open Database Connectivity Abbreviated ODBC. An application program interface (API) from Microsoft that allows a single application to access many types of database and file formats.

ODBC uses Structured Query Language (SQL) for operations that access a database, and the client does not need to know the location of the database or its type or the method used to access the data; all these details are managed by ODBC. Data sources can range from a simple spreadsheet all the way up to very large relational databases.

Open Data-link Interface Abbreviated ODI. A Novell specification, released in 1989, that allows multiple network interface card device drivers and protocols to share a single network interface card without conflict.

ODI defines an interface that separates device drivers from protocol stacks and lets multiple protocol stacks share the same network hardware. Here are the main components:

- The multiple-link interface driver (MLID) manages the sending and receiving of packets to and from the network.
- The link-support layer (LSL) is the interface layer between the device driver and the protocol stacks. Any ODI LAN driver can communicate with any ODI protocol stack via the LSL.
- The MLI (Multiple-Link Interface) communicates with the network interface cards through an MLID and consists of three main components:

- The media-support module (MSM), which manages the details of interfacing ODI MLIDs to the LSL and to the operating system.
- The hardware-specific module (HSM), which is specific to a particular network interface card. It handles adapter initialization, reset, shutdown, packet reception, timeout detection, and multicast addressing.
- The topology-specific module (TSM), which manages operations that are specific to a particular media type, such as Ethernet or token ring.

See also Network Driver Interface Specification; Open Data-link Interface/Network Driver Interface Specification Support.

Open Data-link Interface/Network Driver Interface Specification Support

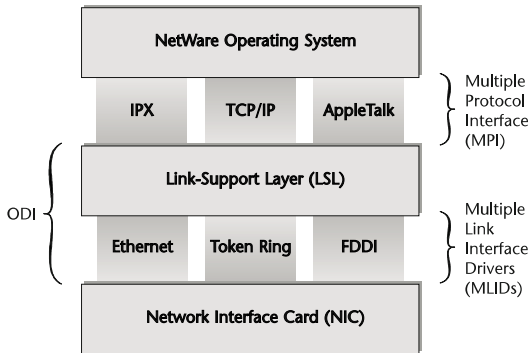
Abbreviated ODINSUP; also written as ODI/NDIS Support. A Novell interface that allows the coexistence of two network driver interfaces:

- Microsoft's NDIS (Network Driver Interface Specification)
- Novell's ODI (Open Data-link Interface)

ODINSUP allows an MS-DOS or Microsoft Windows workstation to connect to dissimilar networks through a single network interface card and to use them as if they were a single network.

ODINSUP also allows NDIS protocol stacks to communicate through the ODI's link-support layer (LSL) and multiple-link interface driver (MLID) so that ODI and NDIS protocol stacks can coexist in the same system, using a single ODI MLID.

OPEN DATA-LINK INTERFACE



Open Desktop A graphical user interface from SCO that provides access to files and system utility functions on the desktop. Files, directories, and applications are represented by icons and displayed in windows.

See also OpenServer; Santa Cruz Operation, Inc.; UnixWare.

Open Directory Services Interface

Abbreviated ODSI. A standard from Microsoft that enables client software to query Internet directories by providing a common API for naming.

OpenDoc A specification for creating compound documents, from Apple, IBM, Borland, and Novell. OpenDoc manages text, spreadsheets, graphics, sound, and video as objects that can be created in one application and then inserted into another. OpenDoc is similar in many respects to Microsoft's Object Linking and Embedding (OLE) specification, but OpenDoc provides for greater network support and also

includes a certification process to ensure that applications work together.

OpenGL *See* Open Graphics Library.

Open Graphics Library Abbreviated OpenGL. A set of graphics libraries originally developed by Silicon Graphics and now supported by IBM, Intel, Microsoft, and many other companies.

OpenGL lets developers create 3-D graphical applications for workstations running the Programmers Hierarchical Interactive Graphics System (PHiGS) extensions to the X Window system.

See also X Window.

Open Group An international consortium of vendors, educational institutions, and government agencies that develops standards for open systems. Formed in 1996 as an umbrella organization to bring together the Open Software Foundation (OSF) and X/Open Company Limited.

For more information on Open Group, see www.opengroup.com.

OpenServer A scalable set of Unix-based products from SCO, including the Desktop System, Host System, and Enterprise System, based on SVR2 but containing many significant SVR4 enhancements.

The Desktop System is a single-user, multi-tasking system with MS-DOS and Windows emulations, TCP/IP, and Internet connectivity.

The Host and Enterprise systems provide high-performance scalable servers for Intel-based platforms, supporting more than 8,000 applications, and with extensive networking support, UPS (uninterruptible power supply), and advanced power management support.

See also Open Desktop.

Open Shortest Path First Abbreviated OSPF. A routing protocol used on TCP/IP networks that takes into account network loading and bandwidth when routing information over the network. Routers maintain a map of the network and swap information on the current status of each network link. OSPF incorporates least-cost routing, equal-cost routing, and load balancing.

See also Routing Information Protocol.

open source software Any software package that includes the original source code from which the product was originally created.

Open source software allows knowledgeable users to make changes to the way the software actually works, unlike products from mainstream software developers which never include the source code. And

while this mainstream software is certainly configurable, it is basically a take-it-or-leave-it package.

See also Free Software Foundation; Linux.

Open Systems Interconnect *See* OSI Reference Model.

OpenView A popular network management package from Hewlett-Packard that includes Network Node Manager, Event Correlation Services, a Java-based Web user interface, Open Database Connectivity (ODBC) integration, and other support services. OpenView is available on HP-UX, a Unix variant, and on Microsoft Windows NT.

operating system Abbreviated OS. The software responsible for allocating system resources, including memory, processor time, disk space, and peripheral devices such as printers, modems, and monitors. All applications use the operating system to gain access to these resources as necessary. The operating system is the first program loaded into the computer as it boots, and it remains in memory throughout the session.

See also kernel; microkernel; network operating system.

Optical Carrier Abbreviated OC. A set of Synchronous Optical Network (SONET) hierarchies that define how digital signals are multiplexed over fiber-optic cable.

SONET is the physical-layer standard used by telephone carriers to connect long-distance services. Table O.1 lists the OC levels and related data rates.

See also digital signal; Synchronous Optical Network; Synchronous Digital Hierarchy.

TABLE O.1 OC LEVELS

OC Level	Data Rate (Mbps)
OC-1	51.84
OC-3	155.52
OC-6	311.04
OC-9	466.56
OC-12	622.08
OC-18	933.12
OC-24	1244.16
OC-36	1866.24
OC-48	2488.32
OC-96	4976.00
OC-192	9952.00

optical character recognition Abbreviated OCR. The computer recognition of printed or typed characters. OCR is usually performed using a standard optical scanner and special software, although some systems use special readers. The text is reproduced just as though it had been typed. Certain advanced systems can even resolve neatly handwritten characters.

See also document management.

optical drive A high-capacity disk-storage device that uses a laser to read and write information.

Because optical drives are relatively slow, they are used for archiving information and for other applications for which high access speed is not critical.

See also high-capacity storage system; jukebox; Magneto-optical storage.

Oracle Corporation A major developer of powerful relational database software, client/server development tools, and other related products for multiuser enterprise computing.

For more information on Oracle, see www.oracle.com.

Oracle NCA See Oracle Network Computing Architecture.

Oracle Network Computing

Architecture Abbreviated Oracle NCA. A set of technologies from Oracle Corporation designed to allow clients to access information from Web servers, database servers, and other systems. NCA includes open Internet standards and special components called *cartridges* that provide a connection between proprietary applications.

Orange Book Lay term for the National Security Agency document called “Trusted Computer System Evaluation Criteria,” or TCSEC, which specifies security levels that vendors must comply with to achieve Department of Defense security standards; so called because of the color of its cover.

This publication details standards for security levels used to control access to computer systems from Class A1, the highest verifiable security level, to Class D, the lowest, which has no security.

Class C2 is the security level most appropriate to the business world; higher levels of security tend to intrude too much into normal commercial working patterns. C2 security requires that the operating system provide individual logins with separate accounts and a verifiable audit trail, that resources have

owners, and that files and other system resources be protected from other processes that might corrupt them.

According to the standard, a C2 compliant workstation cannot be connected to a network. Secure networking is defined in “Trusted Network Interpretation,” which is known as the Red Book.

See also security.

ORB See Object Request Broker.

Organization object In Novell Directory Services (NDS), a container object below the Country object and above the Organizational Unit object that helps to organize other objects in the tree. You must have at least one Organization object, usually the name of your company or department, and you can use multiple Organization objects if you wish.

See also leaf object; Novell Directory Services.

Organizational Role object In Novell Directory Services (NDS), a container object used to specify a role within an organization such as Workgroup Leader or Department Manager who has access to certain NDS objects or files. The Organizational Role object is often used for container administrators.

See also occupant.

Organizational Unit In Microsoft’s Active Directory, an object that can contain other objects, such as other Organizational Units, users, groups, or Distributed File System (DFS) shares.

See also Active Directory.

Organizational Unit object

Organizational Unit object In Novell Directory Services (NDS), a container object below the Organization object. The Organizational Unit object can contain other Organizational Unit objects or leaf objects. This container is not a required container, but its use allows better management of workgroups or project teams.

original equipment manufacturer

Abbreviated OEM. The original manufacturer of a hardware subsystem or component. For example, Canon makes the print engine used in many laser printers, including those from Hewlett-Packard (HP); in this case, Canon is the OEM, and HP is a value-added reseller (VAR).

OS See operating system.

OS/2 A 32-bit multitasking operating system from IBM that runs on Intel processors.

OS/2 was developed jointly by Microsoft and IBM as the successor to MS-DOS, with Windows as a stop-gap measure until OS/2 was ready. When Microsoft chose to back Windows, placing considerable resources behind the breakthrough release of Windows 3, IBM took control of OS/2 development.

The current version of OS/2 Warp includes voice-recognition software, Internet access, and peer-to-peer networking. It supports Novell Directory Services (NDS) and includes IBM Personal Communications/3270, a package used to communicate with Systems Network Architecture (SNA) systems over TCP/IP networks.

IBM's server for network environments is OS/2 Warp Server.

OS/2 client Any computer running OS/2 that connects to a network server. OS/2 client workstations support TCP/IP, IPX/SPX, NetBIOS, and named pipes.

OS/2 Warp Server A network operating system, based on LAN Server 4, from IBM. It provides network services, system management services, remote access, standard file-sharing and network print services, support for up to 1,000 users, and secure Internet access.

As an application server, OS/2 Warp Server supports Microsoft Windows and Windows NT applications and can support OS/2, MS-DOS, Windows, Windows NT, AIX, and Macintosh clients. All the important network protocols are available, including TCP/IP, NetBIOS, and IPX.

OSF Abbreviation for Open Software Foundation. See Open Group.

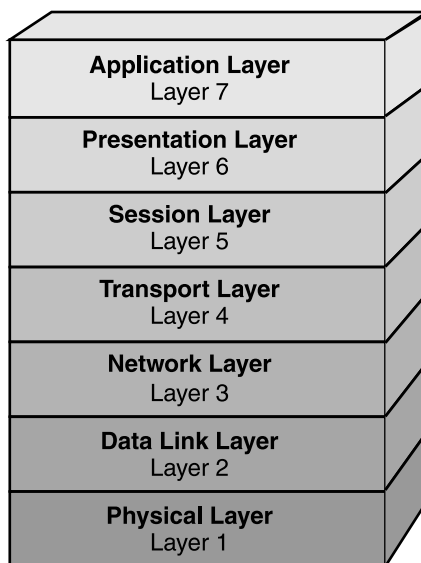
OSI Reference Model A networking reference model defined by the ISO (International Organization for Standardization) that divides computer-to-computer communications into seven connected layers. Such layers are known as a protocol stack.

Each successively higher layer builds on the functions of the layers below, as follows:

- **Application layer 7** The highest level of the model. It defines the manner in which applications interact with the network, including database management, e-mail, and terminal-emulation programs.
- **Presentation layer 6** Defines the way in which data is formatted, presented, converted, and encoded.

- **Session layer 5** Coordinates communications and maintains the session for as long as it is needed, performing security, logging, and administrative functions.
 - **Transport layer 4** Defines protocols for structuring messages and supervises the validity of the transmission by performing some error checking.
 - **Network layer 3** Defines protocols for data routing to ensure that the information arrives at the correct destination node.
 - **Data-link layer 2** Validates the integrity of the flow of data from one node to another by synchronizing blocks of data and controlling the flow of data.
 - **Physical layer 1** Defines the mechanism for communicating with the transmission medium and interface hardware.
- OSPF** See Open Shortest Path First.
- OSx** A version of Unix from Pyramid Technology that includes elements of both AT&T and BSD systems.
-

THE OSI REFERENCE MODEL



out-of-band signaling The transmission of control information on frequencies outside the bandwidth available for a voice or data transfer. The separation is usually accomplished by means of a filter.

Outlook Express Abbreviated OE. In Microsoft Windows, an application used to create, manage, send, and receive e-mail. Outlook Express can also access Internet newsgroups.

output Computer-generated information that is displayed on the screen, printed, written to disk or tape, or sent over a communications link to another computer.

See also input/output.

outsourcing To subcontract a company's data processing operations to outside contractors rather than maintain corporate hardware, software, and staff. Outsourcing is often used as a cost-cutting mechanism, although the cost savings can be difficult to quantify.

See also downsizing.

oversampling A time-division multiplexing (TDM) technique in which each bit from each channel is sampled more than once.

See also time-division multiplexing.

P

P See peta-.

PABX See private automatic branch exchange.

PACE See Priority Access Control Enabled.

Packard Bell NEC Major manufacturer of PCs and servers; manages all NEC Corporation's computer manufacturing outside Japan. Packard Bell has recently established a configure-it-yourself direct sales promotion known as NEC Now.

For more information on Packard-Bell, see www.packardbell.com.

See also NEC Corporation.

packet Any block of data sent over a network or communications link.

Each packet may contain sender, receiver, and error-control information, in addition to the actual message, which may be data, connection management controls, or a request for a service. Packets may be fixed- or variable-length, and they will be reassembled if necessary when they reach their destination. The actual format of a packet depends on the protocol that creates the packet; some protocols use special packets to control communications functions in addition to data packets.

See also datagram; frame; packet switching.

packet assembler/disassembler Abbreviated PAD. A device that is connected to a packet-switched network and that converts a serial data stream from a character-oriented device, such as a bridge or a router, into packets suitable for transmission. It also disassembles packets back into characters for transmission to the character device. PADs are often used to connect a terminal or computer to an X.25 packet-switched network.

Packet Burst Protocol Abbreviated PBP. A Novell NetWare protocol built on top of IPX that speeds up the transfer of multipacket NetWare Core Protocol (NCP) data transfers between a workstation and a server by removing the need to sequence and acknowledge every packet. Using PBP, a workstation or server can transmit a burst of packets before requiring an acknowledgment, thus reducing network traffic.

See also Internetwork Packet eXchange.

packet filtering A process used by bridges to limit protocol-specific traffic to one segment of the network, to isolate e-mail domains, and to perform other traffic-control functions. The network administrator sets the packet-filtering specifications for each bridge. If a packet matches the specifications, the bridge can either accept it or reject it.

See also application-level filter; firewall.

packet-level filter A category of firewall that provides a high degree of convenience, but a relatively low level of security.

A packet-level filter blocks or forwards a packet solely on its merits, without taking into account past history; the filter may also look at the source and destination address information contained in the packet.

This kind of filter is easy to implement, has little effect on network operations, but can be bypassed by encapsulating a blocked protocol within an allowed protocol.

See also application-level filter; firewall.

Packet-level Procedure Abbreviated PAP. An X.25 full-duplex protocol for the transfer of packets between a computer and a modem.

packet signature *See* NCP packet signature.

packet sniffer A program used by an intruder to monitor a data stream for a pattern such as a password or credit card number.

Packet sniffers also have a more salutary purpose when used for network analysis and troubleshooting by the system administrator.

See also network analyzer; protocol analyzer.

packet-switched network A network that consists of a series of interconnected circuits that route individual packets of data over one of several routes, offering flexibility and high reliability.

A packet-switched network may also be called connectionless because it contains

many different possible connections and routes that a packet might take to reach its destination.

The term often refers to networks using the international standard X.25.

See also packet switching.

packet switching A data-transmission method that simultaneously routes and transmits data packets from many customers over a communications channel or telephone line, thus optimizing use of the line.

An addressed packet is routed from node to node until it reaches its destination, although related packets may not all follow the same route to that destination. Because long messages may be divided into several packets, packet sequence numbers are used to reassemble the original message at the destination node.

The standard for packet-switching networks is defined in CCITT recommendation X.25. The Internet is an example of a packet-switching network.

See also Consultative Committee for International Telephony and Telegraphy; packet-switched network.

PAD *See* packet assembler/disassembler.

page A single document available on the World Wide Web or on a corporate intranet. A page can contain any combination of text, graphics, animated graphics, audio, and video and can be static or dynamic.

See also HyperText Markup Language.

page fault In Microsoft Windows, the fault that occurs when Windows attempts

to read to or write from a virtual memory location that is designated not present.

In Microsoft Windows Task Manager, the number of times information is read from disk because it was not present in memory.

page-mode RAM A memory-management technique used to speed up the performance of dynamic RAM (DRAM).

In a page-mode memory system, the memory is divided into pages by specialized DRAM chips. Consecutive accesses to memory addresses in the same page result in a page-mode cycle that takes about half the time of a regular DRAM cycle.

paged memory management unit

Abbreviated PMMU. A specialized chip designed to manage virtual memory. High-end processors, such as the Motorola 68040 and the Intel Pentium, have all the functions of a PMMU built into the chip itself.

See also virtual memory.

paging file In Microsoft Windows 2000, a hidden file used to hold programs and data files that will not currently fit into memory. The paging file, along with an area of physical memory, make up the virtual memory system. In most other operating systems, the paging file is known as the swap file.

PalmPilot The hand-held computer from 3Com Corporation, which has proved to be extremely popular, with more than 1 million units sold to date.

See also 3Com Corporation.

PAP *See* Packet-level Procedure; Password Authentication Protocol; Printer Access Protocol.

parallel communications The transmission of information from computer to computer or from computer to peripheral device, in which all the bits that make up the character are transmitted at the same time over a multiline cable.

See also serial communications.

parallel port An input/output (I/O) port that manages information eight bits at a time; often used to connect a parallel printer.

See also parallel communications; RS-232-C; serial communications; serial port.

parallel processing A computing method that can be performed by systems containing two or more processors operating simultaneously. In parallel processing, all processors work on different aspects of the same program at the same time, in order to share the computational load.

Parallel-processing computers can achieve incredible speeds. The Cray X-MP48 peaks at 1000 million floating-point operations per second (1000 MFLOP) using four extremely powerful processors, and parallel-hypercube systems, first marketed by Intel, can exceed 65,536 processors with speeds of up to 262 billion floating-point operations per second (262 GFLOP).

In all but the most trivial parallel-processing applications, the programmer or the operating system must assign approximate processor loads; otherwise, it is possible for non-optimized systems to fail to take advantage of the power available and, in the worst case,

run more slowly than on single-processor systems.

All this speed is used for applications such as weather forecasting, in which the predictive programs can take as long to run as the weather actually takes to arrive, 3-D seismic modeling, groundwater and toxic flow studies, and the modeling of full-motion dinosaur images used in movies.

See also asymmetrical multiprocessing; symmetrical multiprocessing.

parent directory In a hierarchical directory system, such as that used by MS-DOS, OS/2, Windows, and Unix, the directory immediately above the current directory. The special symbol `..` is shorthand for the name of the parent directory.

See also period and double-period directories.

parent domain In Microsoft Windows 2000 Server, any domain that has another domain subordinate to it.

See also domain; first-layer domain.

parent object In Novell Directory Services (NDS), an object that contains another object. This term is relative, because any particular parent object also has parent objects of its own and can be considered a child object from a certain perspective.

parity In communications, a simple form of error checking that uses an extra or redundant bit after the data bits but before the stop bit or bits. Parity may be set as follows:

- **Odd** Indicates that the sum of all the 1 bits in the byte plus the parity bit must be odd. If the total is already odd, the parity

bit is set to zero; if it is even, the parity bit is set to one.

- **Even** If the sum of all the 1 bits is even, the parity bit must be set to zero; if it is odd, the parity bit must be set to one.
- **Mark** The parity bit is always set to one and is used as the eighth bit.
- **Space** The parity bit is set to zero and is used as the eighth bit.
- **None** If parity is set to none, there is no parity bit, and no parity checking is performed.

The parity settings used by both communicating computers must match. Most online services, such as CompuServe or America Online, use no parity and an 8-bit data word.

See also asynchronous transmission; parity checking; parity error.

parity bit An extra or redundant bit used to detect data transmission errors.

See also parity.

parity checking A check mechanism applied to a character or a series of characters that uses the addition of extra or redundant parity bits.

Parity checking is useful for a variety of purposes, including asynchronous communications and computer memory coordination.

See also parity.

parity error A mismatch in parity bits that indicates an error in transmitted data.

See also parity.

partition 1. A portion of a hard disk that the operating system treats as if it were a

separate drive. In Windows, a hard disk can be divided into several partitions. A primary partition, generally assigned the drive letter C, might contain files that start the computer. You could also create a non-Windows partition and use it for a different operating system. In Novell NetWare, a server must have a NetWare partition defined in order to function as a part of the NetWare file system. Other partitions can manage file systems used by other operating systems.

2. In Novell NetWare, a grouping or collection of objects in the Novell Directory Services (NDS) database. Each partition consists of a container object, all the objects in it, and data about all those objects. Partitions do not include any information about the file system or the directories or files contained there. The data in a partition is also known as a replica.

See also disk mirroring; leaf object; replica; replication; Root object; volume.

partition table **1.** An area of storage on a hard disk that contains information about the partitions the disk contains. This information is usually recorded during the initial preparation of the hard disk before it is formatted.

2. In Novell Directory Services (NDS), a list on each server containing the NDS replicas. For each replica on the server, the partition table contains the partition name, type, time stamp, and partition state.

passive hub A device used in some networks to split a transmission signal, allowing additional hubs to be added to

the network, sometimes at the expense of distance.

See also active hub.

passive termination A method used to terminate a Small Computer System Interface (SCSI) chain of devices. Passive termination is a simple termination method that works best with four or fewer devices on a SCSI daisy chain.

See also active termination; forced perfect termination; Small Computer System Interface.

pass-through authentication In Microsoft Windows NT, an authentication method used when the user account must be authenticated, but the computer used for the logon is neither the domain controller for the account nor the computer on which the user account is defined. In such a case, the computer used for the logon passes the logon information through to the domain controller where the user account is defined.

See also domain; domain controller; user account.

password A security method that identifies a specific, authorized user of a computer system, a network, or a resource by a unique string of characters.

In general, passwords should be a mixture of upper- and lowercase letters and numbers and should be more than six characters. Here are some general guidelines:

- Passwords should be kept secret and changed frequently. The worst passwords are the obvious ones: people's names or initials, place names, phone numbers,

birth dates, and anything to do with computers or Star Trek. There are a limited number of words in the English language, and it is easy for a computer to try them all relatively quickly.

- Change all passwords every 90 days, and change those associated with high-security privileges every month. Some network operating systems require that passwords expire even more frequently. For example, in NetWare 5, passwords expire after 40 days by default.
- Some systems provide default passwords, such as MANAGER, SERVICE, or GUEST, as part of the installation process. These default passwords should be changed immediately.
- Limit concurrent sessions to one per system.
- Do not allow more than two or three invalid password attempts before disconnecting.
- Do not allow generic accounts.
- Promptly remove the accounts of transferred or terminated people, as well as all unused accounts.
- Review the security log files periodically.

See also authentication; Challenge-Handshake Authentication Protocol; Password Authentication Protocol.

Password Authentication Protocol

Abbreviated PAP. A security protocol that requires a user to enter a user name and password before gaining access to a secure server.

See also Challenge-Handshake Authentication Protocol.

password encryption In certain operating systems, the password you enter to gain access to the system is not stored as ordinary text, but is encrypted, and this encrypted form is compared against the encrypted password stored on the server. If the two match, the logon continues; if not, the logon attempt is rejected.

See also Challenge-Handshake Authentication Protocol; Password Authentication Protocol.

password protection The use of one or more passwords to prevent unauthorized access to computer systems.

patch panel A panel, usually located in a wiring closet, that contains rows of telephone-type modular jacks. A patch panel allows the network administrator to connect, disconnect, move, and test network devices by changing these connections.

path The complete location of a directory or a file in the file system. Also called pathname or directory path.

See also Universal Naming Convention.

pathname *See* path.

Payload Type Identifier Abbreviated PTI. In an Asynchronous Transfer Mode (ATM) cell, a field contained in the 5-byte cell header that defines the type of information in the payload area, including user, network, and management information.

See also Cell Loss Priority; Header Error Control; Virtual Channel Identifier; Virtual Path Identifier.

PB *See* petabyte.

PBP See Packet Burst Protocol.

PBX See private branch exchange.

PC98 A personal computer design guide for 1998–99 from Intel, Microsoft, and others, covering the PC system, bus, and devices, including guidelines for various types of mobile PCs.

PC98 also describes requirements for manageability, remote boot support, and specifications for 1394. The basic PC98 should have no ISA (Industry Standard Architecture) devices; PC99 is expected to do away with ISA slots as well.

See also Advanced Configuration and Power Interface; Web-Based Enterprise Management; Wired for Management; Zero Administration for Windows.

PC-based router A router, such as the NetWare Multiprotocol Router or Multiprotocol Router Plus, that operates on a standard Intel-based personal computer.

PC Card A term that describes plug-in cards that conform to the PCMCIA (Personal Computer Memory Card International Association) standard. A PC Card is about the size of a credit card and uses a 68-pin connector with longer power and ground pins that will always engage before the signal pins engage.

Several versions of the standard have been approved by PCMCIA:

- **Type I** The thinnest PC Card, only 3.3 millimeters (0.13 inch) thick, used for memory enhancements, including dynamic RAM, static RAM, and flash memory.

- **Type II** A card used for modems or network interface cards, 5 millimeters (0.2 inch) thick; may also hold a Type I card.

- **Type III** A 10.5 millimeter (0.4 inch) card, used for mini-hard disks and other devices that need more space, including wireless network interface cards; may also hold two Type I or Type II cards.

In theory, each PC Card adapter can support 16 PC Card sockets (if there is enough space), and as many as 255 adapters can be installed in a PC that follows the PCMCIA standard; in other words, PCMCIA allows a maximum of 4080 PC Cards on one computer.

Most PC Card devices are modems, Ethernet and token-ring network adapters, dynamic RAM, and flash memory cards, although mini-hard disks, wireless LAN adapters, and SCSI adapters are also available.

See also PC Memory Card International Association.

PC Card slot An opening in the case of a portable computer, intended to receive a PC Card; also known as a PCMCIA slot.

PC Connection, Inc. A direct marketer of PCs, peripherals, accessories, and networking products to the home, government, business, and educational markets. PC Connection was one of the first companies to provide overnight delivery of products and toll-free technical support.

For more information on PC Connection, see www.pcconnection.com.

PC-DOS See Personal Computer Disk Operating System.

PCI local bus See Peripheral Component Interconnect local bus.

PCI-X A revision to the PCI standard proposed by IBM, Hewlett-Packard, and Compaq that increases the bus width to 64 bits, the bus speed to 133MHz, and the maximum throughput to 1GB per second. This revision is aimed at future workstation and server design.

Several vendors also offer hot-plug PCI slots that allow you to replace a failed component without a system reboot.

See also 1394; local bus; Peripheral Component Interconnect local bus; Plug and Play; Universal Serial Bus.

PCMCIA See PC Memory Card International Association.

PCMCIA slot See PC Card slot.

PC Memory Card International Association Abbreviated PCMCIA. A nonprofit association, formed in 1989, with more than 320 members in the computer and electronics industries, that developed a standard for credit-card-sized, plug-in adapters designed for portable computers.

PCONSOLE A Novell NetWare workstation utility used to set up and manage print queues and print servers on the network.

See also NetWare Administrator.

PCR See Peak Cell Rate.

PCS See Personal Communications Services.

PC Service and Support Certified Professional A certification from Learning Tree designed for the technician. Courses and exams cover basic PC assembly and troubleshooting, installation and configuration of operating systems and peripherals, and optimization of networks.

PDA See personal digital assistant.

PDF See Portable Document Format.

PDN See private data network; public data network.

Peak Cell Rate Abbreviated PCR. In Asynchronous Transfer Mode (ATM), the maximum cell rate that a specific data source on the network can sustain.

See also Asynchronous Transfer Mode; Available Cell Rate; Minimum Cell Rate; Sustainable Cell Rate.

peer-to-peer architecture A network architecture in which two or more nodes can communicate with each other directly, without the need for any intermediary devices. In a peer-to-peer system, a node can be both a client and a server.

See also peer-to-peer network.

peer-to-peer network A LAN in which drives, files, and printers on each PC can be available to every other PC on the network, eliminating the need for a dedicated file server. Each PC can still run local applications.

Peer-to-peer networks introduce their own system management problems, including administration and responsibility for system backup, reliability, and security. Peer-to-peer systems are often used in relatively

small networks, with two to ten users, and can be based on MS-DOS, Windows, or Unix. Performance is not usually as good on a peer-to-peer network as under the client/server model, especially under heavy traffic loads.

See also client/server architecture.

PEM *See* Privacy Enhanced Mail.

Pentium A family of microprocessors introduced by Intel in 1993. The Pentium represents the continuing evolution of the 80486 family of microprocessors and adds several notable features, including instruction code and data caches and a built-in floating-point processor and memory management unit. It also has a superscalar design and dual pipelining, which allow the Pentium to execute more than one instruction per clock cycle, a 32-bit address bus, and a 64-bit data bus.

Available in a range of clock speeds, from 60MHz all the way up to 233MHz, the Pentium is equivalent to 3.1 million transistors, more than twice that of the 80486.

See also Intel Corporation; Pentium II; Pentium III; Pentium Pro.

Pentium II A family of microprocessors from Intel. The Pentium II represents the continuing evolution of the Pentium family of microprocessors and adds several notable features, including integrated L1/L2 caches of up to 2MB that can be accessed at the full clock speed and a built-in floating-point processor and memory management unit. It also has a superscalar design and dual pipelining, which allow the Pentium II

to execute more than one instruction per clock cycle.

Available in a range of clock speeds, from 233MHz all the way up to 450MHz, the Pentium II can use a 100MHz system bus and is equivalent to 7.5 million transistors, more than twice that of the Pentium.

See also Intel Corporation; Pentium; Pentium Pro.

Pentium III A family of microprocessors from Intel. The Pentium III represents the continuing evolution of the Pentium family of microprocessors and adds several notable features, including 50 new floating-point instructions and 8 new registers to speed up floating-point calculations in scientific and engineering calculations, along with 12 new multimedia instructions to increase MPEG-2 performance and speech recognition. The most controversial new feature is the processor serial number, designed to increase network and online shopping security, but feared by many as a threat to privacy.

Available in a whole range of clock speeds, initially from 450MHz to 500MHz versions, the Pentium III can use the Pentium II 100MHz system bus and is equivalent to 9.5 million transistors.

See also Intel Corporation; Pentium; Pentium II; Streaming SIMD Extensions.

Pentium Pro A family of microprocessors introduced by Intel in 1995. The Pentium Pro is optimized for the execution of 32-bit software and is available with clock speeds from 150 to 200MHz. With a 32-bit data bus running at 60 or 66MHz, it supports superscalar

architecture and pipelines and contains the equivalent of 5.5 million transistors.

Dynamic execution (a combination of branch prediction and speculative execution) allows the processor to anticipate and schedule the next instructions for execution. Pentium Pro offers up to 1MB of Level 2 cache that runs at the same speed as the processor.

See also Intel Corporation; Pentium; Pentium II.

PeopleSoft, Inc. A major developer of large enterprise resource planning (ERP) applications for finance, materials and inventory management, distribution, human resources, and manufacturing, all within a single computing environment.

For more information on PeopleSoft, see www.peoplesoft.com.

Performance Monitor In Microsoft Windows NT, a network and server monitoring tool that displays resource use for selected system-level components; you can also use Performance Monitor to troubleshoot performance problems and assess hardware upgrade requirements. Information can be logged to a file for later analysis.

period The . character; pronounced “dot.” Used to indicate the name of the current directory in a pathname and to separate the elements in a domain name, as in www.sybex.com.

period and double-period directories

In a hierarchical directory system, a shorthand way of referring to directories. The period (.) represents the current directory, and the double period (..) represents the di-

rectory immediately above the current directory.

See also parent directory; root directory.

Peripheral Component Interconnect

local bus Abbreviated PCI local bus. A specification introduced by Intel in 1992 for a local bus that allows up to 10 PCI-compliant expansion cards to be plugged into the computer. One of these expansion cards must be the PCI controller card, but the others can include a video card, network interface card, SCSI interface, or any other basic function.

The PCI controller exchanges information with the computer’s processor, either 32 or 64 bits at a time, and allows intelligent PCI adapters to perform certain tasks concurrently with the main processor by using bus-mastering techniques.

PCI is compatible with ISA (Industry Standard Architecture), EISA (Extended Industry Standard Architecture), and MCA (Microchannel Architecture) expansion buses for backward compatibility with older technologies. PCI can operate at a bus speed of 32MHz and can manage a maximum throughput of 132MBps with a 32-bit data path or a rate of 264MBps with a 64-bit data path.

See also 1394; local bus; PCI-X; Plug and Play; Universal Serial Bus.

Perl Acronym formed from Practical Extraction and Report Language. Perl is an interpreted programming language developed by Larry Wall, used to manipulate text, files, and processes and to print reports on the extracted information.

Perl is rapidly becoming the system administrator's answer to all those problems that a C program does not seem to fit. It does not have the arbitrary limitations of other languages, so lines can be of any length, arrays can be of any size, variable names can be as long as you care to make them, and binary data does not cause problems.

permanent swap file A swap file that, once created, is used over and over again. This file is used in virtual memory operations, in which hard-disk space is used in place of RAM.

See also temporary swap file.

permanent virtual circuit Abbreviated PVC. A permanent communications circuit, created and maintained even when no data is being transmitted.

A PVC has no setup overhead and gives improved performance for periodic transmissions that require an immediate connection. Packets are transferred in order over a specific path and arrive at their destination in the same order.

See also switched virtual circuit.

permissions In a network or multiuser environment, the ability of a user to access certain system resources, including files and directories. Permissions are based on the rights given to user accounts by the system administrator.

See also rights.

permuted index A special kind of index used in several of the Unix system manuals. Many of the Unix manuals treat each command on a separate page, and these pages

are not numbered continuously; they are numbered only within each command. This makes it easy to add or remove pages as requirements change, but it can make it difficult to find specific information. The permuted index is the solution.

The permuted index has three columns. The central column, where you start your search, is in alphabetic order. The column to the right lists the command that performs the function and the section number in the man pages where you will find a detailed description, and the column on the left contains additional keywords to help confirm that you have found the correct entry.

See also man pages.

Personal Communications Services

Abbreviated PCS. A digital wireless communications technology that includes voice, data, and video.

PCS competes with the traditional analog cellular phone system, but PCS's digital technology can provide clearer voice quality, better security through encryption, and lower costs, as well as additional services such as messaging, voice mail, and caller ID.

Personal Computer Disk Operating System

Abbreviated PC-DOS. The version of the DOS operating system supplied with PCs made by IBM.

PC-DOS and MS-DOS began as virtually identical operating systems, with only a few minor differences in device driver names and file sizes, but after the release of DOS 6 (MS-DOS 6.2 and PC-DOS 6.1), the two grew much further apart.

See also Microsoft Disk Operating System.

personal digital assistant Abbreviated PDA. A tiny, pen-based, battery-powered computer that combines personal organization software with fax and e-mail facilities into a unit that fits into your pocket. PDAs are available from several manufacturers, including Apple's Newton, and others from 3Com, Casio, Tandy, Toshiba, Motorola, Sharp, Sony, GRiD, and AT&T.

Personal Information Manager Abbreviated PIM. A multipurpose software package that combines a word processor, database, and other accessory modules to allow the user to manipulate data in a less structured way than required by conventional programs. A PIM can store notes, memos, names and addresses, appointments, and to-do lists, and it may be part of the software used in a PDA (personal digital assistant).

Personal NetWare Novell's peer-to-peer network replacement for NetWare Lite, released in 1994, that provides MS-DOS and Microsoft Windows users with the ability to share files, printers, CD-ROMs, and other resources, as well as run standard network applications. Other features include simplified network administration, increased security, and a single login so that users can view or access all network resources at once.

Personal NetWare can manage a maximum of 50 workstations per server and a maximum of 50 servers on each network, giving a maximum of 2500 nodes per network.

personalized menus In Microsoft Windows 2000, a feature that hides infrequently used menu selections.

See also cascading menus.

peta- Abbreviated P. A metric system prefix for one quadrillion, or 10^{15} . In computing, based on the binary system, peta has the value of 1,125,899,906,842,624, or the power of 2 (2^{50}) closest to 1 quadrillion.

petabyte Abbreviated PB. Usually 1,125,899,906,842,624 bytes (2^{50}), but may also refer to 1 quadrillion bytes (10^{15}).

PGP *See* Pretty Good Privacy.

physical address *See* hardware address.

physical device An item of hardware, such as a disk drive or a tape drive, that is physically separate from other devices.

physical drive A real drive in the computer that you can see or touch, as opposed to a conceptual or logical drive. One physical drive may be divided into several logical drives, which are parts of the hard disk that function as if they were separate disk drives.

See also partition; volume.

physical layer The first and lowest of the seven layers in the OSI Reference Model for computer-to-computer communications. The physical layer defines the physical, electrical, mechanical, and functional procedures used to connect the equipment.

See also OSI Reference Model.

physical unit Abbreviated PU. The name used in IBM's Systems Network Architecture (SNA) to indicate a physical device and its associated resources within the network.

See also logical unit.

PIM *See* Personal Information Manager.

pin-compatible A description of a chip or other electronic component with connecting pins exactly equivalent to the connecting pins used by a different device. With a pin-compatible chip, you can easily upgrade a system by replacing the older chip with the newer version.

See also plug-compatible.

Ping Acronym formed from packet internet groper. A TCP/IP command used to test for network connectivity by transmitting a special ICMP (Internet Control Message Protocol) diagnostic packet to a specific node on the network, forcing the node to acknowledge that the packet reached the correct destination. If the node responds, the link is operational; if not, something is wrong. The word *ping* is often used as a verb, as in "ping that workstation to see if it's alive."

Ping is designed for network testing, troubleshooting, and measurement, and because of the large load it can impose on a busy, working network, it should not be used during normal operations, unless the system administrator is tracing a specific problem on the network.

ping of death A very large, specially constructed ICMP packet that violates the

rules for packet size and content, designed to crash the receiving computer.

See also brute-force attack; denial of service attack; dictionary attack.

pinouts The configuration and purpose of each pin in a multipin connector.

pipe A section of memory that can be used by a program or a command to pass information to a second command for processing. The information is stored in a first-in first-out basis and is not altered during transmission. A pipe is opened like a file and is read from or written to in the same way; pipes are also unidirectional in that one pipe is used to read data and another is used to write data.

A special form of pipe, known as a named pipe, originated in the Unix operating system. A named pipe allows two processes to exchange information. This concept has been extended in several network operating systems as a method of interprocess communication, allowing data to be exchanged between applications running on networked computers.

See also mailslots; named pipe; semaphore; shared memory; socket.

pipeline A mechanism used in microprocessors that speeds up the processing of instructions.

The Intel Pentium processor features two pipelines, one for data and one for instructions, and can process two instructions per clock cycle. A processor with two or more pipelines is said to be superscalar.

See also superscalar.

pipeline burst cache A secondary or L2 cache associated with a microprocessor that allows fast data-transfer rates. Pipeline burst cache requires RAM chips that can synchronize with the microprocessor's clock.

pipeline stall A microprocessor design error that leads to delays in the processing of an instruction.

pipelining 1. In processor architecture, a method of fetching and decoding instructions that ensures that the processor never needs to wait; as soon as one instruction is executed, the next one is ready.

2. In parallel processing, the method used to pass instructions from one processing unit to another.

See also parallel processing.

PKI *See* Public Key Infrastructure.

PKUNZIP A very popular file decompression utility available as shareware. PKUNZIP uncompresses files or archives created by PKZIP; both programs are usually available together.

See also PKZIP; WinZip.

PKZIP A very popular file compression utility available as shareware. PKZIP not only compresses files to save disk space or cut modem transmission times, but also combines compressed files to create compressed archives.

See also PKUNZIP; WinZip.

plaintext Text that has not been encrypted in any way and that can be intercepted

and read easily while in transit; usually applied to an unencrypted password.

See also cleartext.

platform 1. An operating system environment, such as a NetWare platform or a Unix platform.

2. A computer system based on a specific microprocessor, such as an Intel-based platform or a PowerPC-based platform.

platform-specific routers Routers based on a specific and proprietary hardware architecture, which is usually vendor-specific.

player A small program launched or used by a Web browser to process a specific type of file that the browser cannot handle. A player is a program that deals with sound files.

See also helper; plug-in; viewer.

plenum cable Cable with a special Teflon coating designed for use in suspended ceilings, in inside walls, or between floors.

The Teflon coating provides low flame-spread and low, nontoxic smoke in the case of an accident. Plenum cables should meet the CMR (Communications Riser Cable) or CMP (Communications Plenum Cable) specifications of the National Electric Code and are often used for cable runs in air-return areas.

See also riser cable.

Plug and Play Abbreviated PnP. A standard from Compaq, Microsoft, Intel, and Phoenix that defines techniques designed to make PC configuration simple and

automatic. A user can plug in a new device, and the operating system will recognize it and configure it automatically when the system is next started.

PnP adapters contain configuration information stored in nonvolatile memory, which includes vendor information, serial number, and checksum information. The PnP chipset allows each adapter to be isolated, one at a time, until all cards have been properly identified by the operating system.

The PnP-compatible BIOS isolates and identifies PnP cards at boot time, and when you insert a new card, the BIOS performs an auto-configuration sequence enabling the new card with appropriate settings.

New PCs with flash BIOS will be easy to upgrade so that they can take advantage of PnP; older systems with ROM-based BIOS will need a hardware change before they can take full advantage of PnP.

See also Peripheral Component Interconnect local bus; Plug and Pray.

Plug and Pray What most of us do when our Plug-and-Play systems do not work automatically.

plug-compatible Any hardware device designed to work in exactly the same way as a device manufactured by a different company. For example, all external serial devices are plug-compatible, because you can replace one with another without changing the cabling or connector.

See also pin-compatible.

plug-in A small program you can link in to your Web browser to add a special capability not originally present or to recognize

new file types or content. Plug-ins are available from a huge number of companies and are usually free.

See also helper.

PMMU *See* paged memory management unit.

PNNI *See* Private Network-to-Network Interface.

PnP *See* Plug and Play.

POH *See* power-on hours.

PointCast, Inc. The largest privately held media company on the Internet, providing online news to corporations. The PointCast broadcast receives more than 120 million hits a day and offers access to a collection of more than 600 leading business sources.

Unlike the World Wide Web and other Internet applications, PointCast uses server push technology, in which the server automatically sends new data to a client without a specific request from that client.

For more information on PointCast, see www.pointcast.com.

point-to-point link A direct connection between two, and only two, locations or nodes.

Point-to-Point Protocol Abbreviated PPP. A TCP/IP protocol used to transmit IP datagrams over serial lines and dial-up telephone point-to-point connections.

PPP allows a PC to establish a temporary direct connection to the Internet via modem and appear to the host system as if it were an Ethernet port on the host's network.

Point-to-Point Tunneling Protocol

PPP provides router-to-router, host-to-router, and host-to-host connections and also provides an automatic method of assigning an IP address so that mobile users can connect to the network at any point.

See also PPP Multilink; Point-to-Point Tunneling Protocol; Serial Line Internet Protocol.

Point-to-Point Tunneling Protocol

Abbreviated PPTP. A proprietary networking protocol proposed by Microsoft that supports virtual private networks, allowing remote users to access Windows NT Server systems across the Internet without compromising security. PPTP allows corporations to use public networks rather than leasing its own lines for wide area communications.

See also encapsulation; Layer 2 Tunneling Protocol; PPP Multilink; tunneling.

point of presence Abbreviated POP. A connection to the telephone company or to long-distance carrier services.

polling A method of controlling the transmission sequence of devices on a shared circuit or multi-drop line by sending an inquiry to each device asking if it wants to transmit. If a device has data to send, it sends back an acknowledgment, and the transmission begins. Three methods are in common use:

- **Roll-call** A master station uses a polling list to locate the next node to poll.
- **Hub** A node polls the next node in sequence.
- **Token-passing** A token is passed to the next node in sequence. This node can

transmit or pass the token to the next device.

polymorphic virus A form of virus that can change its appearance to avoid detection. The virus encrypts itself using a special formula each time an infection occurs. Virus-detecting software uses special scanning techniques to find and remove polymorphic viruses.

See also boot sector virus; file-infecting virus; macro virus; multipart virus; stealth virus; vaccine; virus.

POP *See* point of presence; Post Office Protocol.

port 1. To move a program or an operating system from one hardware platform to another. For example, Windows NT portability refers to the fact that the same operating system can run on both Intel and reduced instruction set computing (RISC) architectures.

2. The point at which a communications circuit terminates at a network, serial, or parallel interface card, usually identified by a specific port number or name.

3. A number used to identify a connection point to a specific Internet protocol.

See also portable; port number.

portability The ability to transfer an application or operating system from one vendor's hardware to another, quickly and easily, without rewriting the software and without affecting its performance.

This can be achieved in several ways:

- Write the program in a portable language, such as C, C++, or Java.

- Use only standard programming language features.
- Use only standard libraries.
- Don't make assumptions about word size or byte ordering.
- Use layers of software to distance the application from operating system or hardware dependencies.

portable Describes the degree to which a program can be moved easily to different computing environments with a minimum number of changes. Applications written for the Unix operating system are often described as portable applications, as are Java applets.

See also Java.

portable computer Any computer light and small enough to be carried easily. There are two types:

- **Laptop computers**, which are small enough to be used in an airplane seat and powerful enough to run major operating systems and popular business applications. Extended battery life is making the laptop a serious alternative to the desktop system.
- **Notebook computers**, which are smaller than laptops and about the size of a textbook or student notebook, but still capable of running major applications. A notebook computer will easily fit into a briefcase.

Major advances in battery life and the use of flash memory are part of the continuing development of portable computers.

See also docking station; port replicator.

Portable Document Format Abbreviated PDF. A file format standard developed by Adobe Systems and others for use in electronic documents. A file in this format usually has the filename extension of .PDF.

Portable Operating System Interface

Abbreviated POSIX. A collection of IEEE standards that defines a complete set of portable operating system services. POSIX is based on Unix services, but it can be implemented by many other operating systems.

Each of the standards defines a specific aspect of an operating system, including such areas as system administration, system security, networking, and the user interface.

When program or operating system service meets the appropriate POSIX standard, it is said to be POSIX-compliant.

See also IEEE standards.

port multiplier A concentrator that provides multiple connections to the network.

port number The default identifier for a TCP/IP (Transmission Control Protocol/Internet Protocol) or Internet process.

For example, ftp (File Transport Protocol), HTML (HyperText Markup Language), and Telnet are all available at preassigned unique port numbers so that the computer knows how to respond when it is contacted on a specific port; Web servers use port 80, and SMTP (Simple Mail Transfer Protocol) e-mail is always delivered to port 25. You can override these defaults by specifying different values in a URL, but whether they will work depends on the configuration on the target system.

A total of 65,535 port numbers are available for use with TCP, and the same number are available for UDP (User Datagram Protocol).

See also port.

port replicator A device containing standard computer ports used to avoid constantly connecting and disconnecting peripherals from a portable computer.

A port replicator duplicates all your computer's ports and may even add a Small Computer System Interface (SCSI) port or a second Universal Serial Bus port. The external monitor, full-sized keyboard, and mouse you use in the office are connected to the port replicator; when it is time to take the portable computer on the road, you simply unplug the port replicator, leaving everything attached to the replicator for your return.

See also docking station.

portal A large Web site that acts as a gateway to the Internet and may also offer search facilities, free e-mail, online chat, instant messaging, as well as other services, including hard news, sports, and personal finance. Portals make money by selling advertising space.

POSIX *See* Portable Operating System Interface.

POST *See* power-on self test.

post An individual article or e-mail message sent to a Usenet newsgroup or

mailing list, rather than a message sent to an individual.

See also posting; Usenet.

Postal Telephone and Telegraph Abbreviated PTT. The official government body that administers and manages the telecommunications systems in many European countries.

posting Sending an article or an e-mail message to a Usenet newsgroup.

See also post; Usenet.

Post Office Protocol Abbreviated POP. An Internet mail server protocol that also provides an incoming mail storage mechanism.

POP works with Simple Mail Transfer Protocol (SMTP), which actually moves the e-mail from one system to another, and the latest version of the standard is POP3.

When a client connects to a POP3 server, all the messages addressed to that client are downloaded; there is no ability to download messages selectively. Once the messages are downloaded, the user can delete or modify messages without further interaction with the server.

In some locations, POP3 is being replaced by another standard, Internet Mail Access Protocol (IMAP) version 4.

See also Internet Mail Access Protocol; Simple Mail Transfer Protocol.

power conditioning The use of protective and conditioning devices to filter out

power surges and spikes and ensure clean power. There are three main types of power-conditioning devices:

- **Suppression**, which protects against sudden destructive transient voltages.
 - **Regulation**, which modifies the power waveform back to a clean sine wave. A UPS (uninterruptible power supply) is a common form of voltage regulator. It may be online, actively modifying the power, or offline and available only after the line voltage drops below a certain level.
 - **Isolation**, which protects against noise.
- These types of devices are often expensive.

Because power conditioning is expensive, usually only the servers or hosts in a network are protected. Surge suppressors may be used with workstations or other important network nodes, such as bridges or routers.

See also blackout; brownout; power surge; spike; surge.

Power Mac A series of computers from Apple Computer, Inc., based on the PowerPC chip. Although Power Macs run on the PowerPC chip rather than on the traditional Motorola chips, they run a version of the Macintosh operating system and look and feel just like 680x0-based computers. They can also run MS-DOS and Windows software under emulation.

power-on hours Abbreviated POH. A cumulative count of the hours since the last time the system was started.

See also mean time between failures.

power-on self test Abbreviated POST. A set of diagnostic programs loaded from ROM before any attempt is made to load the

operating system; designed to ensure that the major system components are present and operating. If a problem is found, the POST firmware displays an error message on the screen, sometimes with a diagnostic code number indicating the type of fault.

PowerPC A family of RISC-based, super-scalar microprocessors jointly developed by Apple, Motorola, and IBM, with a 32-bit address bus and a 64-bit data bus.

- The 601 houses 2.8 million transistors, runs at 110MHz, and is designed for use in high-performance, low-cost PCs.
- The 66MHz 602 is targeted at the consumer electronics and entry-level computer markets.
- The low-wattage 603e is aimed at battery-powered computers.
- The 604 is for high-end PCs and workstations.
- The 64-bit 620 is available in a 133MHz version capable of executing four instruction per clock cycle and is designed for servers and high-performance applications.
- The 750 (also known as the G3) is available in a range of processors running from 333 to 400MHz with an integrated L2 cache of 1MB and is equivalent to 6.35 million transistors.

power supply A part of the computer that converts the power from a wall outlet into the lower voltages, typically 5 to 12 volts DC (direct current), required internally in the computer. PC power supplies are usually rated in watts, ranging from 90 to 300 watts. If the power supply in a computer fails, nothing works—not even the fan.

power surge A sudden, brief, and often destructive increase in line voltage. A power surge may be caused by an electrical appliance, such as a photocopier or elevator, or by power being reapplied after an outage.

See also power conditioning; surge; surge suppressor.

PPP *See* Point-to-Point Protocol.

PPP Multilink An extension to the Point-to-Point Protocol that can provide bandwidth on demand by combining multiple links between two systems; a process also known as *bonding*. PPP Multilink provides the negotiation features and protocols that allow systems to indicate that they can bond. The links can be of different types and different speeds.

See also inverse multiplexing; Point-to-Point Protocol.

PPTP *See* Point-to-Point Tunneling Protocol.

preemptive multitasking A form of multitasking in which the operating system executes an application for a specific period of time, according to its assigned priority. At that time, it is preempted, and another task is given access to the CPU for its allocated time. Although an application can give up control before its time is up, such as during input/output waits, no task is ever allowed to execute for longer than its allotted time period.

See also cooperative multitasking; time-slice multitasking.

presentation layer The sixth of seven layers of the OSI Reference Model for

computer-to-computer communications. The presentation layer defines the way in which data is formatted, presented, converted, and encoded.

See also OSI Reference Model.

Pretty Good Privacy Abbreviated PGP. A popular public-key encryption and digital certificate program, originally written by Phil Zimmermann, available at no charge from certain Internet sites.

PGP uses Diffie-Hellman public-key algorithms, is available for Microsoft Windows and Macintosh platforms, and works with most popular messaging applications such as Microsoft Exchange, Eudora, and Claris EMailer for the Macintosh.

See also Privacy Enhanced Mail; RSA Data Security.

PRI *See* Primary Rate ISDN.

primary domain controller In a Microsoft Windows NT domain, a computer running Windows NT Server that authenticates domain logons and manages the directory database for the domain. All changes to all accounts in the domain are automatically tracked and sent to the primary domain controller. There can be only one primary domain controller in any domain.

See also backup domain controller; domain; domain controller.

primary key *See* key.

primary member One of two members of a mirror set. The primary member contains the original data; the shadow member contains the copy.

See also disk mirroring; shadow member.

Primary Rate ISDN Abbreviated PRI. An ISDN (Integrated Services Digital Network) service that provides 23 B (bearer) channels, capable of speeds of 64Kbps, and one D (data) channel, also capable of 64Kbps. The combined capacity of 1.544Mbps is equivalent to one T1 channel.

See also Basic Rate ISDN.

primary time server In Novell NetWare, a server that provides time information to secondary time servers and to workstations.

A primary time server must synchronize time information with at least one other primary or reference time server.

See also reference time server; secondary time server; single reference time server.

PRINTCON A Novell NetWare workstation utility used to create, view, or modify print-job configurations on the network. Configuration options include the printer to be used, the print queue to process the job through, the print-device mode, the printer form number, and the number of copies.

PRINTDEF A Novell NetWare workstation utility used to create, view, and modify printer definitions on the network.

Printer Access Protocol Abbreviated PAP. The protocol used in AppleTalk networks to manage communications between computers and printers.

See also AppleTalk.

Printer Agent In Novell Distributed Print Services, a printer object that replaces the Print Queue, Printer, and Printer Server objects used in other Novell printing environments.

printer emulation The ability of a printer to change modes so that it behaves like a printer from another manufacturer. For example, many dot-matrix printers offer an Epson printer emulation in addition to their own native mode. Most laser printers offer a Hewlett-Packard LaserJet emulation.

See also emulator; terminal emulation.

print queue A collection of documents waiting to be printed on a particular network printer.

See also Novell Distributed Print Services.

print server A server that handles printing for all users on the network. A print server collects print jobs sent by applications running on other networked PCs, places them in a print queue on the hard disk, and routes them to one or more printers attached to the print server.

See also local printer; Novell Distributed Print Services.

print spooler In an operating system or network operating system, the software that coordinates print jobs sent to a shared printer when that printer is busy. Each print job is stored in a separate file and is printed in turn when the printer becomes free.

Priority Access Control Enabled Abbreviated PACE. A technology from 3Com Corporation designed to deliver on-time multimedia over switched Ethernet networks

with insufficient bandwidth and without the ability to prioritize traffic. This is accomplished by the use of the PACE-enabled switches that allow the switch port and end stations to take turns when transmitting.

Privacy Enhanced Mail Abbreviated PEM. An e-mail standard that uses a patented RSA encryption scheme to provide a confidential method of authentication. PEM is little used due to the proprietary nature of the encryption scheme.

See also Secure MIME; RSA.

private automatic branch exchange

Abbreviated PABX. An automatic telephone system that serves a particular location, such as an office, providing connections from one extension to another, as well as a set of connections to the external telephone network. Many PABXs handle computer data and may include X.25 connections to a packet-switched network.

See also private branch exchange.

private branch exchange Abbreviated PBX. A telephone system, usually owned by the customer, that serves a particular location, such as an office, providing connections from one extension to another, as well as a set of connections to the external telephone network.

See also private automatic branch exchange.

private data network Abbreviated PDN. A highly secure and very expensive network of leased lines built for a single user, usually a corporation.

PDNs are used to transmit highly sensitive data such as banking and other financial information. The service provider guarantees a certain bandwidth will always be available, although some of that bandwidth may go unused during periods of light traffic.

See also Virtual Private Network.

private key One of two keys used in public key encryption. The user keeps the private key secret and uses it to encrypt digital signatures on outgoing messages and to decrypt incoming messages.

See also public key encryption.

private leased circuit A leased communications circuit, available 24 hours a day, 7 days a week, that connects a company's premises with a remote site.

Private Network-to-Network Interface Abbreviated PNNI. A dynamic link-state routing protocol for Asynchronous Transfer Mode (ATM)-based networks. Any given ATM network may include ATM switches from several vendors; PNNI provides a routing protocol to communicate configuration information about the network to these groups of switches.

See also Integrated-Private Network-to-Network Interface.

privileged mode An operating mode supported in protected mode in Intel processors that allows the operating system and certain classes of device drivers to manipulate parts of the system, including memory and input/output ports. Applications cannot be executed in privileged mode.

See also protected mode; privilege level; real mode.

privilege level 1. Those rights granted to a user or a group of users by the network administrator that determine the functions the user can execute. Rights form an important component of network security and can include supervisor rights and read, write, erase, and modify rights, along with several others.

2. A form of protection built into Intel microprocessors. The Intel microprocessor architecture provides two broad classes of protection. One is the ability to separate tasks by giving each task a separate address space. The other mechanism operates within a task to protect the operating system and special processor registers from access by applications. Within a task, four privilege levels are defined. The innermost ring is assigned privilege level 0 (the highest, or most trusted, level), and the outermost ring is privilege level 3 (the lowest, or least privileged, level). Rings 1 and 2 are reserved for the operating system and operating system extensions; level 3 is available to applications. This protection is maintained by complex circuitry in the processor's memory management unit.

PRN In MS-DOS, Windows, and OS/2, the logical device name for a printer, usually the first parallel port, which is also known as LPT1.

See also parallel port.

process In a multitasking operating system, a program or a part of a program. All EXE and COM files execute as processes,

and one process can run one or more other processes.

See also session; thread.

Professional Server Expert Abbreviated PSE. A certification from IBM that offers specialization in a specific network operating system, including Novell NetWare, OS/2 Warp Server, or Windows NT Server. *See also* Professional Server Specialist.

Professional Server Specialist Abbreviated PSS. An introductory hardware certification from IBM designed to assess knowledge of IBM Netfinity and PC server architecture, installation, configuration, and management.

See also Professional Server Expert.

Profile object In Novell NetWare, a special Novell Directory Services (NDS) object used to assign the same login script to a group of users. A Profile login script is executed after the container login script has executed, but before the user login script.

programming language A language used to write a program that the computer can execute. Almost 200 programming languages exist. An example is the popular C language, which is well suited to a variety of computing tasks. With C, programmers can write anything from a device driver, to an application, to an operating system.

Certain kinds of tasks, particularly those involving artificial intelligence (LISP or Prolog), process control (Forth), or highly mathematical applications (Fortran and APL), can benefit from a more specific language.

Programming languages are also divided into low-level languages, such as assembly language, and high-level languages, such as C, C++, and Java.

See also assembly language; compiler; interpreter; machine language.

Project Athena A Massachusetts Institute of Technology project that ran from 1983 to 1991, sponsored by MIT, DEC, and IBM, and developed the X Window system and Kerberos authentication, as well as several other important relational database and network-related systems.

See also Kerberos; X Window.

promiscuous mode A mode in which a network device or interface card captures all the packets on the network, not just those addressed to it specifically.

Network analyzers work in promiscuous mode to monitor network traffic and to perform statistical analyses of the traffic.

See also network analyzer; sniffer.

propagation delay In communications, any delay between the time a signal enters the transmission channel and the time it is received.

This delay is relatively small across a LAN, but can become considerable in satellite communications, in which the signal must travel from one earth station to the satellite and back to earth again. Unusually long delays may require the use of specialized hardware to ensure that the link is not broken prematurely.

property 1. In Novell NetWare, a characteristic of an object in Novell Directory

Services (NDS); also known as an attribute. User object properties include name, login name, password restrictions, e-mail address, and other related information.

2. In Microsoft Windows, a characteristic of an object or device, accessed via that object's Properties dialog box.

property rights In Novell NetWare, characteristics of an object in Novell Directory Services (NDS). Property rights are Add or Delete Self, Compare, Read, Supervisor, and Write.

proprietary software Software developed in-house by a particular business or government agency and never made available commercially to the outside world.

The operating systems used in certain portable computers and PDAs (personal digital assistants) may also be considered proprietary, because they are specific to one system and are not generally available anywhere else.

protected mode In Intel processors, an operating state that supports advanced features. Protected mode in these processors provides hardware support for multitasking and virtual memory management, and it prevents programs from accessing blocks of memory that belong to other executing programs.

In 16-bit protected mode, supported on 80286 and higher processors, the CPU can address a total of 16MB of memory directly; in 32-bit protected mode, supported on 80386 and higher processors, the CPU can address up to 4GB of memory.

Microsoft Windows NT, OS/2, and most versions on Unix running on Intel processors run in protected mode.

See also real mode.

protocol In networking and communications, the formal specification that defines the procedures to follow when transmitting and receiving data. Protocols define the format, timing, sequence, and error checking used on the network.

See also communications protocol; OSI Reference Model; protocol stack.

protocol analyzer A hardware or combined hardware and software product used to analyze the performance data of the network and to find and troubleshoot network problems.

Protocol analyzers vary greatly in complexity. Some use dedicated hardware and can decode as many as 150 protocols; others convert an existing networked PC into a network-specific analyzer.

See also network analyzer; sniffer.

protocol converter A combined hardware and software product that converts from one network protocol to another; used when two dissimilar networks are connected.

See also gateway.

protocol stack The several layers of software that define the computer-to-computer or computer-to-network protocol.

Several companies have developed important proprietary protocol stacks, including

Novell NetWare's IPX/SPX, but the trend these days is moving toward more open systems such as TCP/IP (Transmission Control Protocol/Internet Protocol).

See also OSI Reference Model.

protocol suite *See* protocol stack.

protocol tunneling *See* tunneling.

proxy server A software package running on a server positioned between an internal network and the Internet.

The proxy server filters all outgoing connections so that they appear to be coming from the same machine, in an attempt to conceal the underlying internal network structure from any intruders. By disguising the real structure of the network, the proxy server makes it much more difficult for an intruder to mount a successful attack.

A proxy server will also forward your requests to the Internet, intercept the response, and then forward the response to you at your network node. A system administrator can also regulate the external sites to which users can connect.

See also dual-homed host; firewall.

PSE *See* Professional Server Expert.

PSS *See* Professional Server Specialist.

PSTN *See* Public Switched Telephone Network.

PTI *See* Payload Type Identifier.

PTT *See* Postal Telephone and Telegraph.

PU *See* physical unit.

public data network Abbreviated PDN. Any government-owned or government-controlled commercial packet-switched network, offering wide-area services to data-processing users.

public key encryption An encryption scheme that uses two keys. In an e-mail transaction, the public key encrypts the data, and a corresponding private key decrypts the data. Because the private key is never transmitted or publicized, the encryption scheme is extremely secure.

For digital signatures, the process is reversed; the sender uses the private key to create the digital signature, which can then be read by anyone who has access to the corresponding public key. This confirms that the message really is from the apparent sender.

See also digital signature; private key.

Public Key Infrastructure A proposal to provide a structure for verifying and authenticating users involved in transactions on the Internet or on corporate intranets and extranets.

The proposal involves a set of trusted certificate authorities (CAs) who would publish a person's public key and vouch for the authenticity of the data using a digital signature

or certificate. Other CAs would then vouch for those CAs, and they in turn would be vouched for by other CAs.

See also certificate authority; digital signature.

public network Normal voice telephone systems; also called the direct distance dial (DDD) network.

Public Switched Telephone Network

A designation used by the ITU to describe the local telephone company.

Public trustee In Novell NetWare, a special trustee, used only for trustee assignments.

The Public trustee allows objects in Novell Directory Services (NDS) that do not have any other rights to have the rights granted to the Public trustee. This is similar to the way the user GUEST or the group EVERYONE worked in earlier versions of Novell NetWare.

punch-down block A connecting device used for telephone lines; also known as a quick-connect block. The wires are pushed into metal teeth that strip the insulation away and make a good connection.

push *See* server push.

PVC *See* permanent virtual circuit.

Q

QIC See quarter-inch cartridge.

QoS See quality of service.

quadrature amplitude modulation In communications, a data-encoding technique used by modems. Quadrature amplitude modulation is a combination of phase and amplitude change that can encode multiple bits on a single carrier signal. For example, the CCITT V.42bis standard uses 4 phase changes and 2 amplitudes to create 16 different signal changes.

See also trellis-coded modulation.

quality of service Abbreviated QoS. The network requirements to support a specific application. Different types of networks and network traffic have a different QoS.

QoS includes the ability to guarantee the delivery of time-sensitive data, control the bandwidth, set priorities for specific network traffic, and provide an appropriate level of security.

QoS is often associated with the delivery of data such as live video, while at the same time maintaining sufficient bandwidth for the delivery of normal network traffic, perhaps at a lower data rate.

See also bandwidth on demand; Fast IP; IP Multicast; IP over ATM; IP switching; multicast.

Quantum Corporation A major manufacturer of disk storage systems. Quantum now makes a wide variety of disk types, including IDE, SCSI, solid-state, and Fibre Channel.

For more information on Quantum Corporation, see www.quantum.com.

quarter-inch cartridge Abbreviated QIC. A set of tape standards defined by the Quarter-Inch Cartridge Drive Standards Association, a trade association established in 1987. Several standards are in use today as Table Q.1 shows.

TABLE Q.1 QIC CAPACITIES

QIC Standard	Capacity	Tape Type
QIC-24	60MB	full-sized cartridge
QIC-40	40MB	mini cartridge
QIC-80	80MB	mini cartridge
QIC-100	40MB	mini cartridge

TABLE Q.1 QIC CAPACITIES (CONTINUED)

QIC Standard	Capacity	Tape Type
QIC-120	125MB	full-sized cartridge
QIC-128	128MB	mini cartridge
QIC-150	250MB	full-sized cartridge
QIC-380	380MB	mini cartridge
QIC-525	525MB	full-sized cartridge
QIC-1000	1GB	full-sized cartridge
QIC-1350	1.35GB	full-sized cartridge
QIC-3010	340MB	mini cartridge
QIC-3020	680MB	full-sized cartridge
QIC-4GB	4GB	full-sized cartridge
QIC-5GB	5GB	full-sized cartridge

query language In a database management system, a programming language that allows a user to extract and display specific information from a database.

Structured Query Language (SQL) is an international database query language that allows the user to issue high-level commands or statements, such as SELECT or INSERT, to create or modify data or the database structure.

See also Structured Query Language.

question mark A wildcard character used in many operating systems to represent a single character in a filename or file-name extension.

See also asterisk.

queue A temporary list of items waiting for a particular service, stored on disk in a special directory. For example, a print queue is a list of documents waiting to be printed on a network printer.

See also print queue.

quick-connect block See punch-down block.

QuickTime A cross-platform data format from Apple Computer, Inc., used to display movies. QuickTime synchronizes as many as 32 tracks containing time-based

digital data such as sound, video, MIDI, or other control information. A QuickTime movie created on one platform can be played back on another without modification; versions are available for Macintosh and Microsoft Windows.

R

RAD See Rapid Application Development.

radio frequency interference Abbreviated RFI. Many electronic devices, including radios, televisions, computers, and peripherals, can interfere with other signals in the radio-frequency range by producing electromagnetic radiation. The use of radio frequencies is generally regulated by government agencies.

See also Class A certification; Class B certification; extremely low-frequency emission; Federal Communications Commission.

RADIUS See Remote Authentication Dial In User Service.

RADSL See Rate-Adaptive Digital Subscriber Line.

RAID See redundant array of inexpensive disks.

RAM See random-access memory.

RAM chip A semiconductor storage device, either dynamic RAM or static RAM.

RAM disk An area of memory managed by a special device driver and used as a simulated disk; also called virtual drive. Because the RAM disk operates in memory, it works much faster than a regular hard disk. However, anything stored on a RAM disk

will be erased when the computer is turned off, so contents must be copied onto a real disk to be saved.

See also disk cache.

random access Describes the ability of a storage device to go directly to the required memory address without needing to read from the beginning every time data is requested.

In a random-access device, the information can be read directly by accessing the appropriate memory address. There is nothing random or haphazard about random access; a more precise term is *direct access*.

See also sequential access.

random-access memory Abbreviated RAM. The main system memory in a computer, used for the operating system, applications, and data.

See also dynamic RAM; static RAM.

Rapid Application Development Abbreviated RAD. A set of client/server application-development tools designed to speed up the development of robust applications for SQL databases.

See also Structured Query Language.

RARP See Reverse Address Resolution Protocol.

RAS See Remote Access Server.

Rate-Adaptive Digital Subscriber

Line Abbreviated RADSL. An Asymmetric Digital Subscriber Line (ADSL) service with a provision for testing the line length and quality before starting the service and adjusting the line speed accordingly.

See also Asymmetric Digital Subscriber Line; High-Bit-Rate Digital Subscriber Line; Single-Line Digital Subscriber Line; Very-High-Bit-Rate Digital Subscriber Line.

RBOC See Regional Bell Operating Companies.

RCDD See Registered Communications Distribution Designer.

RCONSOLE A Novell NetWare workstation utility that allows network administrators to manage routers and servers from a remote PC using a modem or from a workstation on the network. RCONSOLE establishes the connection to the server and converts the PC into a virtual server console. This function is also available in NetWare Administrator from the Tools menu.

In NetWare 3, use ACONSOLE to perform this function.

See also NetWare Administrator; REMOTE.

RDP See Remote Desktop Protocol.

read-after-write verification A method of checking that data is written to a hard disk correctly. Data is written to the disk and then read back and compared with the original data still held in memory. If the data read from the disk matches, the data in

memory is released. If the data does not match, that block on the disk is marked as bad, and another attempt is made to write the data elsewhere on the disk.

README file A plain text file that contains information about the software, placed on the distribution disks by the manufacturer.

The filename may vary slightly; it might be READ.ME, README.1ST, README.TXT, or README.DOC, for example. README files may contain last-minute, important information that is not in the program manuals or online help system.

You should always look for a README file when installing a new program on your system; it may contain information pertinent to your specific configuration. You can open a README file in any word processor or text editor because the file does not contain embedded formatting commands or program-specific characters.

read-only Describes a file or other collection of information that can only be read; it cannot be updated in any way or deleted.

Certain important operating system files are designated as read-only to prevent their accidental deletion. Also, certain types of ROM and some devices such as archive backup tapes and CD-ROMs can be read from but not changed.

read-only memory Abbreviated ROM. A semiconductor-based memory system that stores information permanently, retaining its contents when power is switched off. ROMs are used for firmware, such as the BIOS in the PC. In some portable computers,

applications and even the operating system are stored in ROM.

See also flash memory.

RealAudio Technology developed by RealNetworks (previously known as Progressive Networks, Inc.) that lets you play audio files as they are in the process of being downloaded, rather than waiting until the complete file has arrived, which gives a much faster response time. RealAudio uses UDP (User Datagram Protocol) as the delivery mechanism.

See also IP Multicast; multicasting; streaming; User Datagram Protocol.

real mode 1. An operating state supported by all processors in the Intel 80x86 family, and the only operating mode supported by MS-DOS. In real mode, the processor can directly address 1MB of memory. Unlike protected mode, real mode does not offer any advanced hardware features for memory management or multitasking.

2. In Microsoft Windows, an operating mode that runs Windows using less than 1MB of extended memory. Real mode is not used in Windows 3.1 or later.

See also protected mode.

RealNetworks, Inc. Developer of streaming audio and video technology for use over the Internet; previously known as Progressive Networks.

For more information on RealNetworks, Inc., see www.real.com.

See also RealAudio; streaming.

Realtime Streaming Protocol Abbreviated RTSP. A proposed protocol from Netscape and RealNetworks for streaming live and prerecorded audio and video across the Internet and corporate intranets. RTSP also includes features for bidirectional control, support for IP Multicast, and system security.

See also IP Multicast; RealAudio; Real-time Transport Protocol; Resource Reservation Protocol.

Real-time Transport Protocol Abbreviated RTP. A protocol designed for use in online videoconferencing applications involving multiple participants.

RTP can be used with any continuous data stream and so is suitable for use in interactive simulators and with control and measurement applications. RTP uses UDP (User Datagram Protocol) as the delivery mechanism.

See also Realtime Streaming Protocol; Resource Reservation Protocol; User Datagram Protocol.

reboot To restart the computer and reload the operating system, usually after a crash.

In some cases, you may be able to restart the computer from the keyboard; in more severe crashes, you may have to turn the computer off and then back on again.

See also boot.

Receive Data Abbreviated RXD. A hardware signal defined by the RS-232-C standard and used to carry serial data from one device to another.

See also Transmit Data.

record locking A method used to control access to individual records in a database.

In a multiuser environment, there is always the possibility that two users will attempt to update the same record at the same time but with different information. The initial attempt to solve this problem was to use file locking in which the first user to access the file locks out all other users and prevents them from opening the file. After the file is updated and closed again, the next user can gain access.

File locking can seriously degrade overall system performance as many users attempt to access the same files time after time. To avoid this slowdown, many database management systems use record locking, which limits access to individual records within the database files.

See also file and record locking.

Recovery Console In Microsoft Windows 2000, a command-line interface that gives access to a set of commands used to recover a system that will not restart.

See also Emergency Repair Disk; safe mode.

recursion In programming, the ability of a subroutine to call itself.

Recursion is often used when solving problems that repeat the same processing steps. However, some limiting factor must be present; otherwise, the program will never stop.

Red Hat Software A major distributor of Linux. Red Hat Linux is a three-CD package, containing a stable version of the Linux kernel, a large set of operating system

utilities, source code, Apache HTTP Server, sendmail, and Perl, as well as trial versions of commercial software.

SAP, IBM, Compaq, Oracle, Intel, and Novell have all made equity investments in Red Hat Software.

For more information on Red Hat Software, see www.redhat.com.

See also Apache HTTP Server; Linux; Perl.

Red Horde A nickname for Novell, Inc., the leading network operating system software company, as well as the NetWare resellers worldwide. Red is Novell's corporate color.

redirection 1. In Unix and many other operating systems, a shell mechanism that causes the standard input from a program to come from a file rather than from the terminal; it also causes standard output and standard error to go to a file rather than to the terminal. Because Unix is a file-based operating system, and terminals and other devices are treated as though they are files, a program doesn't care or even know if its output is going to a terminal or to a file.

2. A mechanism used by most of the popular Web server software packages that routes clients attempting to access a specific URL to a different URL, either on the same or on a different server. Redirection is a convenient way to avoid dead links.

See also link rot; pipe; Uniform Resource Locator.

redirector A software module loaded onto all the workstations on a network that intercepts application requests for file- and

printer-sharing services and diverts them to the file server for action.

See also NetWare shell; NETx.COM; requester.

reduced instruction set computing

Abbreviated RISC, pronounced “risk.” A processor that recognizes only a limited number of assembly-language instructions.

RISC chips are relatively cheap to produce and debug because they usually contain fewer than 128 instructions. RISC processors are commonly used in workstations, and they can be designed to run up to 70 percent faster than processors that use complex instruction set computing (CISC).

See also complex instruction set computing; IBM RS/6000; instruction set.

redundant array of inexpensive

disks Abbreviated RAID. In networking and mission-critical applications, a method of using several hard disk drives (often SCSI or Integrated Drive Electronics [IDE] drives) in an array to provide fault tolerance in the event that one or more than one drive fails. Each level of RAID is designed for a specific use:

- **RAID 0** Data is striped over one or more drives, but there is no redundant drive. RAID 0 provides no fault tolerance because the loss of a hard disk means a complete loss of data. Some classification schemes omit RAID 0 for this reason.
- **RAID 1** Two hard disks of equal capacity duplicate or mirror each other’s contents. One disk continuously and automatically backs up the other disk. This method is also known as disk mirroring or

disk duplexing, depending on whether one or two independent hard-disk controllers are used.

- **RAID 2** Bit-interleaved data is written across several drives, and then parity and error-correction information is written to additional separate drives. The specific number of error-correction drives depends on the allocation algorithm in use.
- **RAID 3** Bit-interleaved data is written across several drives, but only one parity drive is used. If an error is detected, the data is reread to resolve the problem. The fact that data is reread in the event of an error may add a small performance penalty.
- **RAID 4** Data is written across drives by sectors rather than at the bit level, and a separate drive is used as a parity drive for error detection. Reads and writes occur independently.
- **RAID 5** Data is written across drives in sectors, and parity information is added as another sector, just as if it were ordinary data. This level of RAID can provide faster performance as the parity information is written across all the drives, rather than to a single parity drive.

There is not much difference in speed or quality among these levels. The appropriate level of RAID for any particular installation depends on network usage. RAID levels 1, 3, and 5 are available commercially, and levels 3 and 5 are proving popular for networks.

Several vendors have created their own RAID levels, including 6, 7, 10, 11, and 35. Some of these are actually combinations of existing RAID levels, such as Compaq’s

Level 0+1, which combines RAID levels 0 and 1.

See also backup; disk striping; disk striping with parity; parity; single large expensive disk.

reentrant Describes a programming technique that allows one copy of a program to be loaded into memory and shared. When one program is executing reentrant code, a different program can interrupt and then start or continue execution of that same code.

Many operating system service routines use reentrant code so that only one copy of the code is needed. The technique is also used in multithreaded applications, in which different events are taking place concurrently in the computer.

reference time server In Novell NetWare, a server that provides externally derived time data to secondary time servers and to workstations. Reference time servers take part in time synchronization, but they do not change their time except in response to the external time source.

See also primary time server; secondary time server; single reference time server.

regedit In Microsoft Windows, an application that knowledgeable users can access to edit the contents of the Registry database. In Windows NT, this application is called regedit32.

See also Registry.

Regional Bell Operating Companies

Abbreviated RBOC. The telephone companies formed as a result of the breakup of AT&T, finalized in 1984.

Each of the RBOCs was assigned a specific geographical area, and each of these areas was divided into service areas known as local access and transport areas (LATA).

See also Local Exchange Carrier.

Registered Communications Distribution Designer

Abbreviated RCDD. A certification from BICSI (a telecommunications association) for those involved in the design and installation of low-voltage wiring infrastructures in new construction or in existing buildings. An additional specialization, RCDD: Local Area Network, is available for those involved with designing and installing LAN cabling.

Registry In the Microsoft Windows family of operating systems, a system database containing configuration information.

The operating system continually references the Registry database for information on users and groups, the applications installed on the system and the type of document each can create, what hardware is available and which ports are in use, and property sheets for folders and application icons.

Changes to the Registry are usually made automatically as configuration information is changed using Control Panel applications or the Administrative Tools; however, knowledgeable users can make changes directly using the application regedit or regedit32.

The Registry database replaces the text-based .INI configuration files used in earlier versions of Windows and replaces the MS-DOS configuration files AUTOEXEC.BAT and CONFIG.SYS.

See also regedit.

Registry Editor *See* regedit.

regular expression In Unix, a sequence of characters that can match a set of fixed-text strings used in searching for and replacing text.

Many Unix programs, including vi, ed, emacs, grep, and awk, use regular expressions.

relational database A database model in which the data always appears from the point of view of the user to be a set of two-dimensional tables, with the data presented in rows and columns.

The rows in a table represent records, which are collections of information about a specific topic, such as the entries in a doctor's patient list. The columns represent fields, which are the items that make up a record, such as the name, address, city, state, and zip code in an address list database.

See also database model.

Relative Distinguished Name In Novell Directory Services (NDS), a shortened Distinguished Name that identifies an object by its relationship within the current context.

Relative Distinguished Names do not begin with a period, but you can use a period at the end of the name to move up the NDS tree.

See also Common name; Context; Distinguished Name; NDS tree.

REMOTE A Novell NetWare server utility used to access the server console from a PC or a workstation or by using a modem.

See also RCONSOLE.

remote access A workstation-to-network connection, made using a modem and a telephone line, that allows data to be sent and received over large distances. Remote access and authentication and security for such access is managed differently in different network operating systems.

See also private data network; public data network; Virtual Private Network.

Remote Access Server Abbreviated RAS. In Microsoft Windows NT Server, a software package that allows remote users to connect to the server via modem and access network resources. Users can connect to the server using a telephone line and an analog modem, an ISDN connection, or an X.25 network.

Remote Authentication Dial In User Service Abbreviated RADIUS. A third-party authentication server attached to a network.

Remote users dial in to the server, and the access server requests authentication services from the RADIUS server. The RADIUS server authenticates users and gives them access to network resources. The access server is acting as a client to the RADIUS server.

See also access server; authentication; challenge-response authentication.

remote boot A technique used to boot a workstation from an image file on the file server rather than from a local drive attached directly to the workstation.

remote connection *See* remote access.

remote-control program A program that allows the user to link two PCs together so that one of the computers controls the operation of the other. The connection may be over a dedicated serial line, a local-area network, or a modem-to-modem communications link. Each computer runs a copy of the remote-control program.

Remote-control programs are particularly useful for troubleshooting problems at computers located far from the technical support center, for installing or removing demonstration software without needing to visit the customer site, for training remote users, and for telecommuting.

Popular remote control programs include Symantec's pcAnywhere, Microcom's Carbon Copy, and Traveling Software's LapLink for Windows.

See also mobile computing; wireless communications.

Remote Desktop Protocol Abbreviated RDP. An extension to the International Telecommunications Union (ITU) T.120 protocol that provides the connection between a thin client and the terminal server. RDP works with Microsoft Windows workstations using TCP/IP (Transmission Control Protocol/Internet Protocol).

See also Independent Computing Architecture; thin client.

remote digital loopback test A capability of certain modems that allows the whole circuit to be tested.

See also loopback.

Remote File System Abbreviated RFS. A distributed file system network protocol that allows programs running on a computer to use network resources as though they were local. Originally developed by AT&T, RFS has been incorporated as a part of Unix System V Interface Definition.

Remote Installation Services Abbreviated RIS. In Microsoft Windows 2000 Server, a service that allows the installation of a single workstation image of Windows 2000 Professional to any number of computers while at the same time ensuring that each system has unique security identifiers.

Remote Procedure Call Abbreviated RPC. A set of procedures used to implement client/server architecture in distributed programming.

RPC describes how an application initiates a process on another network node and how it retrieves the appropriate result.

RPCs were first implemented by Sun Microsystems and Hewlett-Packard systems running the Unix operating system.

See also Common Object Request Broker Architecture; Distributed Component Object Model; Object Request Broker.

remote resource Any device not attached to the local node, but available through the network.

Remote Storage In Microsoft Windows 2000 Server, a service that moves in-

frequently used files from local hard disk storage to tape when disk resources drop below a certain level. If a user requests the file after it has been transferred to tape, the file is read back, and a dialog box explains that the file is being retrieved from remote storage.

remote user A user who logs in to the network using a modem and telephone line from a site located some distance from the main network.

A remote user may always dial in from the same location, as in the case of a telecommuter working from home, or may dial in from a different location every time, as in the case of mobile sales people.

See also mobile computing; remote access; wireless communications.

repeater A simple hardware device that moves all packets from one local-area network segment to another by regenerating, retiming, and amplifying the electrical signals.

The main purpose of a repeater is to extend the length of the network transmission medium beyond the normal maximum cable lengths.

See also active hub; bridge; brouter; router.

replica In Novell Directory Services (NDS), a copy of a directory partition.

Replicas are designed to eliminate a single point of failure and to provide faster access to users across a wide-area network.

NetWare contains three major types of replicas:

- **Master** The original replica, created during system installation. A Master replica of the Root partition is stored in a hidden directory on the first file server installed.
- **Read-Write** Used to read or update Novell Directory Database (NDD) information. There will be at least two Read-Write replicas for each partition to ensure that the NDD will continue to function even if some of the servers with replicas are unavailable.
- **Read-Only** Used to display but not modify NDD information.

A fourth replica type, Subordinate Reference, is important to NDS communications. This is a link between a parent partition and a child partition, containing a list of the servers in which replicas of the child partition are stored, their addresses, and replica types, as well as other NDS partition information. Subordinate Reference replicas are maintained by the operating system and cannot be changed by users or the system administrator.

See also Novell Directory Services; NetWare Directory Database; partition; replica synchronization.

replica synchronization In Novell Directory Services (NDS), the process used to exchange information between a partition's replicas to ensure that all information is up-to-date.

When a change is made to a replica, synchronization ensures that the change is

made available to all other replicas as soon as possible.

See also Novell Directory Services; partition; replica.

replication The process of synchronizing data stored on two or more computers either for backup purposes or to make the information more accessible to users at different locations.

In Microsoft Windows NT Server, replication is used to duplicate data to another server; the server holding the master copy of the data is called the export server, and the server receiving the data is known as the import server.

In Novell Directory Services (NDS), replication is used to distribute all or part of the Novell Directory Database to other servers.

requester Special software loaded onto a networked workstation to manage communications between the network and the workstation. This software may also be referred to as a shell, redirector, or client, depending on the networking system in use.

See also NetWare shell; NetWare DOS Requester; NETx.COM.

Request for Comment Abbreviated RFC. A document or a set of documents in which proposed Internet standards are described or defined.

Well over a thousand RFCs are in existence, and they represent a major method of online publication for Internet technical standards.

See also Internet Engineering Task Force.

Request to Send Abbreviated RTS. A hardware signal defined by the RS-232-C standard to request permission to transmit.

See also Clear to Send.

reserved memory In MS-DOS, a term used to describe that area of memory between 640Kb and 1MB, also known as upper memory. Reserved memory is used by MS-DOS to store system and video information.

See also memory management.

reserved word *See* keyword.

resource 1. Any part of a computer system that can be used by a program as it runs. Resources include memory, hard and floppy disks, networking components, the operating system, printers, and other output devices, as well as queues, security features, and other less well defined data structures.

2. In HTML, any URL, directory, or application that the server can access and send to a requesting client.

See also HyperText Markup Language; Uniform Resource Locator.

Resource Reservation Protocol Abbreviated RSVP. An Internet protocol designed to deliver data on time and in the right order over TCP/IP (Transmission Control Protocol/Internet Protocol) networks.

RSVP is a control and signaling protocol, not a routing protocol, and it works by reserving bandwidth from one end system to another; this reduces the bandwidth available to other users.

See also IP switching; Real-time Transport Protocol.

response time The time lag between sending a request and receiving the data.

Response time can be applied to a complete computer system, as in the time taken to look up a certain customer record, or to a system component, as in the time taken to access a specific cluster on disk or for a memory circuit to return data requested by the processor.

restore To copy files from a backup or archival storage to their normal location, especially when the files are being copied to replace files lost by accident.

See also backup.

retensioning A maintenance operation required by certain tape drives to ensure correct tape tension; retensioning fast forwards and then rewinds the entire tape or tape cartridge.

Return key *See* Enter key.

Reverse Address Resolution Protocol

Abbreviated RARP. A part of the TCP/IP (Transmission Control Protocol/Internet Protocol) protocol suite that allows a computer, more specifically a diskless workstation, to obtain an IP address from a server when only the hardware address is known.

See also Address Resolution Protocol; hardware address; IP address.

REXX Acronym formed from Restructured Extended Executor Language. A scripting language from IBM, originally written by Mike Cowlishaw of IBM UK in 1979 for the VM mainframe environment and now available on many operating systems, including OS/2, Unix, and VMS.

REXX is a general-purpose interpreted language and uses English-like words, rather than the sometimes terse syntax of C or C++.

RFC *See* Request for Comment.

RFI *See* radio frequency interference.

RFS *See* Remote File System.

RG-58 A 50-ohm coaxial cable, used in Ethernet networks, that conforms to the IEEE 802.3 10Base2 standard.

RG-59 A 75-ohm coaxial cable used in ARCnet and in television.

RG-62 A 93-ohm coaxial cable used in ARCnet LANs or in IBM 3270 applications.

right angle bracket The > symbol that is used in Unix and other operating systems to direct the output from a command to a file or to a device.

Also commonly used in e-mail messages to indicate the text has been cut from another e-mail message.

rights In a network or multiuser environment, the ability of a user to access certain system resources, including files and directories. Permissions are based on the rights given to user accounts by the system administrator.

See also inherited rights.

rightsizing The process of matching a corporation's goals to the computing and network solutions available to maximize business effectiveness in reaching that goal.

See also downsizing; outsourcing; service bureau.

ring network A network topology in the form of a closed loop or circle, with each node in the network connected to the next.

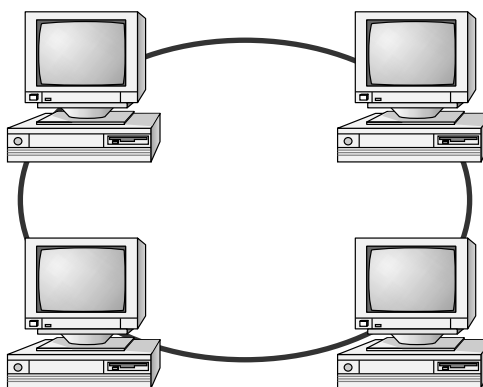
Messages move in one direction around the system. When a message arrives at a node, the node examines the address information in the message. If the address matches the node's address, the message is accepted; otherwise, the node regenerates the signal and places the message back on the network for the next node in the system. It is this re-

generation that allows a ring network to cover greater distances than star networks or bus networks. Ring networks normally use some form of token-passing protocol to regulate network traffic.

The failure of a single node can disrupt network operations; however, fault tolerant techniques have been developed to allow the network to continue to function in the event one or more nodes fail.

See also token-ring network.

RING NETWORK



RIP *See* Routing Information Protocol.

RIS *See* Remote Installation Services.

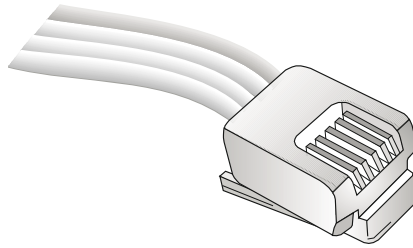
RISC *See* reduced instruction set computing.

riser cable Any cable that runs vertically between floors in a building. Riser cable may be run through special conduits or inside the elevator shaft.

See also plenum cable.

RJ-11 A commonly used modular telephone connector. RJ-11 is a four-wire (two-pair) connector most often used for voice communications.

RJ-12 A commonly used modular telephone connector. RJ-12 is a six-wire (three-pair) connector most often used for voice communications.

RJ-11

RJ-45 A commonly used modular telephone connector. RJ-45 is an eight-wire (four-pair) connector used for data transmission over unshielded twisted-pair (UTP) cable and leased telephone line connections.

RLL encoding An efficient method of storing information on a hard disk. Compared with older, less-efficient methods, such as modified frequency modulation encoding (MFM), RLL encoding effectively doubles the storage capacity of a disk.

See also advanced run-length limited encoding.

rlogin A Unix utility that establishes a terminal to remote host connection on a TCP/IP (Transmission Control Protocol/Internet Protocol) network. Once the connection is established, any commands you enter will run on the remote system.

roaming user profile In Microsoft Windows NT Server, a user profile stored on the server and downloaded to a workstation that follows a user from one computer to another, allowing users to access their profile from any location on the network.

See also local user profile; mandatory user profile.

robot Sometimes abbreviated bot. A World Wide Web application that automatically locates and collects information about new Web sites. Robots are most often used to create large databases of Web sites.

See also spider.

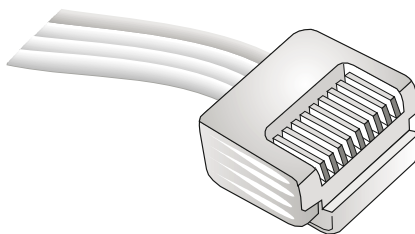
Rockwell Semiconductor Systems A major manufacturer of chip sets for modems, cellular and cordless telephones, fax machines, PC-based videoconferencing systems, and high-speed communications systems.

For more information on Rockwell Semiconductor Systems, see www.rockwell.com.

roll back The ability of a database management system to abort a transaction against the database before the transaction is complete and return to a previous stable condition.

See also roll forward; transaction processing.

RJ-45



roll forward The ability of a database management system to re-create the data in the database by rerunning all the transactions listed in the transaction log.

See also roll back; transaction processing.

ROM *See* read-only memory.

ROM BIOS *See* BIOS.

root 1. In many operating systems, the name of the directory at the top of the directory tree from which all other directories are descended.

2. In Unix, the name of the superuser, user number 0. The system administrator uses this account for certain administrative tasks. One of the main objectives of an intruder on a Unix system is to gain root user status. Once achieved, the intruder has unlimited access to the system.

See also avatar; root directory.

root domain The top-level Domain Name Service (DNS) domain on the Internet. Also, in Microsoft Windows 2000 Server, the top-level domain in a Windows 2000 domain tree.

See also child domain; first-layer domain; parent domain

root directory In a hierarchical directory structure, such as that used in Unix and many other operating systems, the directory from which all other directories must branch. You cannot delete the root directory.

See also parent directory; period and double-period directories.

Root object In Novell Directory Services (NDS), the original container object, created when NDS is first installed.

The Root object contains all other NDS objects and cannot be deleted, renamed, or moved. Sometimes written [Root].

ROT-13 Pronounced “rote-13.” A simple encoding scheme often used to scramble posts to Usenet newsgroups.

ROT-13 works by swapping each alphabetic character with another 13 characters removed from its location in the alphabet, so that a becomes n, and so on; numbers and punctuation are unaffected.

ROT-13 makes the article unreadable until the text is decoded and is often used when the subject matter might be considered offensive. Many newsreaders have a built-in command to unscramble ROT-13 text, and if you use it, don't be surprised by what you read. If you think you might be offended, don't decrypt the post.

You will also find other, inoffensive material encoded by ROT-13, including spoilers that give away the ending of a book or film and answers to puzzles or riddles.

router An intelligent connecting device that can send packets to the correct LAN segment to take them to their destination.

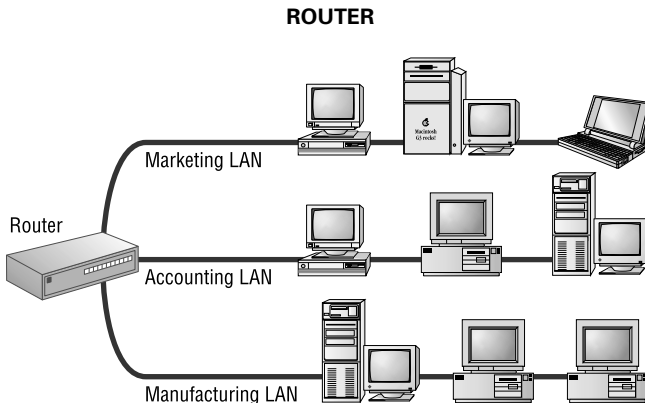
Routers link LAN segments at the network layer of the OSI Reference Model for computer-to-computer communications. The

networks connected by routers can use similar or different networking protocols.

A router may be one or more of the following types:

- **Central** Acts as a network backbone, connecting many LANs.
- **Peripheral** Connects individual LANs to either a central router or to another peripheral router.
- **Local** Operates within its LAN driver's cable-length limitations.
- **Remote** Connects beyond its device driver limitations, perhaps through a modem or remote connection.
- **Internal** Part of a network file server.
- **External** Located in a workstation on the network.

See also bridge; brouter; gateway.



routing The process of directing packets from a network source node to the destination node.

Routing and Remote Access Service

Abbreviated RRAS. In Microsoft Windows 2000 Server, the service used by remote users to access the server. In addition, RRAS provides routing functions. RRAS replaces the Windows NT Remote Access Service (RAS).

Routing Information Protocol Abbreviated RIP. A routing protocol used on TCP/IP (Transmission Control Protocol/Internet Protocol) networks that maintains a list of reachable networks and calculates the degree of difficulty involved in reaching a specific network from a particular location by determining the lowest hop count.

The Internet standard routing protocol Open Shortest Path First (OSPF) is the successor to RIP.

See also Open Shortest Path First.

routing protocol The protocol that enables routing by the use of a specific routing algorithm that determines the most appropriate path between the source and destination nodes.

Routing protocols provide dynamic routing configuration; without routing protocols, system administrators would have to manually configure routing tables.

See also Open Shortest Path First; Routing Information Protocol; routing table.

routing table A table stored in a router; used to keep track of routes to specific network destinations.

See also routing protocol.

RPC *See* Remote Procedure Call.

RPL A Novell NetWare server utility that allows users to boot diskless workstations from files on the server.

See also remote boot.

RRAS *See* Routing and Remote Access Service.

RS-232-C A recommended standard (RS) interface established by the Electronic Industries Association (EIA). Also known as EIA/TIA-232.

The standard defines the specific electrical, functional, and mechanical characteristics used in asynchronous transmissions between a computer (data terminal equipment, or DTE) and a peripheral device (data communications equipment, or DCE). RS is the abbreviation for recommended standard, and the C denotes the third revision of that standard. RS-232-C is compatible with the CCITT V.24 and V.28 standards, as well as ISO IS2110.

RS-232-C uses a 25-pin or 9-pin DB connector. The accompanying illustration shows the pinouts used in a DB-25 male connector. It is used for serial communications between a computer and a peripheral device, such as a printer, modem, or mouse. The maximum cable limit of 15.25 meters (50 feet) can be extended by using high-quality cable, line drivers to boost the signal, or short-haul modems.

See also 1394; High Speed Serial Interface; Universal Serial Bus.

RS-232-C INTERFACE

Signal	Pin number	Pin number	Signal
secondary transmitted data	14	1	protective ground
DCE transmitter signal element timing	15	2	transmitted data
secondary received data	16	3	received data
receiver signal element timing	17	4	request to send
no defined signal designation	18	5	clear to send
secondary request to send	19	6	data set ready
data terminal ready	20	7	signal ground/common return
signal quality detector	21	8	received line signal detector
ring indicator	22	9	+ voltage
data signal rate selector	23	10	- voltage
DTE transmitter signal element timing	24	11	no defined signal designation
no defined signal designation	25	12	secondary received line signal detector
		13	secondary clear to send

RS-422 A recommended standard (RS) interface established by the Electronic Industries Association (EIA). Also known as EIA/TIA-422.

The standard defines the electrical and functional characteristics used in a balanced serial interface, but does not specify a connector. Manufacturers who use this standard use many different types of connectors with nonstandard pin configurations. Serial ports on some Macintosh computers are RS-422 ports.

RS-423 A recommended standard (RS) interface established by the Electronic Industries Association (EIA). Also known as EIA/TIA-423.

The standard defines the electrical and functional characteristics used in an unbalanced

serial interface, but does not specify a connector. Manufacturers who use this standard use many different types of connectors with nonstandard pin configurations.

RS-449 A recommended standard (RS) interface established by the Electronic Industries Association (EIA). Also known as EIA/TIA-449.

The standard defines the specific electrical, functional, and mechanical characteristics used in serial binary data interchange, and it is often used with synchronous transmissions.

RS-449 may be implemented using a 37-pin or 9-pin DB connector; the accompanying illustration shows a DB-37 male connector.

RS-449 INTERFACE

Signal	Pin number	Pin number	Signal
receive common	20	1	shield
no defined signal designation	21	2	signalling rate indicator
send data	22	3	no defined signal designation
send timing	23	4	send data
receive data	24	5	send timing
request to send	25	6	receive data
receive timing	26	7	request to send
clear to send	27	8	receive timing
terminal in service	28	9	clear to send
data mode	29	10	local loopback
terminal ready	30	11	data mode
receiver ready	31	12	terminal ready
select standby	32	13	receiver ready
signal quality	33	14	remote loopback
new signal	34	15	incoming call
terminal timing	35	16	select frequency
standby/indicator	36	17	terminal timing
send common	37	18	test mode
		19	signal ground

RS-485 A recommended standard (RS) interface established by the Electronic Industries Association (EIA). Also known as EIA/TIA-485.

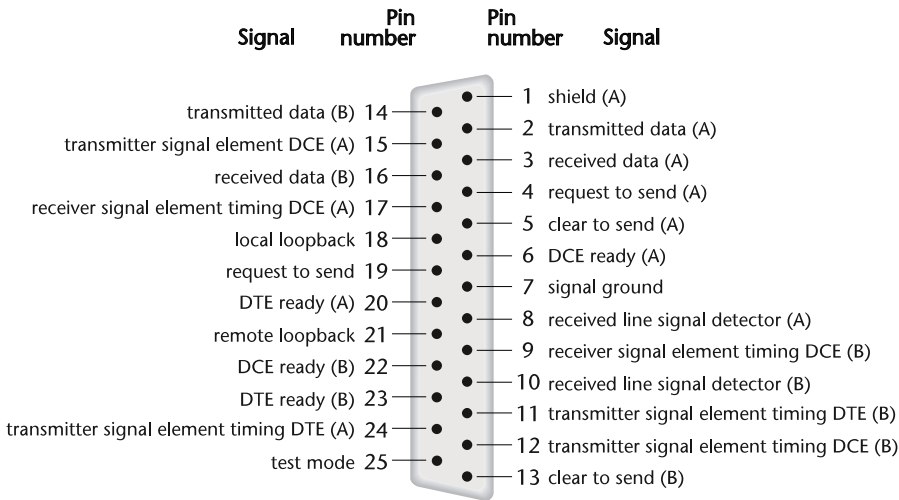
RS-485 is similar to RS-422, except that the associated drivers are tri-state rather than dual-state. RS-485 can be used in multi-point applications, in which one computer controls as many as 64 devices.

RS-530 A recommended standard (RS) interface established by the Electronic Industries Association (EIA). Also known as EIA/TIA-530.

The standard defines the specific electrical, functional, and mechanical characteristics used in transmitting serial binary data, either synchronously or asynchronously, using a 25-pin DB connector.

RS-530 works in conjunction with RS-422 (balanced electrical circuits) or RS-423 (unbalanced electrical circuits) and allows data rates from 20Kbps to 2Mbps. The maximum distance depends on the electrical interface in use. RS-530 is compatible with CCITT V.10, V.11, X26; MIL-188/114, and RS-449.

RS-530 INTERFACE



RS/6000 See IBM RS/6000.

RSA A public key, or asymmetric, encryption scheme invented by and named for three mathematicians—Ron Rivest, Adi Shamir, and Len Adleman.

The theoretical background to RSA is that it is very difficult to find the factors of a very large number that is the product of two prime numbers. RSA has been analyzed closely and is considered very secure provided a sufficiently long key is used.

RSA Data Security A leading publisher of encryption software, founded by mathematicians Ron Rivest, Adi Shamir, and Len Adleman.

The company holds patents on several important encryption schemes and provides encryption and security consulting services.

For more information on RSA Data Security, see www.rsa.com.

RSVP See Resource Reservation Protocol.

RTP See Real-time Transport Protocol.

RTS See Request to Send.

RTSP See Realtime Streaming Protocol.

run-time version A special, limited-capability release of software bundled with a single product that allows that product to run, but does not support any of the other applications capable of running in that same environment. The run-time version provides some but not all the features of the full product.

RXD See Receive Data.

S

S3 Inc. A leader in the field of multimedia and graphics accelerator chips, S3 Inc. sells to companies such as Compaq, Dell, and Diamond Multimedia, who use the chips in their systems and on add-in cards.

For more information on S3 Inc., see www.s3.com.

SAA See Systems Application Architecture.

safe mode In Microsoft Windows, a special way of starting the operating system using only the most essential components, to allow for problem diagnosis and system troubleshooting.

See also Emergency Repair Disk; Recovery Console.

sag A short-term drop in line voltage to between 70 and 90 percent of the nominal voltage.

See also power conditioning; spike; surge.

salvageable files In Novell NetWare, files that have been deleted by users, but that are recoverable.

The FILER utility can display a list of deleted files and can recover the files providing they have not been overwritten in whole or in part.

If the directory containing the deleted file is also deleted, the file is saved in a system directory called DELETED.SAV in the volume's root directory.

SAM See Security Accounts Manager.

SAM database In Microsoft Windows NT, the Security Accounts Manager database that holds all the user accounts, groups, policies, and other information relating to the domain.

The SAM database is maintained by the User Manager for Domains utility and is accessed internally by the Security Accounts Manager.

See also domain; Security Accounts Manager.

same-server migration In Novell NetWare, one of the methods used to upgrade from NetWare 2 to NetWare 4.

The process involves upgrading the file system to the NetWare 3 format first and then upgrading to NetWare 4. The server hardware must be capable of running NetWare 4.

See also across-the-wire migration.

Samsung/AST Research The more than 30 companies in the Samsung Group manufacture everything from consumer electronics to notebook PCs and monitors to ATM switches, and Samsung Electronics is one of the largest producers of DRAM and other semiconductors in the world.

For more information on Samsung, see www.samsung.com, and for more information on AST Research, see www.ast.com.

SAN See storage area network.

sandbox In the Java programming system, an area within which the Java applet may execute. The sandbox is created by the Java Virtual Machine.

The applet can do anything it likes within the sandbox, but it cannot read or alter any data outside the sandbox. This security measure allows users (at least in theory) to run untrusted code without compromising the security of their own environment.

See also Java; Java applet; Java Virtual Machine.

Santa Cruz Operation, Inc. Abbreviated SCO. Developers of several important strains of Unix, including XENIX, SCO UNIX, and the SCO OpenServer series of products.

SCO began shipping Unix products based on Intel processors in 1982, and now SCO accounts for approximately 40 percent of the Unix market, with more than 8,000 applications available from various vendors.

In 1995, SCO bought the rights to Unix from Novell, and recently the company combined UnixWare and OpenServer into a single integrated operating system.

For more information on SCO, see www.sco.com.

See also OpenServer; UnixWare.

Santa Cruz Operation Advanced Certified Engineer Abbreviated SCO ACE. A certification from Santa Cruz Operation based on its SCO Unix products. Three specializations are available: Server Track, Open Server Track, and UnixWare Track.

SAP *See* Service Advertising Protocol.

SAP AG One of the world's largest independent software suppliers, based in Wall-dorf, Germany.

SAP's flagship product goes under the unassuming name of R/3, but the product serves a large part of the corporate client/server world and encompasses accounting and controlling, production and materials management, human resources management, plant maintenance, workflow software, quality management, sales and distribution, and project management.

For more information on SAP, see www.sap.com.

SAR *See* segmentation and reassembly sublayer.

SAS *See* single-attached station.

SATAN *See* Security Administrator Tool for Analyzing Networks.

Scalable Performance Architecture

Abbreviated SPARC. A 32-bit reduced instruction set computing (RISC) processor from Sun Microsystems.

See also SPARCstation; Sun Microsystems, Inc.

scalability The ability of an operating system to add system resources to provide faster processing or to handle increased loads in anticipation of future needs.

In practice, this usually means that an operating system is available on a range of increasingly capable hardware, with only modest increases in price at each level.

Schedule service In Microsoft Windows NT, a system service that performs an

automated task at a specific time using the AT command.

Schema In Microsoft Active Directory, the definition of the objects and their properties that can be stored in the Active Directory database.

The Active Directory Schema is flexible and allows programmers to add new objects and to add new properties to existing objects.

See also Active Directory; Global Catalog; Multimaster Replication; Object

SCO *See* Santa Cruz Operation, Inc.

SCO ACE *See* Santa Cruz Operation Advanced Certified Engineer.

SCOadmin A set of graphical system administrator tools provided with Santa Cruz Operation (SCO) OpenServer.

SCOadmin lets you add or remove users, manage printers and filesystems, and check your network configuration quickly and easily.

SCO OpenServer A set of Unix products from Santa Cruz Operation (SCO). OpenServer includes a journaling filesystem, integrated symmetrical multiprocessing, a set of graphical administration tools, and a Web server. You can also run MS-DOS and Windows applications under software emulation.

See also Santa Cruz Operation, Inc.; SCO Unix.

SCO Unix A popular version of Unix from Santa Cruz Operation (SCO), based on System V, Release 3.2, with many System V, Release 4 enhancements.

SCO Unix includes the Korn shell, X Window, Level C2 security, multiprocessor support, and the ability to run MS-DOS and Windows applications under software emulation.

See also Santa Cruz Operation, Inc.

SCR *See* Sustainable Cell Rate.

script A small program or macro invoked at a particular time.

For example, a login script may execute the same specific set of instructions every time a user logs in to a network. A communications script may send user-identification information to an Internet Service Provider (ISP) each time a subscriber dials up the service.

scripting The process of invoking a script, written in a scripting language, from an HTML document on a Web site.

Scripts can be written in a range of languages, including Perl, Tcl, REXX, JavaScript, JScript, or even Microsoft Visual Basic.

See also Common Gateway Interface; JavaScript; Perl; REXX; Tcl.

SCSI *See* Small Computer System Interface.

SCSI-1 A commonly used name for the first Small Computer System Interface (SCSI) definition, published in 1986; has an 8-bit parallel interface and a maximum data transfer rate of 5MBps.

See also Fast SCSI; Fast/Wide SCSI; SCSI-2; SCSI-3; Small Computer System Interface; Ultra SCSI; Ultra2 SCSI; Wide Ultra SCSI; Wide Ultra2 SCSI.

SCSI-2 A 1994 extension to the Small Computer System Interface (SCSI) definition.

This standard broadened the 8-bit data bus to 16 or 32 bits (also known as Wide SCSI), doubling the data transfer rate to 10 or 20Mbps (also known as Fast SCSI). Wide SCSI and Fast SCSI can be combined to give Fast/Wide SCSI, with a 16-bit data bus and a maximum data-transfer rate of 20Mbps. SCSI-2 is backward compatible with SCSI-1, but for maximum benefit, you should use SCSI-2 devices with a SCSI-2 controller.

SCSI-2 also adds new commands, and although the connector is physically smaller, it uses 68 pins rather than the 50 in SCSI-1. Higher data-transfer rates are achieved by using synchronous rather than asynchronous transfers.

See also Fast SCSI; Fast/Wide SCSI; SCSI; SCSI-1; SCSI-3; Small Computer System Interface; Ultra SCSI; Ultra2 SCSI; Wide Ultra SCSI; Wide Ultra2 SCSI.

SCSI-3 An extension to the Small Computer System Interface (SCSI) standard.

This definition increased the number of connected peripherals from 7 to 16, increased cable lengths, added support for a variety of interfaces including a serial interface, a Fibre Channel interface, a 1394 interface, and support for Serial Storage Architecture and several packet interfaces. Data-transfer rates depend on the hardware implementation, but data rates in excess of 160Mbps are possible.

See also 1394; Fast SCSI; Fast/Wide SCSI; Fibre Channel; Small Computer System Interface; SCSI-1; SCSI-2; Serial Storage

Architecture; Ultra SCSI; Ultra2 SCSI; Wide Ultra SCSI; Wide Ultra2 SCSI.

SCSI bus Another name for the Small Computer System Interface (SCSI) interface and communications protocol.

SCSI terminator The Small Computer System Interface (SCSI) interface must be correctly terminated to prevent signals echoing on the bus.

Many SCSI devices have built-in terminators that engage when they are needed. With some older SCSI devices, you must add an external SCSI terminator that plugs into the device's SCSI connector.

See also active termination; forced perfect termination; passive termination.

scuzzy *See* Small Computer System Interface.

SDH *See* Synchronous Digital Hierarchy.

SDK *See* software development kit.

SDLC *See* Synchronous Data Link Control.

SDRAM *See* synchronous DRAM.

SDSL *See* Single-Line Digital Subscriber Line.

Seagate Desktop Management Suite

A package of network management utilities from Seagate Software, Inc., that includes hardware and software inventory, server monitoring, client monitoring and control, network traffic monitoring, virus protection, remote access, remote control, and print-queue management.

The Desktop Management Suite also uses Seagate's Crystal Reports and includes more than 80 standard reports as well as facilities to create custom reports.

See also Desktop Management Interface.

Seagate Technology, Inc. A major manufacturer of storage technology solutions, including tape drives, hard disks, as well as network information management software and the reporting package, Crystal Reports.

For more information on Seagate Technology, Inc., see www.seagate.com.

search drive A drive that the operating system searches when the requested file is not located in the current directory.

See also current drive.

search engine A special Web site that lets you perform keyword searches to locate Web pages; see Appendix A for a list of popular search engines.

To use a search engine, you enter one or more keywords or, in some cases, a more complex search string such as a Boolean expression. The search engine returns a list of matching Web pages, newsgroups, and FTP archives taken from its database, usually ranked in some way, that contain the expression you are looking for, along with a brief text description of the material.

Searching this database is much faster than actually searching the Internet, but the accuracy and relevance of the information it contains depend on how often the data is updated and on the proportion of the Web that is actually searched for new content.

See also portal.

secondary cache Cache memory located on the motherboard rather than on the microprocessor; also known as L2 cache.

Secondary cache can significantly improve system performance.

second source In computer hardware, an alternative supplier of an identical product. Second sources are a safety net for the buyer, because there are at least two suppliers for one product.

secondary time server In Novell NetWare, a server that does not determine the network time, but receives that information from a primary or single reference time server.

See also primary time server; reference time server; single reference time server.

Secure HTTP Abbreviated S-HTTP, S/HTTP, or HTTP-S. An extension to the Hypertext Transfer Protocol (HTTP), from Enterprise Integration Technology, that allows Web browsers and servers to sign, authenticate, and encrypt an HTTP packet at the application layer.

S-HTTP is not widely used and is being replaced by Secure Sockets Layer (SSL).

See also Secure Sockets Layer.

Secure MIME Abbreviated S/MIME. An extension to the Multipurpose Internet Mail Extension (MIME) e-mail standard that adds security in the form of the RSA public-key algorithm.

Secure Sockets Layer

See also Multipurpose Internet Mail Extension; Pretty Good Privacy; Privacy Enhanced Mail.

Secure Sockets Layer Abbreviated SSL. An interface originally developed by Netscape that provides encrypted data transfer between client and server applications over the Internet.

SSL works at the network level and so can be used by any SSL-compliant application. Applications that use SSL use RSA public key encryption and digital signatures to establish the identity of the two parties in the transaction.

See also RSA; Secure HTTP; SOCKS.

Secure WAN Abbreviated S/WAN. An initiative from RSA Data Security designed to create secure firewall-to-firewall connections over the Internet.

S/WAN creates a Virtual Private Network (VPN) over the Internet, and all data transmitted on this VPN is encrypted to keep it secure.

See also IPsec; SOCKS; Virtual Private Network.

security Operating system controls used by the network administrator to limit users' access to approved areas.

The National Security Agency document called "Trusted Computer System Evaluation Criteria" (TCSEC) specifies security levels that vendors must follow to comply with Department of Defense security standards.

This publication details standards for security levels used to control access to computer systems from Class A1, the highest

verifiable security level, to Class D, the lowest, which has no security.

Class C2 is the security level most appropriate to the business world; higher levels of security tend to intrude too much into normal commercial working patterns. C2 security requires that the operating system provide individual logins with separate accounts and a verifiable audit trail, that resources have owners, and that files and other system resources be protected from other processes that might corrupt them.

An operating system that lets anyone have unfettered access, such as MS-DOS or the MacOS, falls into the D category. C1 and C2 levels can be implemented in a commercial environment. After the B1 level, the computing environment changes radically, and many of the mandatory access-control mechanisms become impractical for normal commercial operations.

See also intruder.

Security Accounts Manager Abbreviated SAM. In Microsoft Windows NT, the security system that manages and provides access to the account or SAM database.

SAM authenticates a user name and password against information contained in the database and creates an access token that includes the user's permissions.

See also SAM database; security identifier.

Security Administrator Tool for Analyzing Networks Abbreviated SATAN. A software package, available free over the Internet, that allows network administrators to identify gaps in their security systems.

Critics of the program argue that SATAN lets hackers exploit the information contained in the program on how to infiltrate these security systems, but so far, the program seems to have acted as a wake-up call for network administrators.

See also intruder; security.

security equivalence In Novell Directory Services (NDS), when an object or trustee receives the same rights given to another object.

See also implied security equivalence; explicit security equivalence.

security groups In Microsoft Windows 2000 Server, the groups used to assign rights and permissions. Security groups replace Windows NT's user groups.

See also distribution groups.

security ID *See* security identifier.

security identifier Abbreviated security ID or SID. In Microsoft Windows NT, a unique name that identifies a logged-on user to the internal security system.

A SID contains a complete set of permissions and can apply to a single user or to a group.

See also Security Accounts Manager.

security log In Microsoft Windows 2000, a system log that records changes to security settings and audited access such as attempts to open files or folders. You can use the Event Viewer to look at the contents of the security log.

See also application log; Event Viewer; system log.

security zone In Microsoft Internet Explorer, a set of classifications that control how a Web-based application can interact with the browser.

The four zones—Internet, Local intranet, Trusted sites, and Restricted sites—place different security restrictions on any Web sites visited and manage ActiveX controls, Java applets, cookies, downloads, and scripting .

segmentation and reassembly

sublayer Abbreviated SAR. One of two sublayers that make up the ATM Adaptation Layer, the other being the convergence sublayer (CS).

The SAR is the lower of the two sublayers, packages variable-sized data packets into fixed-sized cells at the transmitting end, and repackages the cells at the receiving end.

SAR is also responsible for locating and managing lost cells and cells that are received out of sequence.

See also ATM Adaptation Layer; convergence sublayer.

semaphore An interprocess communication signal that indicates the status of a shared system resource, such as shared memory, in a multitasking operating system.

There are several types of semaphores:

Event Allows a thread to tell other threads that an event has occurred and that it is safe for them to resume execution.

Mutual exclusion (mutex) Protects system resources, such as files, data, and peripheral devices, from simultaneous access by several processes.

Multiple wait (muxwait) Allows threads to wait for multiple events to take place or for multiple resources to become free.

See also mailslots; named pipe; pipe; shared memory; socket.

sendmail A Unix system utility program or daemon that acts as a mail transport agent in an e-mail system, receiving messages from a user's e-mail program and then routing the mail to the correct destination. Normal users rarely come into direct contact with sendmail; it is difficult to administer and is a known security problem.

See also Post Office Protocol; Simple Mail Transfer Protocol.

sequel *See* Structured Query Language.

Sequenced Packet eXchange Abbreviated SPX. A set of Novell NetWare protocols implemented on top of IPX (Internetwork Packet eXchange) to form a transport-layer interface.

SPX provides additional capabilities over IPX. For example, it guarantees packet delivery by having the destination node verify that the data were received correctly. If no response is received within a specified time, SPX retransmits the packet. If several retransmissions fail to return an acknowledgment, SPX assumes that the connection has failed and informs the operator. All packets in the transmission are sent in sequence, and they all take the same path to their destination node.

See also Internetwork Packet eXchange.

sequential access An access method used by some storage devices, such as tapes, that requires them to start at the beginning to find a specific storage location. If the information is toward the end of the tape, access can take a long time.

See also random access.

serial communications The transmission of information from computer to computer or from computer to peripheral device one bit at a time.

Serial communications can be synchronous and controlled by a clock, or they can be asynchronous and coordinated by start and stop bits embedded in the data stream. The sending and receiving devices must both use the same baud rate, parity setting, and other communication parameters.

See also 1394; asynchronous transmission; High Speed Serial Interface; RS-232-C; RS-422; RS-423; RS-449; RS-485; RS-530; Serial Storage Architecture; synchronous transmission; Universal Serial Bus.

Serial Line Internet Protocol Abbreviated SLIP. A protocol used to run Internet Protocol over serial lines or telephone connections using modems.

SLIP allows a computer to establish a temporary direct connection to the Internet via modem and to appear to the host system as if it were a port on the host's network.

SLIP is slowly being replaced by PPP (Point-to-Point Protocol).

See also Internet Protocol; Point-to-Point Protocol.

serial port A computer input/output port that supports serial communications, in which information is processed one bit at a time.

RS-232-C is a common protocol used on serial ports when communicating with modems, printers, mice, and other peripherals.

See also parallel port.

Serial Storage Architecture Abbreviated SSA. A high-performance serial interface, originally developed by IBM, used to connect peripherals, such as scanners, disk drives, optical disks, and printers, to workstations or servers.

A typical SSA interface has two ports with a total bandwidth of 80Mbps and can manage two 20Mbps transmissions simultaneously, one in each direction. The shielded four-wire cable is configured as two pairs, and the maximum cable length is 25 meters (82 feet).

SSA is also specified as a physical layer serial interface in SCSI-3.

See also 1394; Fibre Channel; SCSI-3; serial communications.

server Any computer that makes access to files, printing, communications, and other services available to users of the network.

In large networks, a dedicated server runs a special network operating system; in smaller installations, a nondedicated server may run a personal computer operating system with peer-to-peer networking software running on top.

A server typically has a more advanced processor, more memory, a larger cache, and more disk storage than a single-user

workstation. A server may also have several processors rather than just one and may be dedicated to a specific support function such as printing, e-mail, or communications. Many servers also have large power supplies, UPS (uninterruptible power supply) support, and fault-tolerant features, such as RAID technology.

On the Internet, a server responds to requests from a client, usually a Web browser.

See also access server; communications/modem server; file server; print server; Web server.

server application In Object Linking and Embedding (OLE), an application that creates OLE objects.

See also Object Linking and Embedding.

server-based application An application run from the server rather than from a local hard disk.

See also client application; thin client.

server-based computing *See* Terminal services.

Server Manager A Microsoft Windows NT Server utility used to manage domains, workgroups, and computers.

Server Message Block Abbreviated SMB. A distributed file-system network protocol, developed by Microsoft and adopted by many other vendors, that allows a computer to use the files and other resources of another computer as though they were local. For network transfers, SMBs are encapsulated within the NetBIOS network control block packet.

server push A mechanism used on the Internet whereby a client application, usually a Web browser, maintains an open connection to a Web site, and the Web server provides new content to the client automatically, as soon as the new content becomes available. This process continues until the server decides to close the connection.

See also client pull.

server root A directory on a Web server that contains the server program as well as configuration and information files.

Server service In Microsoft Windows NT Server, a service that manages access to and sharing of local resources attached to the network.

server side include Abbreviated SSI. A method used to create dynamic Web pages.

The INCLUDE statement is used to embed commands within an HTML document and tells the Web server to perform specific tasks. A common way of using INCLUDE is to tell the server to load the contents of another external file at a specific point in the original HTML document.

SSIs can degrade server performance due to the number of repeated disk accesses they make, and they can also prove to be a security problem.

See also Extensible Markup Language; HyperText Markup Language.

service In Microsoft Windows NT, a process that performs a particular system function and often provides an API (application programming interface) so that other processes can take advantage of its abilities.

See also Alerter service; Messenger service; Server service.

Service Advertising Protocol Abbreviated SAP. A protocol that provides a method for servers, printers, and other devices to advertise their services on a Novell NetWare network and allows routers to create and maintain a database of current inter-network server information.

SAP packets are normally broadcast every 60 seconds and include the name, network address, and type of service offered by a server.

SAP traffic can be excessive on large networks, and SAP filtering is often used to reduce the traffic; SAP traffic can be severely reduced or even eliminated from rarely used servers.

service bureau A company that provides data processing or business software development services to its customers.

A service bureau might provide typesetting, prepress production, optical document scanning, or other services.

By using a service bureau, a company can avoid the high hardware and personnel costs associated with running its own in-house services.

See also outsourcing.

Service Level Agreement Abbreviated SLA. An agreement between a user and a service provider that defines the terms and conditions of the service, as well as being a means for evaluating the service provided against the service specified.

See also Internet Service Provider; service provider.

service pack A periodic update to the Microsoft Windows family of operating systems.

A service pack includes fixes to bugs and problems reported by both customers and by Microsoft and may also contain additional new features or applications.

Service Profile Identifier Abbreviated SPID. An 8- to 14-character identifier associated with an ISDN connection that defines the services available on that connection.

The SPID is actually stored at the telephone company central office and is accessed by and used to identify your ISDN service. Don't lose it; your ISDN system will not run without it.

See also Integrated Services Digital Network.

service provider A general term used to describe those companies providing a connection to the Internet or to other communications services. Access methods vary from high-speed dedicated access to dialup using SLIP or PPP.

See also Internet Service Provider.

servlet A small application that is written in the Java programming language and runs on a Web server.

SERVMAN A Novell NetWare utility used to manage the server and view network information, including operating system performance parameters and IPX/SPX

(Internetwork Packet eXchange/Sequenced Packet eXchange) settings.

It also provides a convenient front end to the SET command, as well as information about the number of running processes, NetWare Loadable Modules (NLMs) installed, volumes mounted, users logged in, and name spaces loaded.

In NetWare 5, SERVMAN has been merged into MONITOR.

See also MONITOR; SET.

session 1. The time during which a program is running on either a local or a remote computer.

2. An MS-DOS or a Microsoft Windows program run as a separate protected task under certain multitasking operating systems, such as OS/2.

3. In communications, the name for the active connection between a mainframe terminal (or a personal computer emulating a terminal) and the computer itself. Many different transactions or message exchanges may take place during a single session.

See also process; thread.

session layer The fifth of seven layers of the OSI Reference Model for computer-to-computer communications.

The session layer coordinates communications and maintains the session for as long as it is needed, performing security, logging, and administrative functions.

See also OSI Reference Model.

SET A Novell NetWare server command used to establish operating system parameters, including parameters for

communications, the memory pool, file and directory caching, the disk and file system, locking, transaction tracking, the NetWare Core Protocol (NCP), and error handling.

The SET command is also used by many other operating systems, including MS-DOS, OS/2, and Unix, to establish environment values.

setup string *See* control code.

SFT *See* System Fault Tolerance.

sgi *See* Silicon Graphics, Inc.

SGML *See* Standard Generalized Markup Language.

SGRAM *See* synchronous graphics RAM.

shadow member One of two members of a mirror set. The primary member contains the original data; the shadow member contains the copy. The shadow member is read only if the primary member cannot be read.

See also disk mirroring; primary member.

shadow memory The technique of copying the contents of the BIOS ROM into faster RAM when the computer first boots up.

Because RAM is usually two to three times faster than ROM, and the speedier access reduces the time required to execute a BIOS routine, the processor spends more time working and less time waiting.

share In Microsoft Windows NT, a resource such as a printer or a directory, shared by a server or a peer on the network.

shared folder In a networked Macintosh, a folder that is available to other users, either without restriction or through a password. A shared folder in the Macintosh is comparable to a network directory on a PC.

shared memory An interprocess communications technique in which the same memory is accessed by more than one program running in a multitasking operating system. Semaphores or other management elements prevent the applications from colliding or trying to update the same information at the same time.

See also mailslots; named pipe; pipe; semaphore; socket.

sharing In Microsoft Windows, the process of making a network resource, such as a file, a folder, or a printer, available to other network users.

See also share; hidden share.

shell In Unix, the command processor. The shell accepts commands from the user, interprets them, and passes them to the operating system for execution.

The three major shells are the Bourne shell (the original Unix shell from AT&T), the C shell (developed as a part of the BSD Unix efforts), and the Korn shell (also developed by AT&T).

In recent years several public-domain shells have become popular, including Bash (the Bourne-again shell), which is often used on Linux, Tcsh, and Zsh.

See also Bash shell; Bourne shell; C shell; Korn shell.

shielded cable Cable protected against electromagnetic and radio frequency interference (RFI) by metal-backed mylar foil and plastic or PVC.

See also unshielded cable.

shielded twisted-pair cable Abbreviated STP. Cable with a foil shield and copper braid surrounding the pairs of wires.

The wires have a minimum number of twists per foot of cable length; the greater the number of twists, the lower the crosstalk. STP offers high-speed transmission for useful distances, and it is often associated with Token Ring networks, but its bulk quickly fills up wiring conduits.

See also crosstalk; shielded cable; unshielded twisted-pair cable.

short *See* short circuit.

short circuit Often abbreviated to short. A circuit that is accidentally completed at a point too close to its origin to allow normal or complete operation. In cabling, a short circuit often occurs when two stripped wires touch.

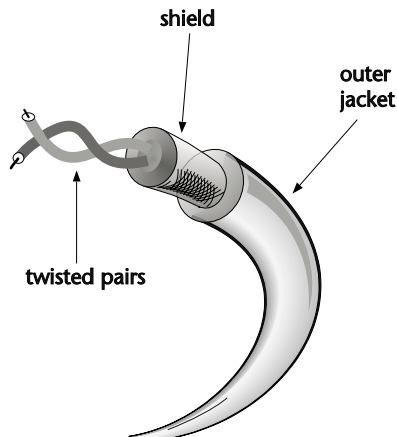
shortcut In Microsoft Windows, a link to an object such as a file, a folder, a Web page, or an application. You can create an unlimited number of shortcuts, and you can place them on the Desktop, on the Start menu, or in a particular folder.

shortcut keystroke *See* key combination.

shortcut trust In Microsoft Windows 2000 Server, an explicit trust used to defeat the trust referral process between directory trees.

See also downlevel trust; explicit trust; two-way transitive trust.

SHIELDED TWISTED-PAIR CABLE



shortest-path routing A routing algorithm in which paths to all network destinations are calculated. The shortest path is then determined by a cost assigned to each link.

short-haul modem A simple, low-cost modem that can transmit information only over short distances, such as from one side of a building to the other side.

See also line driver; long-haul modem.

S-HTTP *See* Secure HTTP.

S/HTTP *See* Secure HTTP.

SID *See* security identifier.

SIG *See* special interest group.

signal-to-noise ratio Abbreviated SNR. The ratio between the desired signal and the unwanted noise at a specific point in a cable; a measure of signal quality.

SNR is particularly important in networks using unshielded twisted-pair cable (UTP). SNR specifications for token-ring networks are much more stringent than those for 10BaseT or ARCnet.

signature A short text file that is automatically added to the end of your e-mail or Usenet posts.

A signature file usually contains your name (or alias) and e-mail address, and some people like to add pithy quotes; whatever your signature file contains, remember to keep it short.

Silicon Graphics Certified IRIX Network Administrator An advanced certification from Silicon Graphics designed to

demonstrate proficiency in administering Silicon Graphics IRIX-based networks.

Silicon Graphics Certified IRIX System Administrator A basic certification from Silicon Graphics designed to demonstrate proficiency in administering Silicon Graphics IRIX systems.

Silicon Graphics, Inc. Sometimes abbreviated sgi. A major manufacturer of Unix and RISC-based graphics workstations and supercomputers.

The company's workstations are used in product engineering, computer simulation, data warehousing, and in Hollywood post-production facilities to add special effects.

For more information on Silicon Graphics Inc, see www.sgi.com.

Simple Mail Transfer Protocol Abbreviated SMTP. The TCP/IP (Transmission Control Protocol/Internet Protocol) protocol that provides a simple e-mail service and is responsible for moving e-mail messages from one e-mail server to another. SMTP provides a direct end-to-end mail delivery, which is rather unusual; most mail systems use store-and-forward protocols.

The e-mail servers run either Post Office Protocol (POP) or Internet Mail Access Protocol (IMAP) to distribute e-mail messages to users.

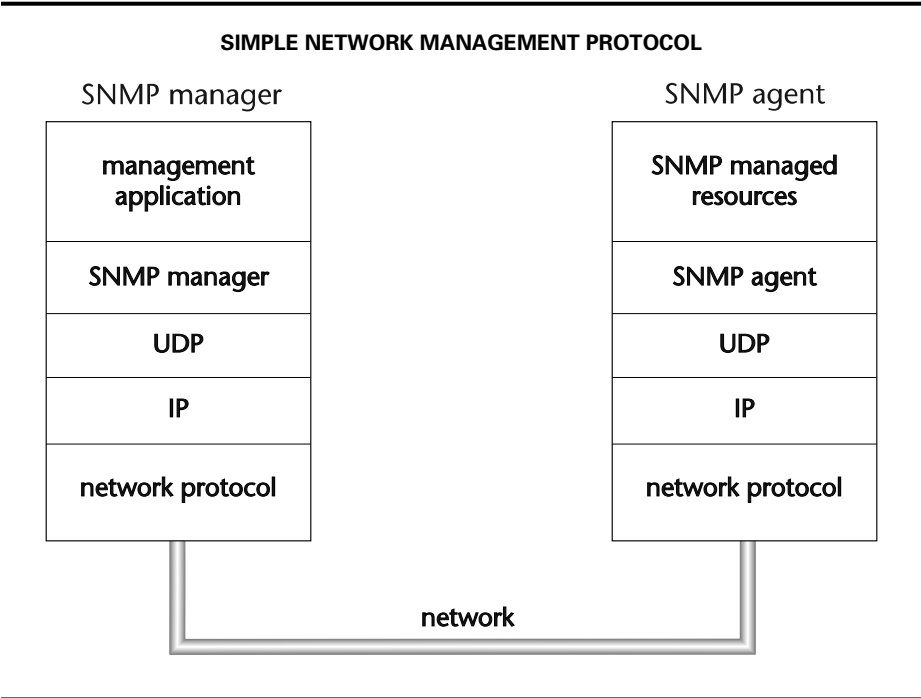
Many third-party vendors sell host software capable of exchanging SMTP e-mail with proprietary e-mail systems, such as IBM's PROFS.

See also Internet Mail Access Protocol; Post Office Protocol; store-and-forward.

Simple Network Management

Protocol Abbreviated SNMP. A standard protocol, part of the TCP/IP (Transmission Control Protocol/Internet Protocol) suite, used to manage and monitor nodes on a network. The accompanying illustration shows how the SNMP manager and agent are organized.

SNMP is a communications protocol for collecting information about devices on the network, including hubs, routers, and bridges. Each piece of information to be collected about a device is defined in a Management Information Base (MIB). SNMP uses UDP (User Datagram Protocol) to send and receive messages on the network.



single-attached station Abbreviated SAS. In the Fiber Distributed Data Interface (FDDI), a device attached to only one of the dual, counter-rotating rings. Noncritical devices, such as workstations, are often connected using SASs, because

they are less expensive than dual-attached stations (DASs).
See also dual-attached station; Fiber Distributed Data Interface.

single-ended SCSI A Small Computer System Interface (SCSI) bus-wiring scheme

single large expensive disk

that uses a single wire for each signal transmitted on the bus. Used more often than differential SCSI.

See also differential SCSI; Small Computer System Interface.

single large expensive disk Abbreviated SLED. The traditional alternative to RAID (redundant array of inexpensive disks), used by most networks.

Single-Line Digital Subscriber Line Abbreviated SDSL; sometimes called Symmetrical Digital Subscriber Line. A symmetrical, bidirectional Digital Subscriber Line service that operates on one twisted-pair wire.

SDSL can provide data rates of up to the T1 rate of 1.544Mbps over a cable length of up to 1,000 feet, and because it operates above the voice frequency, voice and data can be carried on the same connection at the same time.

See also Asymmetric Digital Subscriber Line; Digital Subscriber Line; High-Bit-Rate Digital Subscriber Line; Rate-Adaptive Digital Subscriber Line; Very-High-Bit-Rate Digital Subscriber Line.

single login An authentication process that allows users to log in to a complex network only once rather than requiring them to log in to each separate network element.

Additional connections and drive mappings are managed in the background as a part of the authentication process. Also known as single sign-on.

single-mode fiber Narrow diameter fiber-optic cable in which lasers rather than

LEDs are used to transmit signals through the cable.

Single-mode fiber allows only one route for a light wave to pass through, and it can transmit signals over considerable distances. For this reason, it is often used in telephone networks rather than in local-area networks.

See also multimode fiber.

single reference time server In Novell NetWare, the server that determines the network time and provides that information to workstations and to secondary time servers on the network.

Single reference time servers are not compatible with reference or primary time servers.

See also primary time server; reference time server; secondary time server.

single sign-on *See* single login.

single-user logon In Microsoft Windows NT, a mechanism that allows a user to connect to multiple servers, domains, and applications with a single logon.

See also single login.

single-user system A computer system designed for use by one person at a time, often on a personal computer.

Windows 95 and 98, MS-DOS, the MacOS, OS/2 Warp, and Windows NT Workstation are all examples of single-user operating systems. Unix and most network operating systems are multiuser systems.

Site In Microsoft Active Directory, one or more TCP/IP (Transmission Control

Protocol/Internet Protocol) subnets linked by reliable network connections.

site license A software license that covers all the installed copies of a software package at a specific location or locations.

Some large corporations and government institutions prefer to negotiate a site license rather than try to pay for and keep track of all the individual copies they use. A site license may allow unlimited copies for internal use, or it may limit the number of copies of a program the corporation can use concurrently.

See also application metering; concurrent license.

SLA *See* Service Level Agreement.

slash The / character. In many operating systems, used to separate command-line switches that alter the default settings for an operating system command.

Used in Unix as the name of the root file-system and also to separate elements (files and directories) in a directory pathname.

SLED *See* single large expensive disk.

SLIP *See* Serial Line Internet Protocol.

Small Computer System Interface

Abbreviated SCSI, which is pronounced “scuzzy.”

A high-speed parallel interface defined by the ANSI X3T9.2 committee. SCSI is used

to connect a computer to peripheral devices using only one port. Devices connected in this way are said to be “daisy-chained,” and each device must have a unique identifier or priority number.

SCSI has been standard on the Macintosh since the Mac Plus was introduced, and it is available on personal computers as a single host adapter, using a special connector.

SCSI is often used to connect hard disks, tape drives, CD-ROM drives, and other mass storage media, as well as scanners and printers.

Features of the SCSI definition include:

- The bus can manage simultaneous reads and writes.
- The original standard supports as many as 7 devices on a single host adapter; new standards support as many as 16 devices and a bus length of up to 25 meters (80 feet).
- SCSI devices have their own control circuitry and can disconnect from the host adapter to process tasks on their own, freeing the bus for other purposes.

Table S.1 lists the various SCSI standards, along with the bus speed in Mbps, the bus width in bits, and the maximum number of devices supported.

See also Fast SCSI; Fast/Wide SCSI; SCSI-1; SCSI-2; SCSI-3; Ultra SCSI; Ultra2 SCSI; Wide Ultra SCSI; Wide Ultra2 SCSI.

TABLE S.1 SCSI STANDARDS

SCSI Standard	Maximum Bus Speed	Bus Width	Maximum Devices
SCSI-1	5	8	8

TABLE S.1 SCSI STANDARDS (CONTINUED)

SCSI Standard	Maximum Bus Speed	Bus Width	Maximum Devices
Fast SCSI	10	8	8
Fast/Wide SCSI	20	16	16
Ultra SCSI	20	8	8
Wide Ultra SCSI	40	16	16
Ultra2 SCSI	40	8	8
Wide Ultra2 SCSI	80	16	16

small office/home office Abbreviated SOHO. That portion of the market for computer services occupied by small offices and home-based businesses rather than by large corporate buyers. SOHO is a small but growing market sector characterized by very well informed buyers.

The large number of home offices is the result of many factors in the economy, including corporate downsizing and cheaper, more capable computers and office equipment, and is a trend that is likely to continue.

See also telecommuting.

smart card A credit-card-sized device that stores public and private keys, passwords, and other personal information that must be kept secure. Several operating systems, including Windows 2000, can use smart-card information for certificate-based authentication and log on.

smart hub A concentrator, used in Ethernet or ARCnet networks, with certain network-management facilities built in to firmware that allow the network administrator to control and plan network configurations; also known as an intelligent hub.

In token-ring networks, a smart hub is known as a Controlled Access Unit (CAU). A CAU can determine if nodes are operating, connect and disconnect nodes, and monitor node activity.

See also Controlled Access Unit.

smart terminal *See* intelligent terminal.

SMB *See* Server Message Block.

SMDS *See* Switched Multimegabit Data Services.

smiley A group of text characters used in e-mail and in Usenet posts to indicate humor or some other emotion. You must turn a smiley on its side for it to make sense.

Hundreds of smileys are in common use, and new ones are invented all the time. Two popular favorites are :-), to indicate smiling, and ;-), to indicate winking.

See also emoticon.

S/MIME *See* Secure MIME.

SMM *See* system management mode.

SMS *See* Storage Management Services.

SMTP *See* Simple Mail Transfer Protocol.

SNA *See* Systems Network Architecture.

SNA gateway A hardware and software device that connects an SNA (Systems Network Architecture) mainframe to a local-area network.

SNA gateway products are available from Microsoft and from Novell.

See also Microsoft SNA Server; NetWare for SNA.

snail mail A rude reference to the relatively slow speed of the conventional postal service when compared with the speed of online e-mail systems.

snap-in In Microsoft Management Console (MMC), the software that constitutes the administrative tool.

MMC provides the user interface, and the snap-in actually performs the management or administrative function. Snap-ins can be provided by Microsoft or can be written by third-party software developers.

In Windows 2000, Computer Management is a MMC snap-in.

See also Computer Management; Microsoft Management Console.

sneakernet An informal method of file sharing in which a user copies files on to a floppy disk and then carries the disk to the office of a co-worker.

sniffer A small program loaded onto a system by an intruder, designed to monitor specific traffic on the network.

The sniffer program watches for the first part of any remote login session that includes the user name, password, and host name of a person logging in to another machine.

Once this information is in the hands of the intruder, he or she can log on to that system at will. One weakly secured network can therefore expose not only the local systems, but also any remote systems to which the local users connect.

Sniffer is also the name of a network analyzer product from Network General.

See also intruder; network analyzer; spoofing; Trojan Horse.

SNMP *See* Simple Network Management Protocol.

SNR *See* signal-to-noise ratio.

social engineering A method used by intruders to collect password information from genuine users by false pretenses.

Social engineering can take many forms. For example, a potential intruder telephones a busy technical support department, claiming to be a new service engineer

who has just lost his password. Rather than go through the appropriate procedures, the technical support person assigns a new password over the phone and implements it on the system so that the new engineer can use it immediately.

See also brute-force attack; dictionary attack; sniffer.

socket 1. A general-purpose interprocess communication mechanism, originally developed in the Unix world. Sockets allow processes that are not running at the same time or on the same system to exchange information; pairs of cooperating sockets manage communications between the processes on your computer and those on a remote computer in a networked environment. You can read data from or write data to a socket just as you can to a file. Sockets are now used in many other environments.

2. That part of an IPX (Internetwork Packet eXchange) internetwork node address that represents the destination of an IPX packet. Certain sockets are reserved by NetWare for particular applications. For example, IPX delivers all NetWare Core Protocol (NCP) request packets to socket 451h.

See also mailslots; named pipe; pipe; semaphore; shared memory; WinSock.

socket services Part of the software support needed for PCMCIA (PC Memory Card International Association) hardware devices in a portable computer, controlling the interface to the hardware.

Socket services is the lowest layer in the software that manages PCMCIA cards. It

provides a BIOS-level software interface to the hardware, effectively hiding the specific details from higher levels of software. Socket services also detect when you insert or remove a PCMCIA card and identify the type of card.

See also card services.

SOCKS A proxy protocol that provides a secure channel between two TCP/IP (Transmission Control Protocol/Internet Protocol) systems, usually a Web browser running on an internal corporate intranet and a Web server on the Internet.

SOCKS provides firewall services, as well as fault tolerant features, auditing, and management.

See also firewall; proxy server; Secure Sockets Layer; Secure WAN.

soft modem *See* controllerless modem.

software An application program or operating system that a computer can execute. *Software* is a broad term that can imply one or many programs, and it can also refer to applications that may consist of more than one program.

software development kit Abbreviated SDK. A package that contains useful software development tools, such as editors, compilers, debuggers, libraries, and technical information.

software interrupt An interrupt generated by an instruction in a program, often called a trap.

See also hardware interrupt.

software license A license to use a software package subject to certain conditions. These conditions usually define the rights of the purchaser and limit the liability of the program's publisher.

See also application metering; site license.

software piracy The illegal copying and distribution of copyrighted software. Copying software, like duplicating any copyrighted material, is illegal.

In an attempt to discourage software piracy, the Software Publishers Association (SPA) has run several advertising campaigns, including a billboard showing a pair of handcuffs and the message "Copy software illegally and you could get this hardware absolutely free." The SPA has been successful in persuading companies to inventory their software so that corporations know which software copies they have purchased legally.

Software Publishers Association Abbreviated SPA. An association of software developers and distributors most notable for their tactics in fighting software piracy, including extremely blunt advertising campaigns and unannounced visits to companies suspected of acts of piracy.

software suite A selection of business applications sold as a single integrated package.

The standard version of a software suite usually includes a word processor, a spreadsheet, presentation graphics, and an e-mail program, and the professional version will often add a database program. The cost of the suite is significantly less than the

cost of purchasing each application separately. Additional benefits include inter-application communications, easy installation of the whole package, and reduced training time.

Microsoft Office and Lotus SmartSuite are examples of popular software suites.

SOHO *See* small office/home office.

Solaris A version of Unix from SunSoft that runs on Intel-based PCs and Sun workstations; SunSoft is a subsidiary of Sun Microsystems.

Solaris is based on Unix System V Release 4 and includes networking support, the OpenWindows graphical user interface, and DeskSet, an integrated desktop that includes some 50 productivity tools. Solaris supports a Java Virtual Machine as well as the Common Desktop Environment and WebNFS. Multiprocessor systems are supported, and Solaris uses symmetrical multiprocessing techniques to take advantage of the additional processing power.

See also Common Desktop Environment; Java Virtual Machine; SunOS; WebNFS.

SOM *See* System Object Model.

SONET *See* Synchronous Optical Network

source address The address portion of a packet or datagram that identifies the sender.

See also destination address.

source license A software license that gives the user the right to possess and modify

the original source code from which an application or operating system is created.

In the past, source licenses have either been simply impossible to obtain or have been prohibitively expensive. With the spread of open source software, things are changing.

See also binary license; open source software

source routing IBM's Token Ring method of routing data frames through a network consisting of multiple LANs by specifying the route to be traveled in each frame. The route is actually determined by the end stations through a discovery process supported by source-bridge routers.

IBM bridges can be of two types:

- Single-route broadcasting allows certain bridges to pass the packet so that only a single copy arrives on each ring in the network.
- All-routes broadcasting sends the packet across all the possible routes in the network, so as many copies of the packet arrive at the destination as there are bridges in the network.

SPA *See* Software Publishers Association.

space parity *See* parity.

spanning tree A network segment that is free of logical loops; a network structure that has a root node and one path, usually the shortest distance, that connects all the other nodes.

This tree structure is used in bridged networks to make routing decisions, especially if multiple paths connect nodes, because

these loops could lead to packets looping on their way to their destination.

SP *See* switch processor.

spanning tree algorithm A technique based on the IEEE 802.1 standard that finds the most desirable path between segments of a multilooped, bridged network.

If multiple paths exist in the network, the spanning tree algorithm finds the most efficient path and limits the link between the two networks to this single active path. If this path fails because of a cable failure or other problem, the algorithm reconfigures the network to activate another path, thus keeping the network running.

SPARC *See* Scalable Performance Architecture.

SPARCstation A Sun Microsystems family of Unix workstations based on the SPARC processor.

SPARCstations range from small, diskless desktop systems to high-performance, tower servers in multiprocessor configurations.

See also Scalable Performance Architecture; Sun Microsystems, Inc.

SPEC benchmarks *See* Systems Performance Evaluation Cooperative benchmarks.

special group In Microsoft Windows NT Server, a group whose membership is predefined and automatically updated.

Special groups provide an easy way to describe sets of users. For example, the Everyone group always includes all domain users and is updated automatically as user accounts are added and deleted.

special interest group Abbreviated SIG. A group that meets to share information about a specific topic, such as particular hardware, software, programming languages, or operating systems. A SIG is often part of a user group or other organization.

SPID *See* Service Profile Identifier.

spider A World Wide Web application that automatically locates and collects information about new Web sites. Spiders are most often used to create large databases of Web sites that in turn are accessed by search engines responding to user requests for information.

See also robot; search engine.

spike A short, transient electrical signal, often of very high amplitude.

See also power conditioning; power surge; surge suppressor.

splat A slang expression for the asterisk character (*) that you can yell across a crowded room without fear of being misunderstood.

See also bang.

splitter server In Internet video, a server that receives a video signal and then re-broadcasts that signal across a network.

See also IP multicast; multicast; stream thinning.

spoofing A security breach in which an intruder logs on to the system by pretending to be a genuine user.

The intruder may obtain another person's user name and password in casual

conversation or, in a more concerted attack, may use a network analyzer to monitor and capture network traffic.

See also brute-force attack; dictionary attack; sniffer; social engineering; Trojan Horse.

spooler *See* print spooler.

SPX *See* Sequenced Packet eXchange.

SQE Abbreviation for signal quality error. *See* heartbeat.

SQL *See* Structured Query Language.

SRAM *See* static RAM.

SSA *See* Serial Storage Architecture.

SSE *See* Streaming SIMD Extensions.

SSI *See* server side include.

SSL *See* Secure Sockets Layer.

ST506 Interface A popular hard-disk interface standard developed by Seagate Technologies, first used in IBM's PC/XT computer.

The interface is still used in systems with disk capacities smaller than about 40MB. ST506 has a relatively slow data-transfer rate of 5Mbps.

A later variation of ST506, called ST412, adds several improvements. Because these two interfaces are so closely related, they are often referred to as ST506/412.

See also Enhanced Small Device Interface; Integrated Drive Electronics; Small Computer System Interface.

ST See straight-tip connector.

Standard Generalized Markup

Language Abbreviated SGML. A 1986 standard (ISO 8879) for defining the structure and managing the contents of any digital document.

The standard specifies a definition for formatting a digital document so that it can be modified, viewed, or output on any computer system. Each SGML document consists of two parts:

- The DTD (Document Type Definition) defines the structure of the document.
- The DI (Document Instance) describes the data or text of the document.

HTML, used in World Wide Web documents on the Internet, is a part of SGML.

See also Extensible Markup Language; HyperText Markup Language; Virtual Reality Modeling Language.

Standby Monitor In a Token Ring network, a network node that serves as a

backup to the Active Monitor and can take over in the event that the Active Monitor fails.

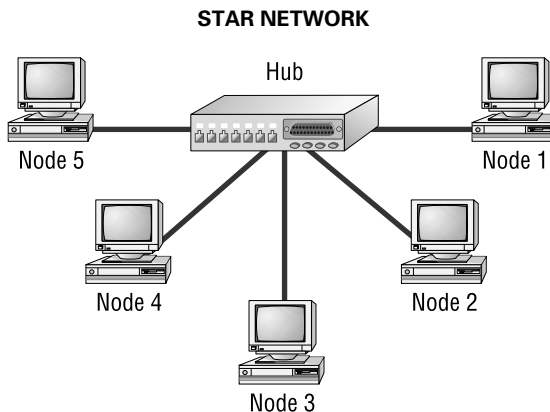
See also Active Monitor.

star network A network topology in the form of a star.

At the center of the star is a wiring hub or concentrator, and the nodes or workstations are arranged around the central point representing the points of the star. Wiring costs tend to be higher for star networks than for other configurations, because each node requires its own individual cable. Star networks do not follow any of the IEEE standards.

See also bus network; ring network; topology.

star-dot-star A commonly available file specification (*.*) that uses the asterisk wildcard character. It is equivalent to specifying any combination of filename and filename extension.



StarLAN A network operating system from AT&T that implements CSMA/CD (Carrier Sense Multiple Access/Collision Detection) protocols on twisted-pair cable transmitting at 1Mbps; a subset of 802.3. In 1988, StarLAN was renamed StarLAN 1, and StarLAN 10 was launched. StarLAN 10 is a 10Mbps Ethernet version that uses twisted-pair cable or fiber-optic cable.

start bit In asynchronous transmissions, a start bit is transmitted to indicate the beginning of a new data word.

See also data bits; parity; stop bit.

STARTUP.NCF A Novell NetWare server configuration boot file that loads the NetWare server's disk driver, along with name spaces and certain SET parameters. *See also* SET.

stateless filter *See* packet-level filter.

static RAM Abbreviated SRAM, pronounced "ess-ram." A type of computer memory that retains its contents as long as power is applied; it does not need constant refreshment, as required by dynamic RAM (DRAM) chips.

An SRAM chip can store only about one-fourth of the information that a DRAM chip of the same complexity can hold. However, SRAM, with access times of 10 to 25 nanoseconds, is much faster than DRAM, at 80 nanoseconds or more, and is often used in caches. SRAM is four to five times as expensive as DRAM.

static routing A method used to preprogram connecting paths between networks into a router by a network administrator.

If a connection fails, the administrator must reprogram the router to use a new path. In most large networks, the delay that this causes is unacceptable, and dynamic routing is used instead. Dynamic routing automatically locates and uses the best available path and recalculates paths if a connection fails.

See also dynamic routing.

statistical multiplexing Abbreviated stat mux. In communications, a method of sharing a transmission channel by using statistical techniques to allocate resources.

A statistical multiplexer can analyze traffic density and dynamically switch to a different channel pattern to speed up the transmission. At the receiving end, the different signals are merged back into individual streams.

See also frequency-division multiplexing; inverse multiplexing; time-division multiplexing.

stat mux *See* statistical multiplexing.

stealth virus A form of virus that attempts to hide from antivirus software and from the operating system by remaining in memory.

See also boot sector virus; file-infecting virus; macro virus; multipart virus; polymorphic virus; Trojan Horse; vaccine.

stop bit In asynchronous transmissions, a stop bit is transmitted to indicate the end of the current data word. Depending on the convention in use, one or two stop bits are used.

See also data bits; parity; start bit.

storage area network Abbreviated SAN. A method used to physically separate the storage function of the network from the data-processing function.

SAN provides a separate network devoted to storage and so helps to reduce network traffic by isolating large data transfers such as backups. Most of the SAN vendors, including StorageTek and Compaq, use a Fibre Channel-based SAN system, although IBM has proposed a proprietary architecture.

See also network attached storage.

Storage Management Services Abbreviated SMS. A set of Novell NetWare Loadable Modules and other software that allows data to be backed up and retrieved from the server and from workstations attached to the network. SMS is independent of both the hardware used to create the backup and also the file systems actually being backed up or restored.

store-and-forward A method that temporarily stores messages at intermediate nodes before forwarding them to the next destination. This technique allows routing over networks that are not available at all times and lets users take advantage of off-peak rates when traffic and costs might be lower.

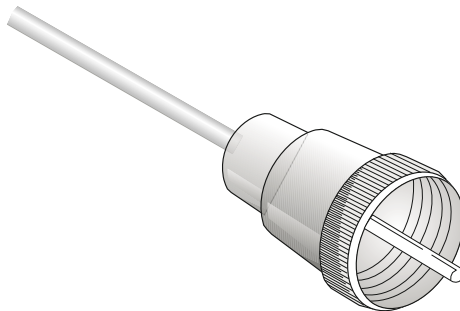
See also message switching.

STP *See* shielded twisted-pair cable.

straight-tip connector Abbreviated ST. A fiber-optic cable connector that maintains the perfect alignment of the ends of the connected fibers, required for efficient light transmission.

stream In Internet video, an end-to-end connection between a client and a server that lets a user start to look at a video clip before the whole file has finished downloading.

STRAIGHT-TIP CONNECTOR



streaming A method used to deliver audio and video content in real time so that the person using the system can interact with the data stream.

A client downloads a portion of the audio/video file, decompresses it, and starts to view the video clip before the rest of the file arrives. Data is built up in a buffer before the playback begins, and the next part of the file is downloaded as the first part plays.

See also IP Multicast multicasting.

Streaming SIMD Extensions Abbreviated SSE. A set of multimedia instructions introduced with Intel's Pentium III.

SSE provides four main improvements over the Pentium II instruction set:

- Eight new directly addressable 128-bit floating-point registers.
- Eight new instructions for streaming data to and from memory.
- Twelve new Single Instruction, Multiple Data (SIMD) integer instructions.
- Fifty new SMID floating-point instructions.

SSE can benefit 3-D graphics, 2-D images, speech recognition, and MPEG-2 encoding, as well as scientific and engineering applications. Most standard office and business applications will see little or no improvement.

See also Pentium III.

Streaming Single Instruction, Multiple Data Extensions *See* Streaming SIMD Extensions.

streaming tape A high-speed tape backup system designed to optimize throughput;

the tape is not stopped during a backup. To use streaming tape, the computer and backup software must be fast enough to keep up with the tape drive.

See also DC-2000; tape cartridge.

STREAMS A Novell NetWare Loadable Module (NLM) that provides a common interface between NetWare and transport protocols such as IPX/SPX, TCP/IP, and SNA.

STREAMS allows services to be provided across the network regardless of the transport protocol in use, because the protocol is transparent to the operating system.

streams A function within Unix that provides flexible communications paths between processes and device drivers.

stream thinning In Internet video, the process of removing video frames to protect the audio feed.

Stream thinning is used to preserve the connection in times of network congestion and to avoid a forced reconnection and its associated disruptive effect on the transmission. Stream thinning restores the full video signal once additional bandwidth becomes available.

StreetTalk The distributed global naming and directory service for Banyan VINES network operating system. The StreetTalk database contains all the necessary information about all nodes and devices on the network, and this database is updated constantly.

Under StreetTalk, all users, printers, and servers have a three-part StreetTalk address in the form of device or user name, domain

name, and organization name. StreetTalk also allows nicknames for nodes and devices. SteetTalk is designed to manage a large number of different environments, including Banyan networks, Windows NT, Unix, and Novell NetWare.

See also domain directory services; Enterprise Network Services; global directory services; NetWare Directory Services; X.500.

string *See* character string.

stripe set A single volume created across multiple hard disk drives and accessed in parallel to optimize disk-access time.

See also disk striping; disk striping with parity.

strong password A password that is specifically designed to be difficult to guess.

Strong passwords are always more than 10 characters and always include punctuation characters and numbers.

See also password; weak password.

structured graphics *See* object-oriented graphics.

Structured Query Language Abbreviated SQL, pronounced “sequel.” In relational database management systems, a query language developed by IBM for use in mainframe applications.

SQL was adopted by Oracle Corporation for use in its database management systems running on all platforms, not just mainframes, and subsequently emerged as a de facto standard for all database management packages.

SQL contains about 60 commands and is used to create, modify, query, and access data organized in tables. It can be used either as an interactive interface or as embedded commands in an application:

- Dynamic SQL statements are interactive, and they can be changed as needed. If you normally access SQL from a command-line environment, you are using dynamic SQL, which is slower than static SQL but much more flexible.
- Static SQL statements are coded into application programs, and as a result, they do not change. These statements are usually processed by a precompiler before being bound into the application.

Many databases implement SQL queries behind the scenes, enabling communication with database servers in systems with client/server architecture.

SQL is an ANSI standard in the United States, as well as a worldwide International Organization for Standardization (ISO) standard.

See also Open Database Connectivity.

structured wiring A planned cabling system for enterprise-wide network communications, including both voice and data. AT&T's Premises Distribution Systems and IBM's Cabling System are both structured wiring designs.

subdirectory A directory contained within another directory. The root directory is the top-level directory, from which all other directories must branch. In common use, subdirectory is synonymous with directory or folder.

See also current directory; parent directory; period and double-period directories.

sub-hive An organizational element in the Microsoft Windows Registry, similar to a subdirectory on a hard disk.

A sub-hive may contain other sub-hives or may contain keys. Microsoft documentation sometimes refers to sub-hives as sub-keys.

See also hive; key; Registry.

sub-key *See* sub-hive.

subnet A logical network created from a single IP address. A mask is used to identify bits from the host portion of the address to be used for subnet addresses.

See also address classes; Classless Inter-Domain Routing; IP address; subnet address; subnet mask.

subnet address The subnet portion of an IP address. In a subnetted network, the host part of the IP address is divided into a subnet portion and a host portion by a subnet mask.

See also address classes; Classless Inter-Domain Routing; IP address; subnet; subnet mask.

subnet mask A number or, more correctly, a bit pattern that identifies which parts of an IP address correspond to the network, subnet, and host portions of the address. Also referred to as an address mask.

See also address classes; Classless Inter-Domain Routing; IP address; subnet; subnet address.

subscribe To post to a Usenet newsgroup or to join a mailing list. This is not a subscription in the sense of a magazine subscription; no money ever changes hands.

Sun Certified Solaris Administrator

A basic certification from Sun Microsystems designed to recognize technical expertise in installing and administering Solaris-based systems.

See also Sun Microsystems, Inc.

Sun Certified Solaris Network

Administrator An advanced certification from Sun Microsystems designed to recognize technical expertise in networking and administering Solaris-based systems and in the security aspects of Solaris.

See also Sun Microsystems, Inc.

Sun Microsystems, Inc. A manufacturer of high-powered workstations and one of the major technical forces in the Unix world.

Sun workstations run Solaris, a version of Unix based on Unix System V Release 4, and range from small, diskless desktop systems to high-performance, tower servers in multiprocessor configurations.

For more information on Sun Microsystems, Inc., see www.sun.com.

See also Java; Jini; Network File System; Solaris.

SunOS A Unix operating system from Sun Microsystems. SunOS is based on BSD Unix; Solaris is based on System V Release 4.

See also Solaris; Sun Microsystems, Inc.

superpipelining A preprocessing technique used by some microprocessors in

which two or more execution stages (such as fetch, decode, execute, or write back) are divided into two or more pipelined stages, giving considerably higher performance.

superscalar A microprocessor architecture that contains more than one execution unit, or pipeline, allowing the processor to execute more than one instruction per clock cycle.

For example, the Pentium processor is superscalar, with two side-by-side pipelines for integer instructions. The processor determines whether an instruction can be executed in parallel with the next instruction in line. If it does not detect any dependencies, the two instructions are executed.

See also complex instruction set computing; reduced instruction set computing.

superserver A computer specifically designed for use as a network server.

A superserver is a very high performance system, often characterized by scalable input/output channels, complex multiprocessing features, and a large price tag. It may have several processors, large amounts of error-correcting memory, cache memory, and hard-disk space, as well as fault-tolerant features, such as redundant power supplies.

superuser A special Unix privilege level, with unlimited access to all files, directories, and commands.

The system administrator must become the superuser to perform certain functions, such as creating new accounts, changing passwords, and other administrative tasks that ordinary users are not allowed to perform

for security reasons. The superuser's login name is usually root, with a user ID of 0.

See also avatar.

surfing To browse your way through various Internet resources, exploring tangents whenever you feel like it.

surge A short, sudden, and often destructive increase in line voltage. A voltage-regulating device, known as a surge suppressor, can protect computer equipment against surges.

See also power conditioning; spike; surge suppressor.

surge protector *See* surge suppressor.

surge suppressor A voltage-regulating device placed between the computer and the AC line connection that protects the computer system from power surges; also known as a surge protector.

See also power conditioning.

Sustainable Cell Rate Abbreviated SCR. In Asynchronous Transfer Mode (ATM), the maximum average cell rate that can be maintained over a virtual connection.

See also Asynchronous Transfer Mode; Available Cell Rate; Minimum Cell Rate; Peak Cell Rate.

SVC *See* switched virtual circuit.

SVID *See* System V Interface Definition.

S/WAN *See* Secure WAN.

swap To temporarily move a process from memory to disk, so that another

process can use that memory space. When space becomes available again, the process is swapped back into memory. This allows more processes to be loaded than there is physical memory space to run them simultaneously.

swapping The process of exchanging one item for another. In a virtual memory system, swapping occurs when a program requests a virtual memory location that is not currently in memory. Swapping may also refer to changing floppy or compact disks as needed when using a single disk drive.

swap space On a hard disk, a file used to store parts of running programs that have been swapped out of memory temporarily to make room for other running programs.

A swap file may be permanent, always occupying the same amount of hard-disk space, even though the application that created it may not be running, or temporary, created as and when needed.

See also permanent swap file; temporary swap file; virtual memory.

switch processor Abbreviated SP. In Cisco Systems 7000 series routers, the processor module that acts as the administrator for all bus activity; also known as the cisco-Bus controller.

See also ciscoBus; Cisco Extended Bus.

Switched 56 A switched four-wire digital data service available from a local exchange carrier that operates at 56Kbps.

See also Virtual Private Network.

Switched Multimegabit Data

Services Abbreviated SMDS. A high-speed metropolitan-area network service based on the 802.6 standard for use over T1 and T3 circuits. SMDS supports Ethernet, Token Ring, and FDDI (Fiber Distributed Data Interface) gateways.

See also Asynchronous Transfer Mode.

switched virtual circuit Abbreviated SVC. A communications circuit that is established for the duration of the session and then disconnected, much like a normal voice telephone call. SVCs are used extensively in X.25 networks

See also permanent virtual circuit.

Sybase, Inc. A leading supplier of database management and applications development software, including PowerBuilder, a rapid application development environment, and PowerJEnterprise, a Java development package.

For more information on Sybase, Inc., see www.sybase.com.

Symantec Corporation A leading developer of utility programs for the PC and the Macintosh whose products include application and system software, security and antivirus packages, remote productivity, and Internet access.

For more information on Symantec Corporation, see www.symantec.com.

Symmetrical Digital Subscriber Line

See Single-Line Digital Subscriber Line

symmetrical multiprocessing A multiprocessing design that assigns a task to a processor in response to system load as the

application starts running. This design makes for a much more flexible system than asymmetrical multiprocessing, in which the programmer matches a specific task to a certain processor while writing the program.

In symmetrical multiprocessing, the overall workload is shared by all processors in the system; system performance increases as more processors are added into the system. The drawback is that symmetrical multiprocessing operating systems are much harder to design than asymmetrical multiprocessing operating systems.

See also asymmetrical multiprocessing.

synchronization The timing of separate elements or events to occur simultaneously. In computer-to-computer communications, the hardware and software must be synchronized so that file transfers can take place.

See also asynchronous transmission; synchronous transmission.

Synchronous Data Link Control Abbreviated SDLC. The data-link protocol most widely used in networks that conform to IBM's SNA (Systems Network Architecture).

SDLC is a bit-oriented synchronous protocol that organizes information into well-defined units known as frames. SDLC is similar to the HDLC (High-level Data Link Control) protocol defined by the International Organization for Standardization (ISO).

See also data-link layer; High-level Data Link Control; Systems Network Architecture.

Synchronous Digital Hierarchy Abbreviated SDH. A set of fiber-optic-based standards from the ITU for use with SONET and ATM in Europe

synchronous DRAM Abbreviated SDRAM. A high-speed memory technology, faster than EDO RAM; used in workstations and servers.

synchronous graphics RAM Abbreviated SGRAM. A type of high-speed dynamic RAM used in video adapters.

Synchronous Optical Network Abbreviated SONET. A set of fiber-optic-based communications standards with transmission rates from 51.84Mbps to 13.22Gbps.

First proposed by Bellcore in the mid-1980s, SONET was standardized by ANSI, and the ITU adapted SONET in creating the worldwide Synchronous Digital Hierarchy (SDH) standard.

SONET uses synchronous transmissions in which individual channels (called tributaries) are merged into higher-level channels using time-division multiplexing techniques. Data is carried in frames of 810 bytes, which also includes control information known as the overhead.

See also Optical Carrier; Synchronous Digital Hierarchy.

SYNCHRONOUS DATA LINK CONTROL

data start field	address field	control field	information field	cyclic redundancy check	data redundancy check	end flag
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synchronous transmission A transmission method that uses a clock signal to regulate data flow.

In synchronous transmissions, frames are separated by equal-sized time intervals. Timing must be controlled precisely on the sending and the receiving computers. Special characters are embedded in the data stream to begin synchronization and to maintain synchronization during the transmission, allowing both computers to check for and correct any variations in timing.

See also asynchronous transmission.

syntax The formal rules of grammar as they apply to a specific programming language or operating system command; in particular, the exact sequence and spelling of command elements required for the command to be interpreted correctly.

syntax error An error in the use of a programming language or operating system command syntax, such as misspelling a keyword or omitting a required space.

SyQuest Technology, Inc. A leading manufacturer of removable storage media, particularly the SCSI-based removable hard disk available for the PC and the Macintosh. For more information on SyQuest Technology, Inc., see www.syquest.com.

System III The release of Unix from AT&T prior to the release of System V.

System III was the first version of Unix to be ported to the Intel family of processors and formed the basis for SCO's release of XENIX.

System IV A version of Unix from AT&T that was never released outside the company, mostly to avoid confusion with the 4.xBSD series of products.

System V The last version of Unix from AT&T, pronounced "System Five." The latest release is known as System V Release 4.2, often abbreviated SVR4 or SVR4.2.

System V Interface Definition Abbreviated SVID. A set of documents released by AT&T that defined the Unix System V interfaces and operating system calls.

system administration The day-to-day administrative and management tasks performed by the system administrator, including:

- Starting up and shutting down the system
- Setting the system time and date
- Assigning and changing passwords
- Adding and removing users and groups

- Installing, upgrading, and removing application packages and installing operating system upgrades
- Backing up the system, storing archives off location, and restoring backups as needed
- Installing and configuring new hardware such as printers, storage devices, and communications systems
- Monitoring system performance and making tuning adjustments as necessary

See also system administrator.

System Administration: Informix Dynamic Server A certification from Informix designed for individuals who configure, maintain, and tune Informix Dynamic Server databases.

See also Database Specialist: Informix-4GL Certified Professional.

system administrator The person charged with the responsibility of managing the system.

In a very large system, the system administrator may in fact be several people or even a department; if you are running Linux on your system at home, you have to be your own system administrator.

See also system administration.

SYSTEM directory In Novell NetWare, the SYS:SYSTEM directory created during installation. This directory contains NetWare operating system files and directories, as well as NetWare Loadable Modules (NLMs) and network administrator utilities.

System Fault Tolerance Abbreviated SFT. In Novell NetWare, a method of duplicating data on several hard disks so that if one disk fails, the data is still available from another disk.

Several levels of hardware and software SFT are available in NetWare, with each level of redundancy decreasing the possibility of catastrophic data loss. For example, SFT I includes HotFix redirection, and SFT II adds disk duplexing and disk mirroring. SFT III uses duplicate servers so that all transactions are recorded on both; if one fails, the other can take over.

See also disk striping; redundant array of inexpensive disks; Transaction Tracking System.

System Information In Microsoft Windows 2000, a system utility that reports detailed hardware and software configuration, hardware resource information, and a system summary. System Information is a Microsoft Management Console (MMC) snap-in located in Computer Management.

See also Computer Management; Microsoft Management Console; snap-in.

system management mode Abbreviated SMM. In the Intel family of microprocessors, a low power consumption mode used to conserve battery power. All recent Intel processors have SMM and so are suitable for use in portable, battery-powered computers.

system log In Microsoft Windows 2000, a default system log that records events generated by Windows 2000 system

services, device drivers, and other system components. You can use the Event Viewer to look at the contents of the system log.

See also application log; Event Viewer; security log.

System Monitor In Microsoft Windows 2000, a network and server monitoring tool that displays resource use for selected system-level components. Replaces the Windows NT Performance Monitor.

See also alert log; counter log; Performance Monitor; trace log.

System Object Model Abbreviated SOM. A specification from IBM that allows objects created in different environments to communicate.

See also Distributed System Object Model.

Systems Application Architecture

Abbreviated SAA. A set of IBM standards, first introduced in 1987, that defines a consistent set of interfaces for future IBM software. Three standards are defined:

- **Common User Access (CUA)** A graphical user interface definition for products designed for use in an object-oriented operating environment. The OS/2 desktop follows CUA guidelines in its design, and Microsoft Windows implements certain CUA features.

- **Common Programming Interface (CPI)** A set of application programming interfaces (APIs) designed to encourage independence from the underlying operating system. The standard database query language is Structured Query Language (SQL).

- **Common Communications Support (CCS)** A common set of communications protocols that interconnect SAA systems and devices.

Systems Network Architecture Abbreviated SNA. IBM's proprietary terminal-to-mainframe protocol, introduced in 1974.

SNA describes a seven-layer system, with each layer building on the services provided by the previous layer. Devices on an SNA system are usually connected using the SDLC protocol, running over serial lines. SNA is not compatible with the seven-layer OSI Reference Model.

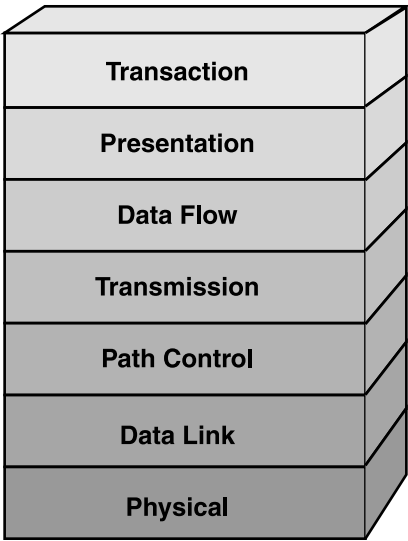
Systems Performance Evaluation

Cooperative benchmarks Abbreviated SPEC benchmarks. A set of ten standardized tests designed to measure workstation performance.

Six of the tests evaluate floating-point performance, and four tests concentrate on integer performance.

Results are reported as SPECmarks, a geometric mean of all ten scores.

SYSTEMS NETWORK ARCHITECTURE



T

T See *tera-*.

T1 A long-distance, point-to-point circuit, providing 24 channels of 64Kbps, giving a total bandwidth of 1.544Mbps.

The standard T1 frame is 193 bits long, made up of twenty-four 8-bit voice samples and one synchronization bit. It transmits 8000 frames per second. When a T1 service is made available in single 64Kbps increments, it is known as fractional T1.

In Europe, the comparable circuit is known as E-1, and it has a speed of 2.054Mbps. T1 has been superseded by the CCITT DS-1 designation.

See also fractional T1; T-carrier.

T1 multiplexer A statistical multiplexer that divides the 1.544Mbps T1 bandwidth into 24 separate 64Kbps channels of digitized data or voice.

See also fractional T1; T1.

T1 small aperture terminal Abbreviated TSAT. A small satellite terminal used for digital communications that can handle T1 data rates of up to 1.544Mbps.

See also fractional T1; T1.

T.120 A group of communications and applications protocols that support real-

time, multipoint data communications over local area networks and ISDN, dial-up, and Internet connections. T.120 became well known after Microsoft incorporated it into the NetMeeting package.

See also H.323; Microsoft NetMeeting.

T2 A long-distance, point-to-point communications service, providing up to 4 T1 channels.

T2 offers 96 channels of 64Kbps, for a total bandwidth of 6.3Mbps. T2 is not available commercially, although it is used within telephone company networks.

See also T1; T-carrier.

T3 A long-distance, point-to-point communications service, providing up to 28 T1 channels.

T3 can carry 672 channels of 64Kbps, for a total bandwidth of 44.736Mbps, and is usually available over fiber-optic cable. T3 is used almost exclusively by AT&T and the regional telephone operating companies, although certain large private corporations are using T3 with digital microwave or fiber-optic networks.

In Europe, T3 has been superseded by the CCITT DS-3 designation.

See also T1; T-carrier.

T4 A long-distance, point-to-point communications service, providing up to 168 T1 channels.

T4 can carry 4032 channels of 64Kbps, for a total bandwidth of 274.176Mbps. T4 can be used for both digitized voice and data transmission.

See also T1; T-carrier.

table In a relational database system, a table is comparable to a database file, but is more highly structured.

The organization of a table is logical, not physical. Each row (or record) in a table contains a unique key, or primary key, so that any item of data in the table can be retrieved by referring only to that key.

Through the process known as normalization, all data items in a row are made to depend only on this primary key. View and data dictionaries in a relational database take the form of two-dimensional tables.

tag An element in HTML used to annotate a document. A tag is text enclosed by angle brackets that tells the client Web browser how to display each part of the document. For example, the tag <H1> indicates the start of a level one heading, and the tag </H1> indicates the end of a level one heading.

See also element; HyperText Markup Language.

tag switching A technology from Cisco Systems, Inc., that integrates data-link layer switching with network layer routing for large-scale networks, allowing the

integration of ATM switches into the Internet and implementing routing on top of those switches.

See also Asynchronous Transfer Mode; IP over ATM; IP switching.

tap A connector that attaches to a cable without blocking the passage of information along that cable; a connection onto the main transmission medium of the network.

tape cartridge A self-contained tape storage module, containing tape much like that in a video cassette. Tape cartridges are primarily used to back up hard-disk systems.

See also DC-2000; quarter-inch cartridge; Zip drive.

tape drive A computer peripheral device that reads from and writes to magnetic tape.

The drive may use tape on an open reel or from an enclosed tape cartridge. Because tape-management software must search from the beginning of the tape every time it wants to find a file (a process called sequential access), tape is too slow to use as a primary storage system; however, tapes are frequently used to back up hard disks.

See also streaming tape.

TAPI *See* Telephony API.

tar In Unix, a utility program that can create, list, add to, and retrieve from an archive file, which is usually stored on tape. The archive file has the filename extension .tar.

The archive created by tar is not compressed and can be further processed by the Unix

compress (extension .Z) or gzip (extension .gz) utilities to give compound extensions such as filename.tar.Z or filename.tar.gz.

task Any independent running program and the set of system resources that it uses. A task may be an operating system process or part of an application.

See also context switching; multitasking; task switching.

Task Manager In Microsoft Windows NT, an application that allows a user to manually inspect the tasks running on the computer and select individual tasks for shutdown. Task Manager is opened by pressing Ctrl+Alt+Del.

task switching To change from one running program to another quickly, either at the direction of the operating system or at the request of the user.

TB *See* terabyte.

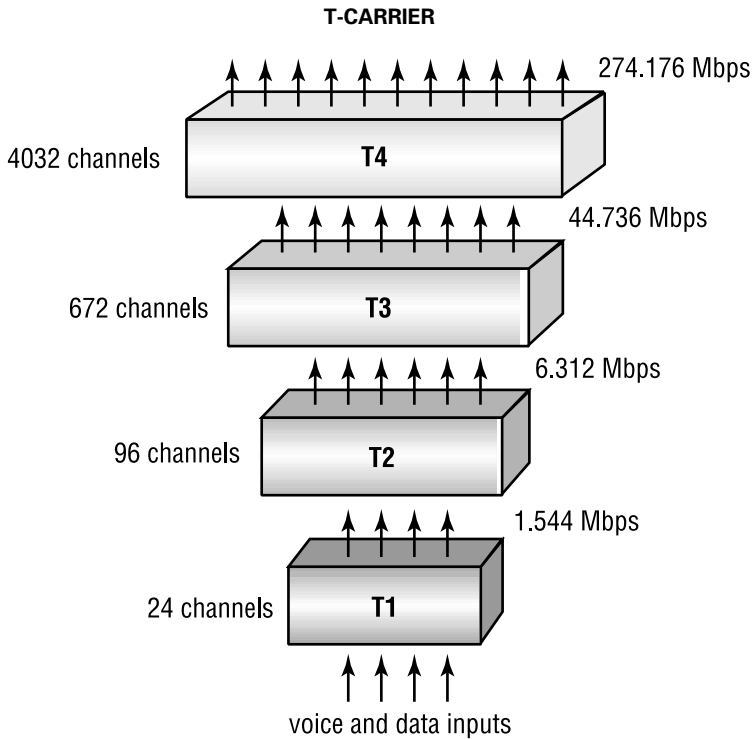
T-carrier A digital communications service from a common carrier for voice or data transmission.

The four-level, time-division multiplexing specification for the U.S. telephone system allows the bit stream of the smaller carriers to be multiplexed into the larger ones.

The following are the four service levels:

- **T1** Provides 24 channels of 64Kbps, giving a total bandwidth of 1.544Mbps. When a T1 service is made available in single 64Kbps increments, it is known as fractional T1.
- **T2** The equivalent of 4 T1 services, T2 offers 96 channels of 64Kbps, for a total bandwidth of 6.3Mbps.
- **T3** The equivalent of 28 T1 circuits, T3 offers 672 channels of 64Kbps, for a total bandwidth of 44.736Mbps. T3 is available commercially, but is not often used for LANs.
- **T4** The equivalent of 168 T1 circuits, T4 provides 4,032 channels of 64Kbps, for a total bandwidth of 274.176Mbps.

See also T1; T2; T3; T4.



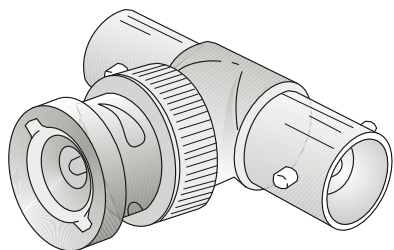
Tcl See Tool Command Language.

TCM See trellis-coded modulation.

TCO See total cost of ownership.

T-connector A T-shaped device, used with coaxial cable, that connects two thin Ethernet cables and also provides a third connector for the network interface card.

T-CONNECTOR



TCP See Transmission Control Protocol.

TCP/IP See Transmission Control Protocol/Internet Protocol.

TCP ports In a TCP/IP network when a computer connects with another computer to access a specific service, and end-to-end

connection is established and a socket is set up at each end of the connection. This socket is created at a particular port number, depending on the application in use. Table T.1 lists the port numbers used for some of the common protocols.

TABLE T.1 PORT NUMBERS FOR COMMON PROTOCOLS

Port Number	Protocol
21	File Transfer Protocol
23	Telnet
25	Simple Mail Transfer Protocol
70	Gopher
79	Finger
80	Hypertext Transfer Protocol
119	Network News Transfer Protocol

TDM See time-division multiplexing.

TDMA See Time Division Multiple Access.

TDR See time-domain reflectometer.

Technical and Office Protocol Abbreviated TOP. An Ethernet implementation for use in an engineering environment, developed by Boeing Corporation.

See also Manufacturing Automation Protocol.

telecommunications A general term for the electronic transmission of all forms of information, including digital data, voice, fax, sound, and video, from one location to another over some form of communications link.

Telecommunications Industry Association Abbreviated TIA. An important trade group, active in the areas of standards development, trade shows, international marketing, and legislative efforts.

The TIA often works in close association with the Electronic Industries Association (EIA).

For more information on TIA, see www.tiaonline.org.

See also Electronic Industries Association.

telecommuting Working at home on a computer connected to the office by modems and telephone lines instead of commuting to the office.

Telecommuting saves time, cuts down on automobile use and pollution, and decreases stress. Some local and state governments actively encourage telecommuting to keep

the number of commuters as low as possible. Most studies indicate that home workers are happier and more productive. However, some jobs do not lend themselves to telecommuting; welding and brain surgery are very difficult to do over the phone.

See also small office/home office.

teleconferencing The use of audio, video, animation, and application sharing, linked by a communications channel, to allow widely separated individuals to take part in a discussion or meeting.

Desktop video and chalkboard programs such as Microsoft NetMeeting are becoming more and more common, and groupware applications such as Lotus Notes are helping people work together.

Telephony API Abbreviated TAPI. A standard telephone interface for Microsoft Windows, developed by Intel and Microsoft, designed to allow applications to set up and control calls.

TAPI does not define the method of data transmission used once a call is in progress. It is completely independent of the telephone network itself and can be used on public-switched telephone networks, ISDN, and IP networks.

See also Telephony Services API.

Telephony Services API Abbreviated TSAPI. A standard telephone interface developed by AT&T and Novell, designed to allow applications to set up and control calls. TSAPI is available as a NetWare Loadable Module for NetWare 4.01 and later.

TSAPI requires that the PBX (private branch exchange) be linked to a server and so has greater control over the call than does Telephony API.

See also Telephony API.

Telnet A terminal emulation protocol, part of the TCP/IP suite of protocols and common in the Unix world, that provides remote terminal-connection services.

The most common terminal emulations are for Digital Equipment Corporation (DEC) VT-52, VT-100, and VT-220 terminals, although many companies offer additional add-in emulations.

See also tn3270.

temporary swap file A swap space that is created every time it is needed. A temporary swap file can consist of several discontinuous pieces of hard-disk space. A temporary swap file does not occupy hard-disk space if the application that created it is not running.

See also swap space; permanent swap file; virtual memory.

ter A term describing a tertiary CCITT recommendation, an alternative or extension to the primary or secondary recommendation.

See also bis.

tera- Abbreviated T. A prefix that means 10^{12} in the metric system, 1,000,000,000,000; commonly referred to as one trillion in the American numbering system, and one million million in the British numbering system.

terabyte Abbreviated TB. In computing, usually 2^{40} , or 1,099,511,627,776 bytes. Terabytes are used to represent extremely large hard-disk capacities.

terminal emulation A method of operation or software that makes a personal computer or a workstation act like a terminal attached to a mainframe, usually for the purpose of telecommunications. Communications programs often include popular emulations, such as ANSI, VT-52, VT-100, VT-200, and TTY.

See also Telnet; tn3270.

Terminal services In Microsoft Windows 2000 Server, an optional component that provides multiple user sessions from the same machine; also known as thin-client computing, multiuser Windows, or server-based computing.

With Terminal services installed, the major portion of all application processing takes place on the server, rather than on the client, allowing you to use cheaper client hardware, install application software on the server rather than on every client, and administer the server from a remote console.

terminate-and-stay-resident

program Abbreviated TSR. A small accessory or utility program that stays loaded in memory, even when it is not actually running. A TSR can be invoked quickly to perform a specific task.

TSR's are often used with single-tasking operating systems such as MS-DOS.

terminator A device attached to the last peripheral device in a series or the last node on a network.

For example, the last device on a SCSI bus must terminate the bus; otherwise, the bus will not perform properly. A 50-ohm resistor is placed at both ends of an Ethernet cable to prevent signals reflecting and interfering with the transmission.

See also active termination; forced perfect termination; passive termination.

T_EX A typesetting language developed by Donald E. Knuth of Stanford University, capable of professional-quality typeset text, particularly of mathematical equations and scientific, Japanese, Chinese, Cyrillic, and Arabic text.

T_EX is not easy for the casual user to master, but several packages are available containing macros designed to solve specific typesetting problems.

text file *See* ASCII file.

text mode A mode in which the computer displays characters on the screen using the built-in character set, but does not show any graphics characters or a mouse pointer. Also known as character mode.

TFTP *See* Trivial File Transfer Protocol.

thick Ethernet Connecting coaxial cable used on an Ethernet network; also known as thicknet.

The cable is 1 centimeter (0.4 inch) thick, almost as thick as your thumb, and can be used to connect network nodes up to a distance of approximately 1006 meters (3300 feet). Thick Ethernet is primarily used for facility-wide installations.

See also thin Ethernet.

thicknet *See* thick Ethernet.

thin client In a client/server network, a client that requires relatively small amounts of local memory and hard-disk space and leaves most of the processing to the server. Sometimes called a Windows terminal.

In some cases, the client operating system, as well as the applications the client runs and the data it manipulates, all reside on the server.

See also NetPC; network computer; total cost of ownership; Zero Administration for Windows.

thin-client computing *See* Terminal services.

thin Ethernet Connecting coaxial cable used on an Ethernet network; also known as thinnet.

The cable is 5 millimeters (0.2 inch) thick, about as thick as your little finger, and can be used to connect network nodes up to a distance of approximately 165 meters (500 feet). Thin Ethernet is primarily used for office installations.

See also thick Ethernet.

thinnet *See* thin Ethernet.

thrashing An excessive amount of disk activity that causes a virtual memory system to spend all its time swapping pages in and out of memory, and no time executing the application.

Thrashing can be caused when poor system configuration creates a swap file that is too small or when insufficient memory is installed in the computer. Increasing the size of the swap file or adding memory are often the best ways to reduce thrashing.

thread 1. A concurrent process that is part of a larger process or program. In a multitasking operating system, a single program may contain several threads, all running at the same time. For example, one part of a program can be making a calculation while another part is drawing a graph or a chart.

2. A connected set of postings to a Usenet newsgroup. Many newsreaders present postings as threads rather than in strict chronological order.

See also multiprocessing; newsgroup; newsreader; Usenet.

threaded newsreader An application used to read the articles posted to Usenet newsgroups.

A threaded newsreader groups the newsgroup posts into threads of related articles, whereas unthreaded newsreaders present them in their original order of posting. Of the two types, threaded newsreaders are much more convenient to use.

See also newsgroup; newsreader; Usenet.

throughput A measure of the data-transfer rate through a complex communications or networking scheme.

Throughput is considered an indication of the overall performance of the system. For example, the throughput of a server depends on the processor type, operating system in use, hard disk capacity, network interface card in use, and the size of the data transfer buffer.

In communications, throughput is usually measured as the number of bits or packets processed each second.

See also bandwidth.

TIA *See* Telecommunications Industry Association.

TIA/EIA structured cabling standards Standards specified by the Electronics Industry Association/Telecommunications Industries Association (EIA/TIA), including:

- ANSI/EIA/TIA-568-1991 Commercial Building Telecommunications Wiring.
- EIA/TIA TSB-36 Additional Cable Specifications for UTP Cables. 1991.
- EIA/TIA TSB-40 Telecommunications Systems Bulletin—Additional Transmission Specifications for UTP Connecting Hardware. 1992.
- ANSI/EIA/TIA-568A 1995 revises the original 568 document and adds material from TSB-36 and TSB-40.
- ANSI/EIA/TIA-569-1990 Commercial Building Standard for Telecommunications Pathways and Spaces.
- ANSI/EIA/TIA-570-1991 Residential and Light Commercial Telecommunications Wiring Standard.
- ANSI/EIA/TIA-606-1993 Administration Standard for the Telecommunications Infrastructure of Commercial Buildings.
- ANSI/EIA/TIA-607-1994 Commercial Building Grounding and Bonding Requirements for Telecommunications.

Local codes and standards may impose additional requirements.

See also cabling standards.

ticket A token within the Kerberos authentication system that contains the user's name and address, as well as the service the user requested, security information, a time deadline, and other authentication information.

See also authentication; Kerberos.

tie line A private circuit, leased from a communications carrier, connecting two or more points in a single organization.

Time Division Multiple Access Abbreviated TDMA. A technique used to allocate multiple channels on the same frequency on a cellular phone or satellite communications system.

See also Code Division Multiple Access.

time-division multiplexing Abbreviated TDM. A method of sharing a transmission channel by dividing the available time equally between competing stations. At the receiving end, the different signals are merged back into their individual streams.

See also frequency-division multiplexing; inverse multiplexing; statistical multiplexing.

time-domain reflectometer Abbreviated TDR. A diagnostic tool used to detect cabling faults.

A TDR calculates the length of a cable by measuring the time it takes for a reflected pulse to return to the TDR and then multiplying that by the nominal velocity of propagation.

time to live Abbreviated TTL. A mechanism used to ensure that misdirected

information doesn't end up traveling a TCP/IP network or the Internet for all eternity.

Each IP datagram contains a TTL value; once this value reaches zero, the datagram is simply assumed to be undeliverable and is discarded.

timeout Many procedures require a device to respond or reply to an inquiry within a certain period of time; if the device does not respond, a timeout condition occurs, thus preventing the procedure from hanging up the computer.

Timeouts are also used in communications to detect transmission failures. Some timeouts are fixed, such as the amount of time during which an operating system will attempt to access a modem or printer; others can be specified by the user.

time-slice multitasking A form of multitasking in which the operating system assigns the same small time period to each process in turn.

See also cooperative multitasking; preemptive multitasking.

time stamp An identification code that includes the time that an event took place.

Most operating systems add a time stamp to indicate a file's create time. Automatic error logging or security auditing processes often add a time stamp to critical events such as changes to passwords or accounts.

time synchronization A method of synchronizing time across all servers on the network so that all servers report the same time.

tn3270 A special version of the Telnet program specifically designed for use with large IBM computers using 3270 and 327x series terminals. Most of the computers on the Internet use Unix, but if you ever encounter an IBM mainframe, you will definitely need tn3270.

So how do you know when to use tn3270 rather than Telnet? It's time to load up tn3270 if you try to connect to an Internet host with Telnet and one of the following happens:

- The on-screen messages are all in upper-case letters rather than the usual Unix mix of uppercase and lowercase letters.
- You see VM or MVS anywhere in the login message. These are both names of IBM operating systems.
- Your session is aborted before it really gets started.

See also Telnet.

token passing A network access method that uses a circulating electronic token to prevent multiple nodes from transmitting on the network simultaneously.

Before a node can transmit, it must be in possession of the token. Fiber Distributed Data Interface (FDDI), Token Ring, and Token Bus networks all use token passing to avoid packet collisions.

See also Carrier Sense Multiple Access/Collision Detection; token-ring network.

Token Ring network IBM's implementation of the token-ring network architecture, which uses a token-passing protocol transmitting at 4 or 16Mbps.

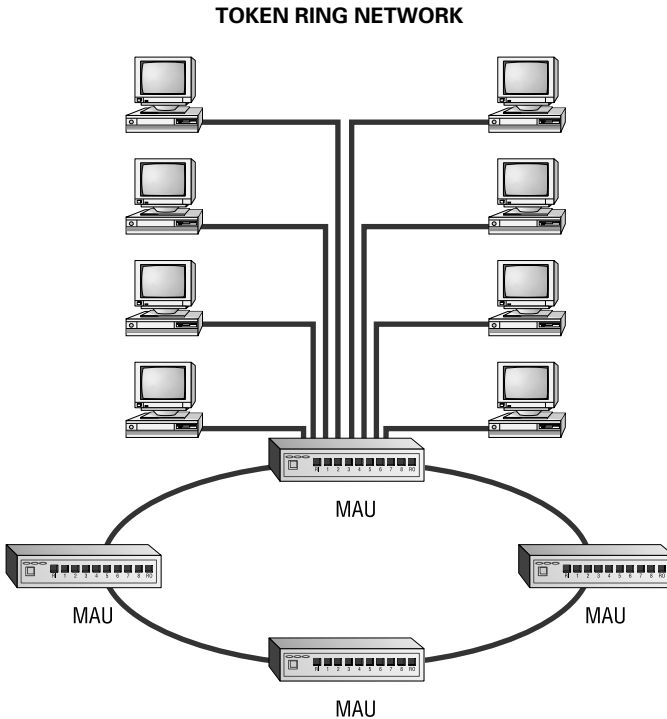
Using standard telephone wiring, a Token Ring network can connect a maximum of 72 devices; with shielded twisted-pair (STP) wiring, each ring can support a maximum of 256 nodes. Although it is based on a closed-loop ring structure, a Token Ring network uses a star-shaped cluster of as many as eight nodes, all attached to the same wiring concentrator or Multistation Access Unit (MAU). The MAUs are then connected to the main ring circuit.

A Token Ring network can include personal computers, minicomputers, and mainframes. The IEEE 802.5 standard defines token-ring networks.

token-ring network A LAN with a ring structure that uses token passing to regulate traffic on the network and avoid collisions.

On a token-ring network, the controlling network interface card generates a token that controls the right to transmit. This token is continuously passed from one node to the next around the network. When a node has information to transmit, it captures the token, sets its status to busy, and adds the message and the destination address. All other nodes continuously read the token to determine if they are the recipient of a message. If they are, they collect the token, extract the message, and return the token to the sender. The sender then removes the message and sets the token status to free, indicating that it can be used by the next node in sequence.

See also 802.5; Carrier Sense Multiple Access/Collision Detection; Token Ring network.



Tool Command Language Abbreviated Tcl, pronounced “tickle.” Developed by John Ousterhout while at the University of California at Berkeley, Tcl is a general-purpose extensible scripting language supplied as a C library on Unix systems.

Tcl is also available in versions for MS-DOS, Windows, and the Macintosh.

See also Perl; REXX; scripting.

TOP *See* Technical and Office Protocol.

top-level domains On the Internet, the highest category of host name, which either

signifies the type of institution or the country of its origin.

In the United States, the most common top-level domains include:

- **com** A commercial organization. Most companies end up in this category.
- **edu** An educational establishment such as a university.
- **gov** A branch of the U.S. government.
- **int** An international organization such as NATO or the United Nations.
- **mil** A branch of the U.S. military.

- **net** A network organization.
- **org** A nonprofit organization.

Most countries also have unique domains named after their international abbreviations; for example, *ca* represents Canada, *uk* represents Great Britain, and *jp* represents Japan.

See also domain; domain name: Domain Name Service.

topology The map of a network. Physical topology describes where the cables are run and where the workstations, nodes, routers, and gateways are located. Networks are usually configured in bus, ring, star, or mesh topologies. Logical topology refers to the paths that messages take to get from one user on the network to another.

TOPS A local-area network (LAN) from TOPS Corporation that uses the LocalTalk protocol to connect Apple computers, PCs, and Sun workstations.

Toshiba Corporation One of the largest manufacturers of electrical equipment in the world, a major supplier of notebook computers, and a co-developer of the digital video disc.

For more information on Toshiba Corporation, see www.toshiba.com.

total cost of ownership Abbreviated TCO. A term first used by the Gardener Group in an attempt to quantify the real cost of a particular computer solution.

TCO encompasses the direct costs of the hardware and the software required and then adds in costs for maintenance and support, costs for the users performing their own technical support rather than their official job, and system productivity costs.

See also NetPC; network computer; thin client; Zero Administration for Windows.

TP *See* twisted-pair cable.

trace log In Microsoft Windows 2000, a System Monitor log used to monitor events such as disk input/output or page faults; when the event occurs, it is logged into the trace log.

See also alert log; counter log; page fault; System Monitor.

tracert A utility used on TCP/IP networks to trace the route that datagrams take between the server and another system. As *tracert* also tells you how long each hop takes, it can be a useful tool in identifying system trouble spots.

trackball A device used for pointing, designed as a space-saving alternative to the mouse.

A trackball contains a movable ball that you rotate with your fingers to move the cursor on the screen. Because it does not need the area of flat space that a mouse needs, trackballs are popular with users of portable computers. The Apple PowerBook includes a trackball as part of the keyboard case, Microsoft has released a small trackball that clips onto the side of a laptop computer, and IBM has developed a dual-button, touch-sensitive pointing stick called the TrackPoint.

traffic The flow of messages and data carried by a communications channel or link. Traffic on a data network is normally measured in bits transferred in a given time period.

transaction

transaction A single activity within a computer system, such as an entry into an airline reservation database, that is executed in real time rather than as a batch process.

transaction processing A processing method in which transactions are executed immediately when they are received by the system, rather than at some later time as in batch-processing systems. Airline reservation databases and automatic teller machines are examples of transaction-processing systems.

See also on-line transaction processing; roll back; roll forward; two-phase commit.

Transaction Tracking System Abbreviated TTS. A fault-tolerant feature of Novell NetWare that maintains the integrity of databases by backing out or rolling back incomplete transactions that result from a failure in a network component.

transceiver A contraction of transmitter/receiver. A device capable of both transmitting and receiving data.

The data may be located on the network interface card that connects a workstation to a network, or it may be on a separate device. A transceiver can convert between an AUI (Attachment Unit Interface) Ethernet connection and another type of cabling, such as fiber-optic, coaxial, or unshielded twisted pair (UTP).

transfer rate *See* data-transfer rate.

transient *See* surge.

transitive trust *See* two-way transitive trust.

Transmission Control Protocol Abbreviated TCP. The transport-level protocol used in the TCP/IP suite of protocols. It works above IP in the protocol stack and provides reliable data delivery over connection-oriented links.

TCP adds a header to the datagram that contains the information needed to get the datagram to its destination. The source port number and the destination port number allow data to be sent back and forth to the correct processes running on each computer. A sequence number allows the datagrams to be rebuilt in the correct order in the receiving computer, and a checksum verifies that the data received is the same as the data sent. In addition to these fields, the TCP header contains the following information:

- **Acknowledgment number** Indicates that the data was received successfully. If the datagram is damaged in transit, the receiver discards the data and does not send an acknowledgment to the sender. After a specified timeout expires, the sender retransmits data for which no acknowledgment has been received.
- **Offset** Specifies the length of the header.
- **Reserved** Variables set aside for future use.
- **Flags** Indicate that this packet is the end of the data or that the data is urgent.

- **Window size** Provides a way to increase packet size, which can improve efficiency in data transfers.
- **Urgent pointer** Gives the location of urgent data.
- **Options** Reserved for future use or for special options as defined by the protocol.
- **Padding** Ensures that the header ends on a 32-bit boundary.

The data immediately follow this header information.

See also Internet Protocol; Transmission Control Protocol/Internet Protocol; User Datagram Protocol.

Transmission Control Protocol/

Internet Protocol Abbreviated TCP/IP. A set of communications protocols first developed by the Defense Advanced Research Projects Agency (DARPA) in the late 1970s. The set of TCP/IP protocols encompasses media access, packet transport, session communications, file transfer, e-mail, and terminal emulation.

TCP/IP is a widely published open standard, and while completely independent of any specific hardware or software company, it is supported by a huge number of vendors and is available on many different computers, from PCs to mainframes, run-

ning many different operating systems. Many corporations, universities, and government agencies use TCP/IP, and it is also the basis of the Internet.

TCP/IP is separated from the network hardware and will run over Ethernet, token-ring, X.25 networks, and dial-up connections. It is a routable protocol, so datagrams can be sent over specific routes, and it has reliable and efficient data-delivery mechanisms.

TCP/IP uses a common expandable addressing scheme, so any system can address any other system, even in a network as large as the Internet, and new networks can be added without service disruptions.

The popularity that the TCP/IP family of protocols enjoys today did not arise just because the protocols were available or even because the U.S. government mandated their use. They are popular because they are robust, solid protocols that solve many of the most difficult networking problems and do so in an elegant and efficient way.

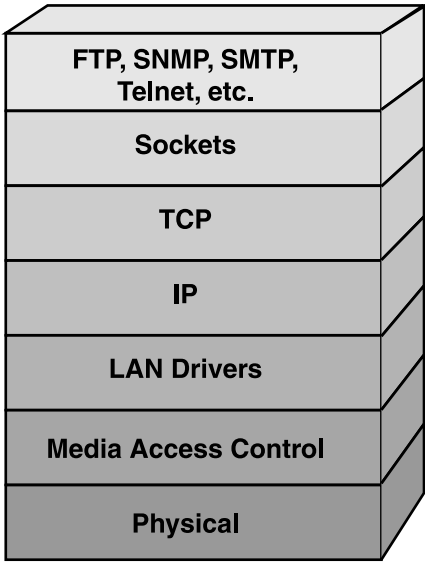
See also Address Resolution Protocol; File Transfer Protocol; Hypertext Transfer Protocol; Internet Control Message Protocol; Internet Mail Access Protocol; Internet Protocol; Post Office Protocol; Simple Mail Transfer Protocol; Simple Network Management Protocol; Telnet; tn3270; User Datagram Protocol.

transmission medium

TCP HEADER



TRANSMISSION CONTROL PROTOCOL/INTERNET PROTOCOL



transmission medium The physical cabling used to carry network information, such as fiber-optic, coaxial, shielded twisted-pair (STP), and unshielded twisted-pair (UTP) cabling.

transmission mode The manner in which communications can take place between a sender and a receiver.

Several modes are defined, as follows:

- **Simplex** Communications can go in only one direction, so the sender can use the whole of the available bandwidth.
- **Half-duplex** Communications can go in two directions, but only in one direction at a time.
- **Full-duplex** Communications can go in two directions simultaneously.
- **Echo-plex** A rare mode in which characters are retransmitted to the sender for error-checking purposes.

Transmit Data Abbreviated TXD. A hardware signal defined by the RS-232-C standard that carries information from one device to another.

See also Receive Data.

transport layer The fourth of seven layers of the OSI Reference Model for computer-to-computer communications. The transport layer defines protocols for message structure and supervises the validity of the transmission by performing some error checking.

See also OSI Reference Model.

trap *See* software interrupt.

trap door An entry point in a computer network, through which an intruder can gain access without authentication.

tree In Microsoft Active Directory, a hierarchy of domains linked via trust rela-

tionships that share the same namespace, Directory Schema, and Global Catalog.

See also Active Directory; Directory Schema; Global Catalog.

Tree object In Novell Directory Services (NDS), the hierarchical organization of all the objects on the network into a single structure.

See also container object; Novell Directory Services; Organization object.

trellis-coded modulation Abbreviated TCM. A form of quadrature amplitude modulation used in modems that operate at 9600 bits per second or higher. TCM encodes data as a set of bits associated with both phase and amplitude changes.

Trivial File Transfer Protocol Abbreviated TFTP. A little-used and simplified version of the TCP/IP file transfer protocol that does not include password protection.

Because it has no security associated with it, most system administrators do not support its use and recommend File Transfer Protocol instead.

See also File Transfer Protocol.

Trojan Horse A type of computer virus that pretends to be a useful program, such as a game or a utility, to entice you to use it, when in reality it contains special code that will intentionally damage any system onto which it is loaded.

See also logic bomb.

trusted domain In Microsoft Windows NT Server, a domain that a trusting domain will allow to authenticate logons.

See also trusting domain; trust relationship.

trustee In Novell NetWare, a user or group object that has been granted rights to work with a directory, an object, or a file.

See also Public trustee; trustee assignment.

trustee assignment In Novell NetWare, a mechanism that determines how a user can access an object, a directory, or a file. Also known as trustee rights.

For example, trustee rights regulate whether a user can read a file, change it, change its name, delete it, or control other users' trustee rights to it. Trustee rights are assigned to individual users, and one user's rights can be different from another user's rights to the same directory

See also Public trustee; trustee.

trustee rights *See* trustee assignment.

trusting domain In Microsoft Windows NT Server, a domain that lets users and groups in the trusted domain use its resources.

See also trusted domain; trust relationship.

trust relationship In Microsoft Windows NT Server, a link between domains that allows pass-through authentication, in which a trusting domain allows the logon authentication of a trusted domain.

With the right trust relationships, a user with one user account in one domain has

the potential to access the whole network. Global groups and user accounts defined in the trusted domain can be assigned rights and permissions in a trusting domain, even though those accounts do not exist in the trusting domain's directory database.

See also authentication; pass-through authentication; trusted domain; trusting domain; user account.

TSAPI *See* Telephony Services API.

TSAT *See* T1 small aperture terminal.

TSR *See* terminate-and-stay-resident program.

TTL *See* time to live.

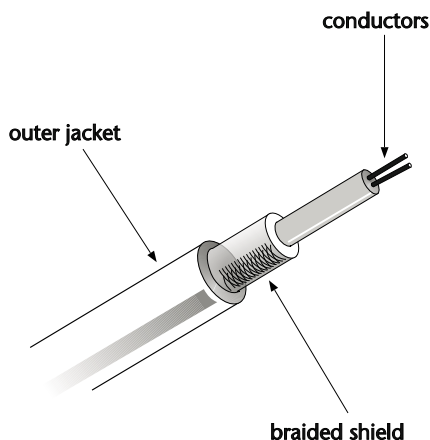
TTS *See* Transaction Tracking System.

tunneling The encapsulation of one protocol within another, often used to transport a local-area network protocol across a backbone that does not support that particular protocol.

Tunneling is also used to create pseudo-connections across connectionless networks such as the Internet and may be referred to as protocol encapsulation or synchronous pass-through.

See also encapsulation; Point-to-Point Tunneling Protocol.

twinaxial cable A cable with two coaxial cables inside a single insulating shield. Twinaxial cable is used with IBM AS/400 minicomputers.

TWINAXIAL CABLE

twisted-pair cable Abbreviated TP. Cable that comprises two or more pairs of insulated wires twisted together, at six twists per inch.

In twisted-pair cable, one wire carries the signal and the other is grounded. The cable may be shielded or unshielded. Telephone wire installed in modern buildings is often twisted-pair wiring.

two-phase commit A method used in transaction processing to ensure data is posted to shared databases correctly by dividing the writing of data into two steps.

Each of the steps must receive a verification of completeness from the shared databases; otherwise, the transaction-processing system rolls back the transaction and tries again.

See also on-line transaction processing; roll back; roll forward; transaction processing.

two-way transitive trust In Microsoft Windows 2000 Server, an automatic trust relationship created between two Windows 2000 domains.

If domain A trusts domain B, and domain B trusts domain C, then domain A trusts domain C, and domain C also trusts domain A. In Windows NT, all trust relationships were created and managed manually.

See also downlevel trust; explicit trust; shortcut trust;

TXD *See* Transmit Data.

Type 1–9 cable IBM Cabling System specifications, as follows:

- **Type 1** Shielded, twisted, dual-pair cable with 22-gauge solid conductors

and a braided shield. Used with Token Ring networks.

- **Type 2** Two-pair, shielded cable with solid conductors and a braided shield. Type 2 also includes four pairs of unshielded voice-grade lines, giving a total of six pairs in the same sheath.
- **Type 3** Four unshielded, solid, twisted pairs, used for voice or data. IBM's variant of twisted-pair telephone wire.
- **Type 4** No published specification.
- **Type 5** Dual 100/140 micron fiber-optic cable; IBM now recommends 125-micron fiber-optic cable, which is the current industry standard for fiber-optic cable.
- **Type 6** Shielded, two-pair, braided cable used for patch cables. Type 6 is more flexible than Type 1 cable.
- **Type 7** No published specification.
- **Type 8** Shielded, dual-pair cable with no twists, housed in a flat jacket; commonly used under carpets.

- **Type 9** Shielded, dual-pair, plenum cable with solid or braided conductors and a fire-resistant outer coating, for use between floors in a building.

See also American Wire Gauge; cabling standards; plenum cable; riser cable; structured wiring.

type-ahead buffer *See* keyboard buffer.

typefull naming In Novell Directory Services (NDS), a formal method used to name objects that includes name types for each part of the name. For example, .CN=Gary.OU=Marketing.O=Sybex.

See also typeless naming.

typeless naming In Novell Directory Services (NDS), a common method used to name objects that does not include name types for each part of the name. For example, .Gary.Marketing.Sybex.

See also typefull naming.

U

UART *See* universal asynchronous receiver/transmitter.

UBR *See* Unspecified Bit Rate.

UDP *See* User Datagram Protocol.

Ultra SCSI An extension of the SCSI-2 standard that increases the data-transfer rate to 20Mbps independent of the bus width. Ultra SCSI supports four to eight devices depending on cable type and length.

See also SCSI-2; Small Computer System Interface.

Ultra2 SCSI An extension of the SCSI-2 standard that increases the data transfer rate to 40Mbps over an 8-bit bus. Ultra2 SCSI supports a maximum of 8 devices.

See also SCSI-2; Small Computer System Interface.

Ultra Wide SCSI An extension of the SCSI-2 standard that increases the data-transfer rate to 80Mbps over a 16-bit bus. Ultra2 SCSI supports a maximum of 16 devices.

See also SCSI-2; Small Computer System Interface.

Ultrix A version of Unix from Digital Equipment Corporation that looks and works like BSD Unix.

See also BSD Unix.

UMB *See* upper memory block.

unauthorized access To gain entry to a computer system using a stolen or guessed password.

See also hacker; intruder.

unbundled software 1. Software sold with a computer system that is priced separately, rather than included as part of a package.

2. A feature in an application repackaged and sold by itself at a lower price.

See also bundled software.

UNC *See* Universal Naming Convention.

uncompress The process of restoring a compressed file to its original form.

See also unzip.

undelete To recover an accidentally deleted file. Many operating systems include commands you can use to recover a deleted file; however, once the file has been overwritten on your hard disk by a new file, the original is lost, and the only way to get it back is to reload it from a recent backup.

See also file recovery.

unicast The broadcast of individual audio or video signals from a server to individual clients to provide an on-demand video service.

See also IP multicast; multicast.

Unicode

Unicode A 16-bit character code, defined by the Unicode Consortium and by ISO 10646, that supports a maximum of 65,536 unique characters rather than the 256 characters available in the current ASCII character set.

By using two bytes to represent each character, Unicode allows almost all the world's written languages to be represented in a single character set; for example, the Chinese language defines almost 10,000 basic ideographs. When universally adopted, Unicode will make multilingual software much easier to write and maintain.

Products such as Novell NetWare and Microsoft Windows NT provide Unicode support.

See also American Standard Code for Information Interchange; double-byte character set; Extended Binary Coded Decimal Interchange Code.

Uniform Resource Characteristic Abbreviated URC. A mechanism designed to provide additional information about an Internet site or a Web page.

A URC lists pairs of attributes and their values and might contain author information,

keywords, prices, a copyright statement, or even a digital certificate, along with their associated Uniform Resource Identifiers.

See also Uniform Resource Identifier; Uniform Resource Locator.

Uniform Resource Identifier Abbreviated URI. In the HTTP message header, a set of characters that identifies a resource such as a file from anywhere on the Internet.

The URI includes Uniform Resource Names and Uniform Resource Locators to identify the file by type and location.

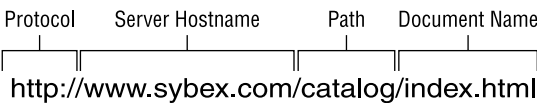
See also Uniform Resource Name; Uniform Resource Locator.

Uniform Resource Locator Abbreviated URL. An address for a resource on the Internet.

URLs are used as a linking mechanism between Web pages and as a method for Web browsers to access Web pages.

A URL specifies the protocol to be used to access the resource (such as HTTP or FTP), the name of the server where the resource is located (as in www.sybex.com), the path to that resource (as in `/catalog`), and the name of the document to open (`/index.html`).

UNIFORM RESOURCE LOCATOR



Uniform Resource Name Abbreviated URN. A proposal from the IETF (Internet Engineering Task Force) for a naming scheme that will identify Internet resources by name, irrespective of where they are located.

See also Uniform Resource Locator.

UniForum A nonprofit organization founded in 1980 dedicated to improve Unix through the open exchange of ideas and information between users and developers.

Previously known as /usr/group, UniForum has several hundred thousand members in many countries throughout the world.

uninterruptible power supply Abbreviated UPS; pronounced “you-pea-ess.” An alternative power source, usually consisting of a set of batteries, used to power a computer system if the normal power service is interrupted or falls below acceptable levels.

A UPS system is usually applied only to the most critical devices on the network, such as servers, routers, gateways, and independent hard disks. They are of two main types:

- An online UPS continuously monitors and modifies the power flowing through the unit. If an outage occurs, the UPS continues to provide regulated power.
- A standby UPS monitors the AC level, but only switches in when the power drops below a preset level; it contains circuitry capable of switching to backup power in 5 milliseconds or less.

See also power conditioning; UPS monitoring.

universal asynchronous receiver/transmitter Abbreviated UART; pronounced “you-art.” An electronic module that combines the transmitting and receiving circuitry needed for asynchronous communications over a serial line.

See also asynchronous transmission.

Universal Coordinated Time In Novell NetWare, the standard system time based on Greenwich Mean Time (GMT).

Local time on a NetWare server is defined in terms of the difference from GMT; for example, Pacific Standard Time is GMT minus eight hours.

Universal Naming Convention Abbreviated UNC. In Microsoft Windows NT Server, a scheme used to gain access to a shared resource.

The general form of a UNC is:

`\\servername\sharename\path\filename`

The `servername` portion identifies the name of the server where the resource is located, and `sharename` identifies the name of the shared file resource under which the information has been shared. This is followed by the path and filename of the resource.

Universal Serial Bus Abbreviated USB. A standard from Intel and Microsoft for a high-speed peripheral bus designed to remove the need for almost all the connectors on the back of a personal computer.

USB defines the ports and bus characteristics with data transfer rates of up to 12Mbps over a single cable of up to 5 meters (16 feet). USB is capable of supporting a maximum of

63 devices such as external CD-ROM drives, printers, external modems, mice, and the keyboard and also supplies power to some devices so there is no need for separate power cords or batteries. Most personal computers will have two USB ports.

See also 1394.

universal synchronous receiver/transmitter Abbreviated USRT. An electronic module that combines the transmitting and receiving circuitry needed for synchronous communications over a serial line.

See also synchronous transmission.

Unix Pronounced “yoo-nix.” A multiuser, multitasking operating system, originally conceived in 1969 at AT&T’s Bell Labs by Ken Thompson and Dennis Ritchie. Since then, Unix has gone on to become the most widely used general purpose operating system in the world.

Over the last 25 years, there have been three major strands in Unix development:

- Original AT&T Unix from Versions 1 to 7, and Systems III to V.
- Microsoft/SCO XENIX.
- Berkeley releases from 1BSD to 4.4BSD.

In addition, a large number of commercial Unix-related systems have been released by developers such as Apple, Digital Equipment Corporation, Hewlett-Packard, IBM, SCO, Silicon Graphics, and Sun Microsystems. Unix is available on a huge range of hardware, from a PC to a supercomputer.

Several free (or almost free) versions of Unix are available in the Intel world, including Linux, FreeBSD, and NetBSD; since

they contain no proprietary code, they are not affected by licensing agreements other than the GNU General Public License.

Unix today is a very different animal from the Unix of the 1980s. A typical system then consisted of a minicomputer serving a collection of dumb terminals. Unix today is more likely to be a graphical workstation on a network or a Web server on the Internet and is a large and complicated commercial offering, serving a wide range of applications.

See also Advanced Interactive Executive; A/UX; Berkeley Unix; BSD Unix; FreeBSD; Interactive Unix; Linux; NetBSD; SCO Unix; Solaris; System V Interface Definition; UnixWare.

Unix client Any computer running Unix that connects to the network.

Unix shell In Unix, a program that acts as a user interface, interpreting commands typed at the keyboard and passing them on to the operating system.

The shell sets up standard input, standard output, and standard error, lets you customize your Unix session environment, and gives access to a shell programming language for creating shell scripts.

Some versions of Unix provide only one shell, while others provide a selection from which you can choose to use the one you like best.

Common shells include the following:

- **Bourne** Very compact and simple to use; the original Unix shell.
- **Korn** Perhaps the most popular shell; an upward compatible extension to the

Bourne shell with a history file, command-line editing, aliases, and job control.

- **C** The first BSD shell, the C shell uses C-like syntax and offers a history mechanism, aliasing, and job control.
- **Bash** The Bourne-again shell from the Free Software Foundation extends the capabilities of the Bourne shell in a similar way to the Korn shell.
- **Rc** A small, compact, and elegant shell with a strong C flavor but without command-line editing and job control.
- **Tcsh** An enhanced version of the C shell.
- **Zsh** A large shell that seems to offer all the features available in all the other shells.

See also Bash shell; Bourne shell; C shell; Korn shell.

UnixWare A release of the Unix operating system, originally from Novell, now available from SCO.

UnixWare is a 64-bit operating system that combines SCO's OpenServer system with previous versions of UnixWare and includes Netscape's FastTrack Server for Web site creation. A separate software development kit, which includes a C/C++ compiler, a debugger, and other development tools, is also available.

See also Santa Cruz Operation, Inc.

unmoderated newsgroup A Usenet newsgroup or mailing list in which posts are not subject to review before distribution. You will find the discussions in unmoder-

ated newsgroups to be wildly spontaneous, but they may also contain more than their share of flames and flame wars.

See also alt newsgroups; moderated newsgroup.

unshielded cable Any cable not protected from electromagnetic interference or radio frequency interference (RFI) by an outer foil shield.

unshielded twisted-pair cable Abbreviated UTP. Cable that contains two or more pairs of twisted copper wires.

The greater the number of twists, the lower the crosstalk. UTP is offered in both voice grade and data grade. The advantages of UTP include ease of installation and low cost of materials. Its drawbacks are limited signaling speeds and shorter maximum cable-segment lengths.

See also crosstalk; shielded twisted-pair cable.

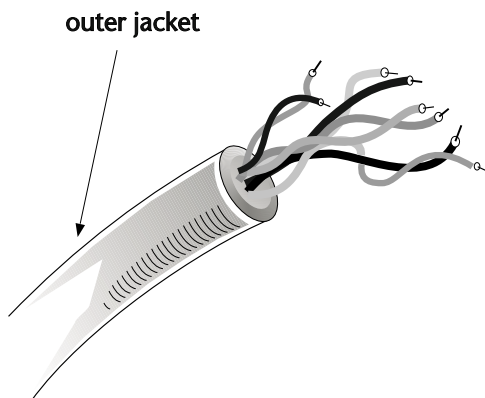
Unspecified Bit Rate Abbreviated UBR. A type of Asynchronous Transfer Mode (ATM) service that provides spare bandwidth to noncritical services such as file transfers.

See also Asynchronous Transfer Mode; Available Bit Rate; Constant Bit Rate; Variable Bit Rate.

unsubscribe To remove the name of a Usenet newsgroup from the list of newsgroups maintained by your newsreader. If you change your mind, you can always subscribe to the newsgroup again in the future.

See also newsgroup; subscribe.

UNSHIELDED TWISTED-PAIR CABLE



unzip The process of uncompressing an archive created by PKZIP, WinZip, compress, or gzip.

See also uncompress.

update sequence number Abbreviated USN. In Microsoft Active Directory, a unique 16-bit number used to track updates to Active Directory properties.

upgradable computer A computer system specifically designed to be upgraded as technology advances.

The amount of circuitry that must be changed when you make the upgrade and the method of upgrading differ from one upgradable computer to another. At a minimum, you must replace the processor; at most, you need to change nearly all the circuitry installed in the computer. In some systems, the use of a ZIF socket to hold the main processor makes an upgrade easy; in

other systems, replacing the main processor can be extremely difficult.

upgrade 1. The process of installing a newer and more powerful version of a software or hardware product. For example, you may upgrade to a newer and more capable version of a software package, such as Microsoft Word, or to a larger hard disk. In the case of hardware, an upgrade is often called an upgrade kit.

2. A new and more powerful version of software (a noun referring to the improved software itself).

uplink The transmission of information from an earth station to a communications satellite.

See also downlink.

upload In communications, sending a file or files from one computer to another over

a network or via a modem. For example, you might upload a file to a network server.

See also download.

UPN *See* user principal name.

upper memory *See* reserved memory.

upper memory block Abbreviated UMB. The memory between 640KB and 1MB in a personal computer running MS-DOS. This area was originally reserved for system and video use; however, not all the space is used. The unused portions are the UMBs.

With an Intel 80386 (or later) processor, as much as 120KB of additional memory can be gained by accessing UMBs. This space can be used to load device drivers and terminate-and-stay-resident programs.

UPS *See* uninterruptible power supply.

UPS monitoring The process that a server uses to make sure that an attached UPS (uninterruptible power supply) system is functioning properly.

See also power conditioning.

uptime The length or percentage of time during which a computer system is functioning and available for use.

See also downtime; power-on hours.

upward compatibility The design of software that incorporates the capability to function with other, more powerful products likely to become available in the near future. Adherence to design standards makes upward compatibility possible.

URC *See* Uniform Resource Characteristic.

URI *See* Uniform Resource Identifier.

URL *See* Uniform Resource Locator.

URN *See* Uniform Resource Name.

USB *See* Universal Serial Bus.

Usenet Contraction of user network; sometimes written as UseNet or USENET, pronounced “yooz-net.” An international, noncommercial network, linking many thousands of Unix sites.

Although there is a very close relationship between the Internet and Usenet, they are not the same thing by any means. Usenet predates the Internet; in the early days, information was distributed by dial-up connections and UUCP software. Not every Internet computer is part of Usenet, and not every Usenet system can be reached from the Internet.

Like the Internet, Usenet has no central governing body; it is run by the people who use it. With tens of thousands of newsgroups, Usenet is accessed by millions of people every day, in more than 100 countries.

See also Usenet newsgroups.

Usenet articles An individual e-mail message sent to a Usenet newsgroup.

Usenet newsgroups The individual discussion groups within Usenet.

Usenet newsgroups contain articles posted by Internet and Usenet subscribers; very few of them actually contain hard news.

Most newsgroups are concerned with a single subject, and the range of subjects

throughout Usenet is simply phenomenal; if people are interested in a topic, you will find a newsgroup for that topic.

user Any person allowed to access a computer system or network.

user account A security mechanism used to control access to a network or to a multiuser computer system, established and maintained by the network administrator. Elements of a user account include password information, rights and permissions, and information about the groups to which the user belongs.

User Datagram Protocol Abbreviated UDP. The connectionless, transport-level protocol used in the TCP/IP suite of protocols, usually bundled with IP-layer software. Because UDP does not add overhead, as does connection-oriented TCP, UDP is often used with SNMP (Simple Network Management Protocol) applications.

Multicast applications, such as Mbone and the Real-time Transport Protocol, that deliver audio and video streams use UDP as their delivery mechanism because the acknowledgment and retransmission services offered by TCP are not needed and add too much overhead. If a packet of audio data is lost, retransmission is neither practical nor desirable.

See also multicast backbone; Real-time Transport Protocol; Transmission Control Protocol.

user group A group of users of a specific computer or software package who meet to share tips and listen to industry experts.

Some PC user groups hold large, well-attended monthly meetings, run their own Web sites, and publish newsletters of exceptional quality.

See also special interest group.

User Manager for Domains In Microsoft Windows NT Server, an administrative tool used to manage accounts, groups, and security policies throughout the domain.

See also user rights policies.

User object In Novell Directory Services (NDS), an object that represents a user with access to the network.

See also Leaf object; Organization object; Root object; user template.

user principal name Abbreviated UPN. In Microsoft Active Directory, a name for a user. UPN consists of the user's logon name and the Domain Name Service (DNS) name of the domain where the user object resides, separated by an at sign. For example, billg@microsoft.com.

user rights policies In Microsoft Windows NT Server, a mechanism used to determine the rights that users and groups possess when trying to perform network tasks. User rights policies are administered by the User Manager for Domains tool.

See also User Manager for Domains.

user template In Novell Directory Services (NDS), a special User object that assigns default property values and rights when a new user is created.

A user template speeds up the process of creating a large number of new users at the same time, especially if these users share details such as locations, account restrictions, and so on.

See also User object.

USN *See* update sequence number.

U.S. Robotics Corporation A major manufacturer of modems and creator of the PalmPilot personal digital assistant; merged with 3Com Corporation in 1998 in one of the largest deals in telecommunications-industry history.

For more information on U.S. Robotics Corporation, see www.usr.com.

USRT *See* universal synchronous receiver/transmitter.

UTP *See* unshielded twisted-pair cable.

uucp Pronounced “you-you-cee-pee.” A standard set of Unix utilities used to manage the transmission of information between Unix systems and to Usenet newsgroups, using serial connections and regular telephone lines. The name is derived from “Unix-to-Unix copy.”

These utilities were originally developed by Mike Lesk at Bell Labs in the 1970s, and a version known as HoneyDanBer uucp was developed in the 1980s by P. Honeyman, D. A. Nowitz, and B. E. Redman. An even newer version, known as Taylor uucp, is

available on many systems, particularly Linux.

See also Network News Transport Protocol.

uudecode Pronounced “you-you-de-code.”

1. To convert a text file created by the Unix uuencode utility back into its original binary form. Graphical images and other binary files are often sent to Usenet newsgroups in this form, because the newsgroups can only handle text and don’t know how to manage binary files.

2. The name of the utility program that performs a text-to-binary file conversion. Originally a Unix utility, uudecode is now available for most operating systems.

See also uucp; uuencode.

uuencode Pronounced “you-you-en-code.”

1. To convert a binary file such as a graphical image into a text file so that the file can be sent over the Internet or to a Usenet newsgroup as a part of an e-mail message. When you receive a uuencoded text file, you must process it through the Unix uudecode utility to turn it back into a graphical image that you can view.

2. The name of the utility that performs a binary-to-text file conversion. Originally a Unix utility, uuencode is now available for most operating systems.

See also uucp; uudecode.

V

V.17 A CCITT (ITU) Group 3 fax modulation standard for transmitting fax data at up to 14,400bps, with a fallback to 12,000bps as line conditions deteriorate.

V.21 A CCITT (ITU) standard for 300bps modems using full-duplex transmission over dial-up lines. This standard is not compatible with the Bell 103 standard widely used in the United States.

V.22 A CCITT (ITU) standard for 600bps and 1200bps full-duplex modems over two-wire, dial-up, or leased lines. V.22 uses phase-shift keying modulation.

V.22 bis A CCITT (ITU) standard for 2400bps full-duplex modems over dial-up and two-wire leased lines, with fallback to 1200bps and then 600bps operation. V.22 bis uses quadrature amplitude modulation.

V.23 A CCITT (ITU) standard for 600bps or 1200bps synchronous or asynchronous half-duplex modems used on dial-up lines.

V.24 A CCITT (ITU) definition of the interface between a modem and a computer system. V.24 is functionally equivalent to the RS-232-C standard, but does not specify connectors or pin assignments; those are defined in ISO 2110.

V.25 A CCITT (ITU) standard for automatic calling and answering circuits over dial-up lines using a parallel interface. V.25

includes the disabling of echo suppression on manually dialed calls.

V.25 bis A CCITT (ITU) standard for automatic calling and answering circuits over dial-up lines with three modes: asynchronous, character-oriented synchronous, and bit-oriented synchronous (HDLC/SDLC). V.25 bis does not include modem configuration commands.

V.26 A CCITT (ITU) standard for 1200bps, full-duplex modems used over four-wire leased lines.

V.26 bis A CCITT (ITU) standard for 1200bps and 2400bps full-duplex modems used on dial-up lines.

V.27 A CCITT (ITU) standard for 4800bps, full-duplex modems used with four-wire leased lines, with a manual equalizer.

V.27 bis A CCITT (ITU) standard for 2400bps or 4800bps, full-duplex modems used with four-wire leased lines. The main advance over V.27 is the addition of an automatic adaptive equalizer for use on leased circuits.

V.27 ter A CCITT (ITU) standard for 2400bps or 4800bps, full-duplex modems used with dial-up lines. Used in some CCITT Group 3 fax transmissions.

See also CCITT Groups 1–4.

V.29 A CCITT (ITU) standard for 9600bps modems used with point-to-point, four-wire leased lines. This standard has been adopted for CCITT Group 3 fax transmissions over dial-up lines at 9600bps and 7200bps.

See also CCITT Groups 1–4.

V.32 A CCITT (ITU) standard for 9600bps modems, with fallback to 4800bps, used over two-wire, dial-up lines or two-wire or four-wire leased lines, with echo canceling to remove any telephone-line echo, and quadrature amplitude modulation.

V.32 encodes four data bits for each baud to give an effective throughput of 9600bps and includes trellis-coded modulation error-correcting techniques. V.32 is the first standard for 9600bps modems using standard lines anywhere in the world.

V.32 bis A CCITT (ITU) standard extending V.32 to 7200, 12,000, and 14,400 bits per second. V.32 bis uses trellis-coded modulation.

V.32 terbo A pseudo-standard proposed by AT&T and others that supports transmission at up to 19,200bps.

The name is a pun and does not represent an ITU standard; the next revision of the V.32 standard after V.32 bis will be V.32 ter, and V.32 terbo has been replaced by the ITU standard V.34.

See also V.fast.

V.33 A CCITT (ITU) standard for 12,000bps and 14,400bps modems used

over four-wire, leased circuits, with time-division multiplexing available for line sharing.

V.34 A CCITT (ITU) standard for 28.8Kbps modems using trellis-coded modulation techniques and advanced data compression.

See also V.fast.

V.42 A CCITT (ITU) standard for error correction rather than for a modem. V.42 uses LAP-M (Link Access Procedure-Modem) as the primary error-correcting protocol, with MNP (Microcom Networking Protocol) classes 2 through 4 as an alternative.

V.42 can be used with V.22, V.22 bis, V.26 ter, V.32, and V.32 bis.

V.42 bis A CCITT (ITU) standard that adds a British Telecom Lempel-Ziv data-compression technique to V.42 error correction, usually capable of achieving a compression ratio of 3.5 to 1.

V.54 A CCITT (ITU) standard that specifies the loopback tests incorporated into modems for testing the telephone circuit and isolating any transmission problems.

V.56 bis A CCITT (ITU) standard that defines a network transmission model used to evaluate modem performance over two-wire voice-grade circuits.

V.90 A CCITT (ITU) standard for modems; also known as the 56K modem standard. V.90 describes an asymmetric

connection, with theoretical speeds of up to 56Kbps downstream and an upstream connection rate of up to 33.6Kbps.

V.90 modems attain their high speed by assuming the circuit is a digital circuit and reducing the number of analog-to-digital conversions they perform, except for the conversion that takes place for outbound traffic at the user's modem.

Whether you actually achieve these rates depends on the quality of the phone line and the distance to the local telephone company central office. If the other end of the connection is not digital, the modem switches into full analog mode at 28.8 or 33.6Kbps.

In order to reduce crosstalk between adjacent lines, the FCC has placed restrictions on maximum signal strength levels, and so 54Kbps is the theoretical maximum data rate.

See also K56Flex; X2.

V.110 A CCITT (ITU) standard that specifies how DTE (data terminal equipment) using synchronous or asynchronous serial interfaces is supported on an ISDN (Integrated Services Digital Network).

V.120 A CCITT (ITU) standard that specifies how DTE (data terminal equipment) using synchronous or asynchronous serial interfaces is supported on an ISDN (Integrated Services Digital Network) using a protocol to encapsulate the transmitted data.

vaccine A utility program designed to protect files from viruses. By adding a small

amount of code to an existing file, the vaccine program causes an alert to be generated if a virus does attack.

See also antivirus program; boot sector virus; file-infecting virus; infection; macro virus; multipart virus; polymorphic virus; stealth virus; Trojan Horse; virus.

Value Added Network Abbreviated VAN. Commercially available turn-key data networks from companies such as CompuServe, GE Information Services, and Infonet Services Corporation.

VANs are available on a leased-line or a dial-up rate and save an organization the trouble of setting up the equipment and contracting for the lines and service. VANs can also provide additional services such as message routing, resource management, and protocol-conversion services.

value-added process Abbreviated VAP. In Novell NetWare 2.x, an application that adds functions to the network operating system, such as print server or communications server software.

NetWare Loadable Modules (NLMs) provide a similar function in later versions of NetWare.

See also Virtual Loadable Module.

value-added reseller Abbreviated VAR. A company that adds value to a system, repackages it, and then resells it to the public.

This added value can take the form of better documentation, user support, service support, system integration, or sometimes just a

new nameplate on the box. For example, Canon makes the print engine used in many laser printers, including those from Hewlett-Packard (HP); in this case, Canon is an OEM (original equipment manufacturer) and HP is the VAR.

value entry In Microsoft Windows, the actual data in the Registry, stored in the keys. Each value entry has a name, a data type (which determines the length and the format of the value), and the value itself.

See also hive; key; Registry; sub-hive; sub-key; volatile key.

values In Novell Directory Services (NDS), the actual data stored in the properties of an NDS object. Properties can have more than one value; some are required, and others are optional.

vampire tap A connector used to connect one cable segment to another. A needle on the connector pierces the cable insulation to make a connection to the wire within.

VAN *See* Value Added Network.

VAP *See* value-added process.

vaporware A slang term for a product that has been announced but has missed its release date, often by a large margin, and so is not actually available.

VAR *See* value-added reseller.

Variable Bit Rate Abbreviated VBR. A connection-oriented Asynchronous Transfer Mode service for real-time applications for which minor data loss is acceptable, and

for non-real-time VBR applications such as transaction processing.

See also Asynchronous Transfer Mode; Available Bit Rate; Constant Bit Rate; Unspecified Bit Rate.

VAX Digital Equipment Corporation's (DEC's) popular line of minicomputers and workstations, first introduced in 1977.

See also DECnet; VMS.

VBA *See* Visual Basic for Applications.

VBR *See* Variable Bit Rate.

VBScript A version of Microsoft Visual Basic used as a scripting language in Microsoft Internet Explorer Web browser and in Microsoft Internet Information Server.

See also scripting.

VCC *See* Virtual Channel Connection.

VCI *See* Virtual Channel Identifier.

VDSL *See* Very-High-Bit-Rate Digital Subscriber Line.

vector graphics *See* object-oriented graphics.

vendor The person or company that manufactures, supplies, or sells computer hardware, software, or related services.

Vendor Independent Messaging

Abbreviated VIM. An e-mail API originally developed by Lotus, and supported by Apple, IBM, MCI, Novell, and Oracle and other e-mail vendors. Developers use VIM to add e-mail capabilities to their applications.

Microsoft supports its own e-mail API, Messaging API (MAPI), and a VIM-to-MAPI dynamic link library makes it possible to exchange messages between the two interfaces.

See also Messaging API.

version number A method of identifying a particular software or hardware release, assigned by the developer, that often includes numbers before and after a decimal point; the higher the number, the more recent the release.

The number before the decimal point indicates the major revision levels (as in Microsoft Windows NT 3 and Windows NT 4), and the part after the decimal indicates a minor revision level (as in Windows NT 3.5 and Windows NT 3.51). In some cases, a minor revision can produce a significant difference in performance.

Many people steer clear of any release labeled 1.0, because this number implies the first release of a product that may not have had extensive real-world use. Microsoft has avoided this issue with the release of Windows systems tied to the year, as in Windows 98 and Office 2000.

vertical application An application specifically created for a narrow and specialized market or profession. Software designed for veterinary hospital management is an example of a vertical application.

vertical bar The | symbol. Used in Unix and in other operating systems to pipe the output of one command into the input of another.

Very-High-Bit-Rate Digital Subscriber Line Abbreviated VDSL. A higher-speed version of Asymmetric Digital Subscriber Line (ADSL).

VDSL is asymmetrical with a higher downstream data-transfer rate than its upstream rate. Upstream rates can be from 1.6Mbps to 2.3Mbps, and downstream rates range from 12.96Mbps to 51.84Mbps, depending on the distance involved.

See also Asymmetric Digital Subscriber Line; Digital Subscriber Line; High-Bit-Rate Digital Subscriber Line; Rate-Adaptive Digital Subscriber Line; Single-Line Digital Subscriber Line.

very low-frequency emission Abbreviated VLF. Radiation emitted by a computer monitor and other common household electrical appliances, such as televisions, hair dryers, electric blankets, and food processors.

VLF emissions range from 2 to 400kHz and decline with the square of the distance from the source. Emissions are not constant around a computer monitor; they are higher from the sides and rear, and weakest from the front of the screen.

Sweden is the only country to have defined a set of standards for monitor emissions. In 1990, Mat Oct Provadet (MPR), the Swedish National Board for Meteorology and Testing, revised its guidelines for acceptable VLF emissions as less than or equal to 25 nanoTesla (nT). A nanoTesla is a unit of measurement for small magnetic fields.

See also extremely low-frequency emission; radio frequency interference.

very small aperture terminal Abbreviated VSAT. A small satellite terminal used for digital communications, from 1 to 3 meters (3.3 to 10 feet) in diameter, capable of managing digital transmissions of up to 56Kbps. Satellites that can handle T1 data rates of up to 1.544Mbps are known as TSATs.

VESA See Video Electronics Standards Association.

VESA local bus Abbreviated VL bus; also known as VL local bus. A local bus architecture introduced by VESA (Video Electronics Standards Association), in which as many as three VL bus adapter slots are built into the motherboard. The VL bus allows for bus mastering. The most common VL bus adapters are video adapters, hard-disk controllers, and network interface cards.

See also Peripheral Component Interconnect local bus; PC Card.

V.everything A marketing term used by some modem manufacturers to describe modems that comply with both the K56Flex and X2 proposed standards as well as with the adopted CCITT (ITU) V.90 standard.

A V.everything modem should be compatible with any other modem capable of operating at the same speed.

V.Fast An interim modem standard for uncompressed data-transfer rates of up to 28.8Kbps over dial-up voice-grade phone lines. Made obsolete by the publication of the CCITT (ITU) V.34 standard.

See also V.34.

vi A popular Unix screen editor, originally written by Bill Joy as part of the BSD Unix systems. Pronounced “vee-eye.”

video adapter An adapter that provides the text and graphics output to the monitor. Some later video adapters, such as the SVGA, are included in the circuitry on the motherboard rather than as separate plug-in boards.

video conferencing A method used to allow people at remote locations to join in a conference and share information. Originally done with analog video and expensive satellite links, video conferencing is now performed with compressed digital video transmitted over a local-area network or the Internet.

From an application standpoint, video conferencing has gone way beyond looking at a picture of a person; users can look at and update charts, make drawings or sketches on a chalkboard, update spreadsheets, and so on, all online.

A video camera and a speakerphone are linked to a PC at each site, and the PC in turn is linked to the network.

See also CU-SeeMe; desktop video; H.323; Microsoft NetMeeting; T.120.

Video Electronics Standards

Association Abbreviated VESA. An association of video graphics adapter and monitor manufacturers that sets standards for personal computer video. VESA is most notable for its role in standardizing Super VGA (SVGA) hardware and the development of the VESA local bus.

See also VESA local bus.

video RAM Abbreviated VRAM, pronounced “vee-ram.” Special-purpose RAM with two data paths for access (conventional RAM has just one). These two paths let a VRAM board manage two functions at once: refreshing the display and communicating with the processor. VRAM does not require the system to complete one function before starting the other, so it allows faster operation for the whole video system.

viewer An application launched by a Web browser to view a file that the browser cannot display by itself. Sometimes called a helper application.

A viewer displays video clips and animation files.

See also helper; player; plug-in.

ViewSonic Corp. A manufacturer of high-performance monitors, flat-panel LCD and LCD projectors, and professional monitors for CAD/CAM and graphic design.

For more information on ViewSonic Corp., see www.viewsonic.com.

VIM *See* Vendor Independent Messaging.

VINES *See* Banyan VINES.

Virtual Channel Connection Abbreviated VCC. In Asynchronous Transfer Mode (ATM), a logical connection between two end stations. A VCC can be either switched or dedicated.

See also Asynchronous Transfer Mode; Virtual Path Connection.

Virtual Channel Identifier Abbreviated VCI. In an Asynchronous Transfer Mode (ATM) cell header, a 16-bit field used to identify virtual channels between users or between users and networks.

See also Asynchronous Transfer Mode.

virtual circuit A temporary shared communications path that appears to the user as a dedicated connection. A virtual circuit is maintained only for as long as the customer requires a connection; the next time a call is placed, a different virtual circuit may be used.

See also bandwidth on demand; permanent virtual circuit; switched virtual circuit.

virtual console A feature found in Linux, UnixWare, and other operating systems that allows you to log on to the same terminal as though you were different users. Also known as a virtual terminal.

To open a new virtual console, hold the Alt key as you press one of the F1 to F8 function keys; as you press each function key, you see a new screen complete with its own login prompt. Each virtual console displays its output on its own virtual screen.

virtual data network A method used to provide full interconnection of all LAN segments without using dedicated circuits so that customers pay only for the services they actually use. Also known as a virtual LAN.

See also bandwidth on demand; virtual circuit.

virtual directory A directory that appears to a user to be located on the service they are accessing, but which is actually located on a linked server. Virtual directories are often used on Web servers where they are used to balance the overall load, but still present a single, simple view of the information present.

virtual drive *See* RAM disk.

virtual LAN *See* virtual data network.

Virtual Loadable Module Abbreviated VLM. A Novell NetWare modular program that runs on each MS-DOS client and allows that workstation to communicate with the server.

The NetWare DOS Requester consists of several VLMs that replace and provide backward compatibility with the NetWare shells used in earlier versions of NetWare.

See also NetWare DOS Requester; NetWare Loadable Module.

virtual machine An environment created by the operating system that gives each executing application the illusion that it has complete control of an independent computer and can access all the system resources that it needs.

For example, the Intel 80386 (and higher) processor can run multiple MS-DOS applications in completely separate and protected address spaces using virtual 8086 mode.

In the Java environment, a Java applet executes inside the Java Virtual Machine.

See also Java Virtual Machine.

virtual memory A memory-management technique that allows information in physical memory to be swapped out to a hard disk if necessary.

This technique provides applications with more memory space than is actually available in the computer. True virtual-memory management requires specialized hardware in the processor for the operating system to use; it is not just a matter of writing information out to a swap area on the hard disk at the application level.

In a virtual memory system, programs and their data are divided into smaller pieces called *pages*. When more memory is needed, the operating system decides which pages are least likely to be needed soon (using an algorithm based on frequency of use, most recent use, and program priority), and it writes these pages out to disk. The memory space that they used is now available to the rest of the system for other applications. When these pages are needed again, they are loaded back into real memory, displacing other pages.

See also permanent swap file; swapping; swap space.

Virtual Path Connection Abbreviated VPC. In Asynchronous Transfer Mode (ATM), a set of logical Virtual Channel Connections (VCCs) between two end stations. All channels in a specific VPC connect the same two end stations.

See also Asynchronous Transfer Mode; Virtual Channel Connection.

Virtual Path Identifier Abbreviated VPI. In an Asynchronous Transfer Mode

Virtual Private Network

(ATM) cell header, an 8-bit field used to identify virtual paths between users or between users and networks.

See also Asynchronous Transfer Mode.

Virtual Private Network Abbreviated VPN. Originally, a private network for voice and data built from traditional carrier services. More recently, a VPN is an encrypted private tunnel across the Internet.

Virtual Reality Modeling Language

Abbreviated VRML. A specification and file format for three-dimensional rendering used in conjunction with Web browsers and that uses existing Web protocols and works within the Internet framework.

VRML is a subset of Silicon Graphics' Inventor File Format, created by Mark Pesce and Tony Parisi in 1994.

virtual root The root directory that a user sees when connected to a Web server. The virtual root is an alias to an actual location on a hard disk, either on the Web server or on a different server. This technique allows one URL to represent the Web site, while, at the same time, allowing the Webmaster to change the root directory without affecting the URL.

See also Uniform Resource Locator.

Virtual Telecommunications Access

Method Abbreviated VTAM. IBM software that runs on a mainframe computer running the MVS or VM operating system. VTAM controls communications in an SNA (Systems Network Architecture) environment. VTAM supports a range of

network protocols, including SDLC (Synchronous Data Link Control) and Token Ring.

virtual terminal *See* virtual console.

virus A program intended to damage a computer system without the user's knowledge or permission.

A virus clones itself from disk to disk or from system to system over a network. Numbers are hard to come by, but certain authorities claim that there are approximately 30,000 known viruses, with 400 new ones appearing each month.

A virus may attach itself to a program or to the partition table or boot track on a hard disk. When a certain event occurs, a date passes, or a specific program executes, the virus is triggered into action.

See also boot sector virus; file-infecting virus; macro virus; multipart virus; polymorphic virus; stealth virus; Trojan Horse; vaccine.

Visual Basic for Applications Abbreviated VBA. A version of Microsoft Visual Basic included with Microsoft Windows 98 applications such as Excel; it is used to write everything from simple macros to complex custom applications.

VL bus *See* VESA local bus.

VLF *See* very low-frequency emission.

VL local bus *See* VESA local bus.

VLM *See* Virtual Loadable Module.

VM *See* virtual machine.

VMS A 32-bit, multiuser, multitasking, virtual memory operating system from Digital Equipment Corporation (DEC) for the VAX line of computers.

See also DECnet; VAX.

voice mail A computerized store-and-forward system for voice messages.

A voice-mail system uses prerecorded messages to route the caller to the correct person, department, or mailbox and then digitizes the incoming messages and stores them on disk for review by the recipient. Users can often forward voice-mail messages to another department or person after attaching their own comments.

See also electronic mail.

volatile key In Microsoft Windows, a Registry key whose contents are constructed at the time the system boots, rather than being permanently stored in a hive file.

See also hive; key; Registry; sub-hive; sub-key; value entry.

volume In the Novell networking world, a volume is the highest level of the file server directory and file structure.

Large hard disks can be divided into several volumes when the network operating system is first installed, and with volume spanning, volumes can include multiple physical drives.

volume label In many operating systems, the name assigned to a disk by the user, displayed on the first line of a directory listing. The Macintosh, on which disks are

often referred to by name, uses the term *volume name* instead.

volume name *See* volume label.

volume reference number *See* volume serial number.

volume serial number In several operating systems, a unique number assigned to a disk during the formatting process. This number is displayed at the beginning of a directory listing. The MacOS assigns a similar number, known as a volume reference number, that programs can use when referring to disks.

volume spanning In Novell NetWare, a mechanism that allows a volume to include several physical hard disks.

See also volume.

VPC *See* Virtual Path Connection.

VPI *See* Virtual Path Identifier.

VP/ix An MS-DOS emulation package from Sun Microsystems that allows you to access programs and data files under the MS-DOS partition and use them in the Unix environment as a task under Unix. To the user, the sessions look exactly as they would if you changed partitions and booted MS-DOS.

VPN *See* Virtual Private Network.

VRAM *See* video RAM.

VRML *See* Virtual Reality Modeling Language.

VSAT *See* very small aperture terminal.

VT-52, VT-100, VT-200 A series of asynchronous terminals manufactured by Digital Equipment Corporation (DEC) that uses a specific set of control codes for display management. Many communications and

terminal-emulation packages include emulations of these terminals.

VTAM See Virtual Telecommunications Access Method.

W

W3 *See* World Wide Web.

W3C *See* World Wide Web Consortium.

WABI *See* Windows Application Binary Interface.

WAIS *See* Wide Area Information Service.

wait state A clock cycle during which no instructions are executed, because the processor is waiting for data from memory.

Static RAM chips and paged-mode RAM chips are becoming popular because they can store information without being constantly refreshed by the processor, thus eliminating the wait state. A computer that can process information without wait states is known as a zero-wait-state computer.

WAN *See* wide-area network.

warm boot A reboot performed after the operating system has been running for some period of time, by pressing Ctrl+Alt+Del rather than cycling the power to the computer.

Warp *See* OS/2.

WATS *See* Wide Area Telephone Service.

wavelength division multiple access

Abbreviated WDMA. A technique that manages multiple transmissions on a fiber-optic cable system using wavelength division multiplexing (WDM).

WDMA divides each channel into a set of time slots using time-division multiplexing, and data from different sources is assigned to a repeating set of time slots.

See also time-division multiplexing; wavelength division multiplexing.

wavelength division multiplexing

Abbreviated WDM. A frequency-division multiplexing (FDM) technique that allows a single fiber-optic cable to carry multiple light signals rather than a single light signal. WDM places each signal on a different frequency.

See also frequency-division multiplexing; inverse multiplexing; time-division multiplexing; statistical multiplexing.

WBEM *See* Web-Based Enterprise Management.

WDM *See* wavelength division multiplexing.

WDMA *See* wavelength division multiple access.

weak password A password that is easy to guess, such as an English word, any reference to Star Trek or Star Wars, or a word or number that relates to the user, such as the name of a family member or a birth date.

See also password; strong password.

Web *See* World Wide Web.

Web-Based Enterprise Management

Abbreviated WBEM. A platform-independent management standard, originally proposed by BMC Software, Cisco Systems, Compaq, Intel, and Microsoft and now supported by more than 70 companies, to allow network, systems, and applications software data from a variety of sources to be accessed and reported by management applications regardless of the original source of that data.

See also Desktop Management Interface; Simple Network Management Protocol; total cost of ownership; Wired for Management.

Web browser A client application that lets you look at hypertext documents, follow links to other HTML documents, and download files on the Internet or on a corporate intranet.

When you find something that interests you as you browse through a hypertext document, you can click on that object, and the system automatically takes care of accessing the Internet host that holds the document you requested; you don't need to know the IP address, the name of the host system, or any other details.

A Web browser will also display the graphics in a Web page, play audio and video clips, and execute small Java or ActiveX programs called applets, although certain older Web browsers may need helper, or plug-in, applications to perform some of these tasks.

Netscape Navigator and Microsoft Internet Explorer are examples of popular Web browsers.

See also ActiveX; helper; HyperText Markup Language; player; plug-in; Uniform Resource Locator; viewer; World Wide Web.

Webmaster The person responsible for maintaining and administering a Web site.

WebNFS A technology from Sun Microsystems that extends the Network File System (NFS) to the Internet.

WebNFS makes file access across the Internet as easy as local file access, provides enhanced download performance, and can automatically resume a file transfer after a broken connection is restored.

Web page Information placed on a Web server for viewing with a Web browser. A Web page can contain text, graphics, audio or video clips, and links to other Web pages.

See also Web browser.

Web portal *See* portal.

Web server A hardware and software package that provides services to client computers running Web browsers.

Clients make requests in the form of HTTP messages; the server responds to these messages, returning Web pages or other requested documents to the client. Most Web servers run a version of Unix or Microsoft Windows NT Server.

See also Hypertext Transfer Protocol; Web browser; Web page; World Wide Web.

Web site A group of HTML documents and associated scripts supported by a Web server on the World Wide Web.

Most Web sites have a home page used as a starting point or index into the site, with other Web pages or even other Web sites connected by links. To connect to a Web site, you need an Internet connection and a Web browser.

See also HyperText Markup Language; Web browser; Web page; World Wide Web.

wedged A slang expression for a frozen or hung terminal.

Western Digital Corp A major manufacturer of storage systems, including 3.5-inch Enhanced IDE drives for PCs, high-capacity SCSI drives for workstations and servers, and RAID systems.

For more information on Western Digital Corp, see www.wdc.com.

WfM *See* Wired for Management.

whiteboard An application that lets several network users look at and share images, data, and text simultaneously, as they all participate in a common conference call.

Each person's comments and suggestions are labeled and separated from the comments made by others participating in the call.

White Pine Software Distributors of the popular Cu-SeeMe application, a video conferencing and videophone product that works over the Internet.

For more information on White Pine Software, see www.wpine.com.

whitespace A collective name used for groups of spaces, tabs, and newlines, those printable characters that only produce blank spaces.

Wide Area Information Service Abbreviated WAIS, pronounced "ways." A service used to access text databases or libraries on the Internet.

WAIS uses simple natural-language queries and takes advantage of index searches for fast retrieval. Unlike Gopher, which searches only the names of Gopher resources, WAIS can search the content of all documents retrievable from WAIS databases. WAIS is particularly adept at searching through collections of Usenet newsgroups, electronic texts, and newspaper archives. WAIS has largely been replaced by the search engines now available on the World Wide Web.

See also search engine.

wide-area network Abbreviated WAN. A network that connects users across large distances, often crossing the geographical boundaries of cities or states.

See also local-area network; metropolitan-area network.

Wide Area Telephone Service Abbreviated WATS. A discounted toll service available from all long-distance and local telephone companies that provides unlimited use of a telephone circuit for a fixed charge.

wideband In communications, a channel capable of handling more frequencies than a standard 3kHz voice channel.

wideband transmission See broadband network.

Wide SCSI A version of the SCSI-2 standard that provides data transfer rates of up to 20MBps over a 16-bit data bus.

See also Fast SCSI; Fast/Wide SCSI; SCSI-1; SCSI-2; SCSI-3; Small Computer System Interface; Ultra SCSI; Ultra2 SCSI; Wide Ultra SCSI; Wide Ultra2 SCSI.

Wide Ultra SCSI A version of the SCSI-2 standard that provides data transfer rates of up to 40MBps over a 16-bit data bus.

See also Fast SCSI; Fast/Wide SCSI; SCSI-1; SCSI-2; SCSI-3; Small Computer System Interface; Ultra SCSI; Ultra2 SCSI; Wide SCSI.

Wide Ultra2 SCSI A version of the SCSI-2 standard that provides data transfer rates of up to 80MBps over a 16-bit data bus.

See also Fast SCSI; Fast/Wide SCSI; SCSI-1; SCSI-2; SCSI-3; Small Computer System Interface; Ultra SCSI; Ultra2 SCSI; Wide SCSI; Wide Ultra SCSI.

wildcard character A character that represents one or more unknown characters. In many operating systems, a question mark (?) represents a single unknown character in a filename or filename extension, and an asterisk (*) represents any number of unknown characters.

See also star-dot-star.

WinCGI A Common Gateway Interface (CGI) that allows HTTP applications to be written in Visual Basic. WinCGI has largely

been replaced by other types of Web server extensions.

See also Common Gateway Interface; Hypertext Transfer Protocol.

window In a graphical user interface, a rectangular portion of the screen that acts as a viewing area for applications.

Windows can be tiled so that they are displayed side by side, or they can be cascaded so that their individual title bars are always visible. They then can be individually moved and sized on the screen. Some programs can open multiple document windows inside their application window to display several word-processing or spreadsheet data files at the same time.

See also X Window.

Windows See Microsoft Windows.

Windows 3.1 See Microsoft Windows 3.1.

Windows 95 See Microsoft Windows 95.

Windows 98 See Microsoft Windows 98.

Windows Application Binary Interface Abbreviated WABI. A specification developed by Sun Microsystems that defines how Microsoft Windows applications run on Unix workstations. The WABI interface translates the system calls made by the application into system calls that the underlying Unix system can provide.

Windows CE See Microsoft Windows CE.

Windows client Any computer running Windows that connects to the network.

Windows Internet Naming Service

Abbreviated WINS. A Microsoft Windows NT Server service that maps NetBIOS computer names used in Windows networks to IP addresses used in TCP/IP-based networks. WINS is almost completely automated; it builds its own database and manages updates to the database.

See also network basic input/output system; IP address.

Windows Network Transit Protocol

See WinSock.

Windows NT Diagnostics A utility program from Microsoft that is used to look at and troubleshoot a Windows NT Server system; also known as winmsd.

Windows NT Diagnostics details transport information and network statistics, as well as service, resource, environment, and device driver information.

Windows NT Server *See* Microsoft Windows NT Server.

Windows NT Workstation *See* Microsoft Windows NT Workstation.

Windows Sockets *See* WinSock.

Windows terminal *See* thin client.

winmsd *See* Windows NT Diagnostics.

WINS *See* Windows Internet Naming Service.

WinSock An API for writers of TCP/IP-related software. Also known as Windows

Sockets or Windows Network Transit Protocol.

WinSock is designed to provide the same kinds of services that Berkeley sockets provide in the Unix world. In Windows, WinSock is implemented as a dynamic link library (DLL) that sits below the application but above the TCP/IP stack. Developers write programs that communicate with the DLL, which converts the commands and messages and passes them to the TCP/IP protocol stack below.

The latest version, WinSock 2, supports multiple communications protocols, including IPX (Internetwork Packet eXchange), DECnet, and XNS (Xerox Network Services), as well as real-time multimedia communications across a wide range of communications circuits.

Wintel A contraction of Windows and Intel. Pertaining to an Intel-based computer that runs Microsoft Windows.

WinZip A popular Windows file compression and decompression program from Nico Mak Computing, Inc.

Wired for Management Abbreviated WfM. A specification from Intel to make Intel-based systems more manageable.

WfM supports the Desktop Management Interface (DMI) and includes asset management, remote wake-up, power management, and system configuration and reconfiguration.

See also Advanced Configuration and Power Interface; Web-Based Enterprise Management.

wireless communications A method of connecting a node or a group of nodes into the main network using a technology other than conventional cabling.

The following methods are in use:

- **Infrared line of sight** High-frequency light waves are used to transmit data between nodes up to 24.4 meters (80 feet) apart using an unobstructed path; infrared beams cannot pass through masonry walls. Data rates are relatively high, in the tens of megabits per second range.
- **High-frequency radio** High-frequency radio signals transmit data to nodes from 12.2 to 39.6 meters (40 to 130 feet) apart, depending on the nature of obstructions separating them; the signal can penetrate thin walls but not supporting masonry. Data rates are usually less than 1Mbps.
- **Spread-spectrum radio** A small set of frequencies are available for wireless LANs without FCC approval. The 902 to 928Mhz band is known as the Industrial, Scientific, Medical (ISM) band and is not regulated. The 2.4 to 2.483Ghz band is regulated and requires an FCC license for use. Spread-spectrum nodes can be up to 243.8 meters (800 feet) apart in an open environment, and these radio waves can pass through masonry walls. However, in an environment with fully enclosed offices, distances are limited to 33.5 meters (110 feet). Data rates are usually less than 1Mbps.

Wireless LANs are not always completely wireless and may be used to replace the cabling on certain network segments or to

connect groups of networks that use conventional cabling.

See also mobile computing.

wiring closet A location where cables are gathered together to connect to the central wiring in an office or building.

word length 1. The standard data unit in a particular computer. The most common words are 8, 16, 32, or 64 bits in length.

2. In communications, the number of data bits in a data word.

workflow software Software that allows users to move and manage information among themselves, combining the functions of e-mail, imaging, and document management.

A document moves through various stages of processing as it is edited, signed, or validated by the various members of the workgroup. Each stage is orchestrated and validated by the workflow software.

See also workgroup.

workgroup A group of individuals who work together and share the same files and databases over a local-area network. Special software coordinates the workgroup and allows users to edit and exchange files and update databases as a group.

See also workflow software.

WorkGroup Director A Simple Network Management Protocol (SNMP) software package from Cisco Systems.

WorkGroup Director runs on Unix systems either as a stand-alone application or as an element in a larger SNMP-based network management system.

working directory Jargon for the current working directory; also known as the current directory.

workstation 1. In networking, any personal computer attached to the network.

2. A high-performance computer optimized for graphics applications, such as computer-aided design (CAD), computer-aided engineering (CAE), or scientific applications.

World Wide Web Abbreviated WWW, W3, or simply the Web. A huge collection of hypertext pages on the Internet.

World Wide Web concepts were developed in Switzerland by the European Laboratory for Particle Physics (known as CERN), but the Web is not just a tool for scientists; it is one of the most flexible and exciting tools in existence.

Hypertext links connect pieces of information (text, graphics, animation, audio, and video) in separate HTML pages located at the same or at different Internet sites, and you explore these pages and links using a Web browser such as Netscape Navigator or Microsoft Internet Explorer.

You can also access a Web resource directly if you specify the appropriate URL (Uniform Resource Locator).

World Wide Web traffic is growing faster than most other Internet services, and the reason for this becomes obvious once you try a capable Web browser; it is very easy

and a lot of fun to access World Wide Web information.

See also HyperText Markup Language; Internet; portal; search engine; Web browser.

World Wide Web Consortium Abbreviated W3C. An international consortium founded in 1994 to develop protocols for the continuing evolution of the World Wide Web.

For more information on the World Wide Web Consortium, see www.w3.org.

WORM Acronym for Write Once Read Many. A high-capacity optical storage device that can only be written to once, but that can be read a number of times.

WORM devices can store huge amounts of data, as much as 1 terabyte, are highly reliable, and are well suited to archival and other nonchanging storage.

write-back cache A technique used in cache design for writing information back into main memory.

In a write-back cache, the cache stores the changed block of data, but only updates main memory under certain conditions, such as when the whole block must be overwritten because a newer block must be loaded into the cache or when the controlling algorithm determines that too much time has elapsed since the last update. This method is rather complex to implement, but is much faster than other designs.

See also cache; write-through cache.

Write Once Read Many *See* WORM.

write-through cache

write-through cache A technique used in cache design for writing information back into main memory.

In a write-through cache, each time the processor returns a changed bit of data to the cache, the cache updates that information in both the cache and in main memory. This

method is simple to implement, but is not as fast as other designs; delays can be introduced when the processor must wait to complete write operations to slower main memory.

See also cache; write-back cache.

WWW *See* World Wide Web.

X

X See X Window.

X2 A modem technology from U.S. Robotics (now part of 3COM) that provides a data rate of up to 56Kbps downstream and up to 40Kbps upstream. Replaced by the V.90 standard.

See also K56Flex; V.90.

X.21 A CCITT (ITU) standard that defines a protocol used in a circuit-switching network.

X.25 A CCITT (ITU) standard, developed in 1976, that defines the connection between a terminal and a public packet-switched network.

X.25 describes the electrical connections, the transmission protocol, error detection and correction, and other aspects of the link. X.25 standards parallel the lowest three levels of the OSI Reference Model for computer-to-computer communications: the physical layer, data-link layer, and network layer.

X.25 gateway *See* X.75.

X.28 A CCITT (ITU) standard, developed in 1977, that defines a DTE/DCE (data terminal equipment/data communications equipment) interface for accessing a PAD (packet assembler/disassembler) in a public data network that does not cross an international boundary but is confined within one country.

X.29 A CCITT (ITU) standard, developed in 1977, for user data and the exchange of control information between a PAD (packet assembler/disassembler) and packet-mode DTE (data terminal equipment) or another PAD.

X.75 A CCITT (ITU) standard that defines the procedures used to connect two separate packet-switched networks, such as those located in separate countries; often referred to as an X.25 gateway.

X.200 A CCITT (ITU) standard that documents the seven-layer OSI Reference Model for computer-to-computer communications.

X.400 A CCITT (ITU) recommended standard, released in 1984 and revised several times since then, for public or private international e-mail distribution systems, defining how messages will be transferred across the network or between two or more connected heterogeneous networks.

X.400 defines the components of an electronic address as well as the details of the envelope surrounding the message and the rules to follow when converting between message types, such as text or fax.

See also X.500; X.509.

X.500 A CCITT (ITU) recommended standard, first released in 1988 and revised several times since then, for a global directory system for locating e-mail users, to be used with the X.400 e-mail services. X.500

is similar to a hierarchical worldwide telephone book.

The Lightweight Directory Access Protocol (LDAP), now widely accepted in the Internet community, is a subset of X.500, and products such as Novell Directory Services (NDS) are based on X.500.

See also Lightweight Directory Access Protocol; Novell Directory Services; X.400; X.509.

X.509 A CCITT (ITU) suggested standard for authentication services, including digital signatures, based on X.500.

X Consortium A group of vendors that develops products based on the X Window specifications.

Originally formed in 1988 at MIT by Apple, AT&T, Digital Equipment Corporation, Hewlett-Packard, and Sun Microsystems, in 1993 the X Consortium became an independent, nonprofit organization.

XCSE *See* Xylan Certified Switching Expert.

XCSS *See* Xylan Certified Switching Specialist.

XENIX A version of Unix developed from AT&T Version 7 as a joint venture between Microsoft and Santa Cruz Operation (SCO) and released in 1980.

XENIX was intended to support commercial applications on the IBM PC—versions were also developed for the Motorola 68000 and Zilog Z8000 processors—and for a while XENIX was the most successful Unix in terms of sheer numbers sold. Microsoft sold its interest in XENIX to SCO in 1987.

Xerox Network Services Abbreviated XNS. A multilayer communications protocol, first developed by Xerox, and later used by Novell and other network software suppliers. XNS also supports a distributed file system that lets users access other computers' files and printers as if they were local.

XFree86 A version of the X11R6 X Window system freely available for Intel-based Unix systems such as Linux and FreeBSD.

XML *See* Extensible Markup Language.

Xmodem A popular file transfer protocol available in many off-the-shelf and shareware communications packages.

Xmodem was originally developed by Ward Christiansen for early PCs using the CP/M operating system. Xmodem divides the data for a transmission into blocks. Each block consists of the start-of-header character, a block number, 128 bytes of data, and a checksum. An acknowledgment byte is returned to the sender if the checksum calculation is identical to the sender's checksum; however, this requirement to acknowledge every transmitted block can lead to poor performance.

An extension to Xmodem, called Xmodem-CRC, adds a more stringent error-checking method by using a cyclical redundancy check (CRC) to detect transmission errors rather than Xmodem's simple additive checksum.

Another variation is Xmodem-1K, which transfers data in 1024-byte blocks.

See also Kermit; Ymodem; Zmodem.

XMS *See* Extended Memory Specification.

XNS *See* Xerox Network Services.

XON/XOFF In asynchronous transmissions between two PCs, a simple method of flow control.

The receiving PC sends an XOFF control character (ASCII 19, Ctrl+S) to pause the transmission of data when the receive buffer is full and then sends an XON character (ASCII 17, Ctrl+Q) when it is ready to continue the transmission.

See also flow control; handshaking.

X/Open *See* Open Group.

XT *See* crosstalk.

X terminal A high-quality graphics terminal with a large amount of memory; designed for use with the X Window system. X terminals usually contain X server software in read-only memory.

See also X Window.

X Window A windowing environment developed at MIT for Unix workstations. Often referred to simply as X.

X Window is an open and nonproprietary bit-mapped graphics system, designed to be independent of both the display hardware and the underlying operating system. It is supported by all the major workstation vendors.

X Window implements a client/server environment, but with the sense of the terms reversed from today's common usage.

See also X terminal.

Xylan Certified Switching Expert Abbreviated XCSE. An advanced certification from Xylan for system professionals who design, maintain, and troubleshoot switched networks.

See also Xylan Certified Switching Specialist.

Xylan Certified Switching Specialist

Abbreviated XCSS. A basic certification from Xylan for system professionals who develop solutions for switched networks that covers basic concepts of frame and cells switching.

See also Xylan Certified Switching Expert.

Y

Y2K problem The inability of some older computer programs to process dates correctly after midnight on December 31, 1999.

Computer programmers in the 1960s and '70s abbreviated the date field to two digits, partly to save space and partly because they were convinced that the software would be rewritten before the abbreviated date ever became a problem. So Y2K isn't really a bug; the programmers and designers did it on purpose.

This means that older software reads 99 as 1999, and as the year 2000 begins, it will interpret 00 as the first day of 1900.

Dates and calculations based on dates are used in almost all business, sales, accounting, and commercial software, much of which was written in computer languages no longer in common use. Without expensive and time-consuming modifications and testing, these systems will at best provide unpredictable results after January 1, 2000; at worst, they will simply fail to operate.

Year 2000 bug *See* Y2K problem.

Yellow Pages A name for the security and file-access databases on Unix systems. These databases are now known as the Network Information Service (NIS).

See also Network Information Service.

Ymodem A popular file transfer protocol available in many off-the-shelf and shareware communications packages.

Ymodem, a variation of the Xmodem protocol, divides the data to be transmitted into blocks; each block consists of the start-of-header character, a block number, 1KB of data, and a checksum. Ymodem's larger data block means less overhead for error control when compared with Xmodem, but if the block has to be retransmitted because the protocol detects an error, there is more data to resend. Ymodem also incorporates the capabilities to send multiple files in the same session and to abort file transfer during the transmission.

See also Kermit; Zmodem.

Z

.Z Filename extension that identifies a Unix file compressed by the gzip or the compact utility.

.z Filename extension that identifies a Unix file compressed by the compress utility.

ZAK See Zero Administration Kit.

ZAW See Zero Administration for Windows.

Z.E.N.works See Novell Z.E.N.works.

Zero Administration Kit Abbreviated ZAK. A collection of tools, scripts, and methodologies used by system administrators to simplify the implementation of secure management on the Windows family of operating systems.

ZAK allows system administrators to perform three main functions quickly and easily:

- To centralize configuration
- To eliminate local access to the desktop computer
- To allow storage of data and applications on the server rather than on the client

See also Advanced Configuration and Power Interface; Desktop Management Interface; NetPC; network computer; thin client; total cost of ownership; Wired for Management; Zero Administration for Windows.

Zero Administration for Windows

Abbreviated ZAWS. A Microsoft-led initiative designed to lower the total cost of ownership of Windows-based network clients in the corporate world.

ZAW has evolved over time and now focuses on three main areas:

- Installing an operating system on a new computer or on a new hard disk in an existing computer
- Deploying applications from central servers to the desktop
- Distributing modifications or upgrades to applications

See also NetPC; network computer; thin client; total cost of ownership.

Zero Insertion Force socket A specially designed chip socket that makes replacing a chip easier and safer.

To change a chip in a ZIF socket, you raise a lever beside the socket to free the original chip's pins from the socket. You then slide the old chip out and slide in the replacement chip, taking care to align the pins and holes. Finally, you lower the lever again. A ZIF socket minimizes damage to the delicate pins that connect the chip to the rest of the system.

zero-slot LAN A local-area network that uses one of the existing serial or parallel ports on the computer rather than a special network interface card plugged in to the computer's expansion bus.

Because zero-slot LANs can only transmit as fast as the computer's output port, they are considerably slower than networks that use network-specific hardware and software. The maximum length of each cable segment is also severely limited, so zero-slot LANs can network only two or three computers. The advantage of a zero-slot LAN is its low cost compared with dedicated network systems.

See also peer-to-peer network.

zero-wait-state computer A computer that can process information without processor wait states, which are clock cycles during which no instructions are executed because the processor is waiting for data from a device or from memory.

Static RAM chips and paged-mode RAM chips are becoming popular because they can store information without being constantly refreshed by the processor, thus eliminating the wait state.

See also wait state.

ZIF socket *See* Zero Insertion Force socket.

.ZIP Filename extension that identifies a file compressed by the PKZIP or WinZip utilities.

See also PKZIP; WinZip; ZIP file.

Zip Drive A popular removable storage device from Iomega Corporation, capable of storing 100MB on relatively cheap, portable, 3.5-inch disks.

Zip Drives have emerged as the de facto standard personal computer backup device.

See also Iomega Corporation.

ZIP file A file whose contents have been compressed by one of the popular file-compression utilities, such as PKZIP, WinZip, or other comparable program; the filename extension is .ZIP.

A ZIP file can contain a single compressed file or a whole collection of archives. A file compressed in this way is said to have been zipped.

To uncompress a ZIP file, use the same utility that compressed it originally. Some ZIP files are self-extracting and can uncompress themselves when you click their icon.

See also file compression; PKZIP; WinZip; .ZIP.

zipped *See* ZIP file.

Zmodem A popular file transfer protocol available in many off-the-shelf and shareware communications packages.

Zmodem is similar to Xmodem and Ymodem but is designed to handle larger data transfers with fewer errors. Zmodem also includes a feature called checkpoint restart, which allows an interrupted transmission to resume at the point of interruption, rather than starting again at the beginning of the transmission.

See also Kermit; Xmodem; Ymodem.

zombie In Unix and other operating systems, a dead process that has not yet been deleted from the process table.

Most zombies disappear almost immediately, although from time to time, you may find one that is impossible to delete without rebooting the system. Zombies do not

consume any system resources, other than their slot in the process table.

zone On a local-area network such as AppleTalk, a logical subgroup of users within a larger group of interconnected networks.

Appendix A: Information on the Internet

This appendix contains a selection of the networking resources available on the World Wide Web, broken out by category to help you find the information you are looking for quickly and easily. To be consistent with the rest of this book, I have not specified the protocol used to access each Web site; unless a different protocol is specified, you can simply assume that HTTP will work in all cases. Just add `http://` to the beginning of each Web address in your browser when you access a site.

The Web is in a constant state of flux as URLs change and Web sites disappear. The better-organized sites will simply post a link to the new location if they make substantive changes, and you can use that new link to go right to the new or reorganized site. Other sites reorganize themselves periodically as a part of their housekeeping; the information you want is still available, but you have to look in another place to find it or use the site’s built-in search engine to find it.

When all else fails, use one of the Web sites listed in the “Portals and Search Engines” section of this appendix. In it you will find a list of the sites you can use to search the Web for documents using keywords, phrases, or even Boolean expressions. See each individual search engine site for details.

Hardware Companies

This section lists the major manufacturers and suppliers of networking and computer hardware.

Acer Group	www.acer.com.tw
Adaptec, Inc.	www.adaptec.com
Advanced Micro Devices, Inc.	www.amd.com
Apple Computer, Inc.	www.apple.com
Ascend Communications, Inc.	www.ascend.com
AST Research	www.ast.com

Appendix A: Information on the Internet

Bay Networks, Inc.	www.baynetworks.com
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Cabletron Systems	www.cabletron.com
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Canon, Inc.	www.canon.com
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Cirrus Logic, Inc.	www.cirrus.com
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Cisco Systems, Inc.	www.cisco.com
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Compaq Computer Corporation	www.compaq.com
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Cyrix	www.cyrix.com
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Dell Computer Corporation	www.dell.com
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Dialogic Corporation	www.dialogic.com
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Digital Equipment Corporation	www.digital.com
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Gateway, Inc.	www.gateway.com
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Hewlett-Packard Company	www.hp.com
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Integrated Device Technology	www.idt.com
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Intel Corporation	www.intel.com
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IBM	www.ibm.com
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Iomega Corporation	www.iomega.com
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Lexmark International, Inc.	www.lexmark.com
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Lucent Technologies, Inc.	www.lucent.com
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Microcom, Inc.	www.microcom.com
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Micron Technology	www.micronpc.com
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Appendix A: Information on the Internet

Motorola, Inc.	www.motorola.com
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National Semiconductor Corporation	www.national.com
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NEC Corporation	www.nec.com
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Packard-Bell	www.packardbell.com
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Quantum Corporation	www.quantum.com
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Rockwell Semiconductor Systems	www.rockwell.com
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S3, Inc.	www.s3.com
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Samsung Electronics	www.samsung.com
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Seagate Technology, Inc.	www.seagate.com
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Silicon Graphics, Inc.	www.sgi.com
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Sun Microsystems, Inc.	www.sun.com
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SyQuest Technology, Inc.	www.syquest.com
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Toshiba Corporation	www.toshiba.com
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U.S. Robotics Corporation	www.usr.com
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ViewSonic Corp.	www.viewsonic.com
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Western Digital Corp.	www.wdc.com
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Software and Service Companies

This section lists the major providers of networking operating systems, applications software, and online and other services.

America Online	www.aol.com
Baan Company	www2.baan.com
Banyan Systems, Inc.	www.banyan.com
Caldera, Inc.	www.caldera.com
CompuServe, Inc.	www.compuserve.com
Computer Associates International	www.cai.com
Gartner Group, Inc.	www.gartner.com
Informix Software, Inc.	www.informix.com
Inprise Corporation	www.inprise.com
Learning Tree International	www.learningtree.com
Marimba, Inc.	www.marimba.com
Microsoft, Inc.	www.microsoft.com
Netscape Communications Corporation	www.netscape.com
Network Associates, Inc.	www.networkassociates.com
Novell, Inc.	www.novell.com
Oracle Corporation	www.oracle.com
PC Connection	www.pcconnection.com

Appendix A: Information on the Internet

PeopleSoft, Inc.	www.peoplesoft.com
PointCast, Inc.	www.pointcast.com
RealNetworks, Inc.	www.real.com
Red Hat Software	www.redhat.com
RSA Data Security	www.rsa.com
SCO	www.sco.com
SAP	www.sap.com
Sybase, Inc.	www.sybase.com
Symantec Corporation	www.symantec.com
White Pine Software	www.wpine.com

Internet Organizations

Here you will find URLs relating to the main Internet organizations.

Internet Society	www.isoc.org
Internet Architecture Board	www.iab.org/iab
Internet Engineering Steering Group	www.ietf.org/iesg.htm
Internet Engineeirng Task Force	www.ietf.org
Internet Research Task Force	www.irtf.org
International Ad Hoc Committee	www.iahc.org

Internet Assigned Numbers Authority	www.isi.edu/iana
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World Wide Web Consortium	www.w3.org
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Federal Networking Council	www.fnc.gov
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Standards Groups and Trade Organizations

This section provides URLs for the major network-related standards groups and trade associations.

Active Group	www.activex.org
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ANSI	www.ansi.org
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ATM Forum	www.atmforum.com
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CERN	www.cern.ch
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CERT	www.cert.org
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EIA	www.eia.org
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FCC	www.fcc.gov
-----	--

IEEE	www.ieee.org
------	--

IrDA	www.irda.org
------	--

ISO	www.iso.ch
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ITU	www.itu.ch
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NCSC	www.ncsc.com
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Object Management Group	www.omg.org
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Appendix A: Information on the Internet

Open Group	www.opengroup.org
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TIA	www.tiaonline.org
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Portals and Search Engines

The distinction between a Web portal and a search engine continues to blur, so they are grouped together in this section.

Alta Vista	www.altavista.digital.com
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America Online	www.aol.com
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Chickclick	www.chickclick.com
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ESPN	www.espn.com
------	--

Excite	www.excite.com
--------	--

GoTo.com	www.goto.com
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Hotbot	www.hotbot.com
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Infoseek	www.infoseek.com
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Ivillage	www.ivillage.com
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Lycos	www.lycos.com
-------	--

Netcenter	www.netcenter.com
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Northern Light	www.nlsearch.com
----------------	--

Snap!	www.snap.com
-------	--

Yahoo!	www.yahoo.com
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Open Source Software

This section contains details on where to find open-source software, as well as URLs for some of the organizations supporting open source and free software.

Open Source Initiative	www.opensource.org
Free Software Foundation	www.fsf.org
Slashdot	www.slashdot.org
O'Reilly Open Source Center	opensource.oreilly.com
The Apache Project	www.apache.org
GNU	www.gnu.org
Linux Online	www.linux.org
The Linux Kernel Archives	www.kernel.org
The Linux Documentation Project	metalab.unc.edu/LDP
Linux Web pages	www.linux-center.org
The Perl Institute	www.perl.org
Netscape's Mozilla Project	www.mozilla.org
XFree86	www.xfree86.org
FreeBSD Project	ftp.freebsd.org
Open BSD	www.openbsd.org

Appendix A: Information on the Internet

Local Exchange Carriers

This section lists the companies providing local carrier services throughout the United States.

Ameritech, Inc.	www.ameritech.com
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Bell Atlantic Corp	www.bell-atl.com
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Bell South	www.bellsouth.com
------------	--

GTE Service Corp	www.gte.com
------------------	--

NYNEX Corp	www.nynex.com
------------	--

Pacific Bell	www.pacbell.com
--------------	--

SBC Communications, Inc.	www.sbc.com
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U.S. West Communications Group	www.uswest.com
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Interexchange Carriers

This section lists the carriers that provide long-distance service in the United States.

AT&T Corp	www.att.com
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LDDS WorldCom, Inc.	www.wcom.com
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GTE Service Corp.	www.gte.com
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MCI Telecommunications Corp.	www.mci.com
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Sprint Communications Company	www.sprintbiz.com
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MFS Communications Company	www.mfsdatanet.com
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Norlight Telecommunications, Inc.	www.norlight.com
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Appendix B: Certification Programs

This appendix presents information on the many and varied computer and networking certification programs available in the industry. Table B.1 lists the abbreviations and complete names for the most common certification programs, and Table B.2 lists all the network-related certification programs offered by each company or computer-industry organization.

TABLE B.1 CERTIFICATION PROGRAM ABBREVIATIONS

Abbreviation	Certification Program
A+	A+ Certification
AASE	Associated Accredited Systems Engineer
ABCP	Associate Business Continuity Professional
ACE	Advanced Certified Engineer
ACTE	Ascend Certified Technical Expert
ASE	Accredited Systems Engineer
BAC	Baan Advanced Certification
BBC	Baan Basic Certification
BNCE	Bay Networks Certified Expert
BNCS	Bay Networks Certified Specialist
CATE	Certified Advanced Technical Expert
CBCP	Certified Business Continuity Professional
CBE	Certified Banyan Engineer
CBS	Certified Banyan Specialist
CC	Certified Consultant

Appendix B: Certification Programs

TABLE B.1 CERTIFICATION PROGRAM ABBREVIATIONS *(CONTINUED)*

Abbreviation	Certification Program
CCIE	Cisco Certified Internetworking Expert
CCP	Certified Computing Professional
CDA	Certified Database Administrator
CDIA	Certified Document Image Architect
CE	Certified Expert
CINA	Certified IRIX Network Administrator
CISA	Certified Information System Auditor
CISA	Certified IRIX System Administrator
CISSP	Certified Information Systems Security Professional
CJD	Certified Java Developer
CJP	Certified Java Programmer
CLP	Certified Lotus Professional
CLS	Certified Lotus Specialist
CNA	Certified Novell Administrator
CNE	Certified Novell Engineer
CNI	Certified Novell Instructor
CNP	Certified Network Professional
CNX	Certified Network Expert

TABLE B.1 CERTIFICATION PROGRAM ABBREVIATIONS *(CONTINUED)*

Abbreviation	Certification Program
CPDA	Certified Powerbuilder Developer Associate
CPDP	Certified Powerbuilder Developer Professional
CPTS	Certified Performance and Tuning Specialist
CS	Certified Specialist
CSA	Certified Solaris Administrator
CSE	Certified Solutions Expert
CSE	Certified Systems Expert
CSE	Certified Switching Expert
CSNA	Certified Solaris Network Administrator
CSS	Certified Switching Specialist
CU	Certified User
CUE	Certified Unicenter Engineer
MBCP	Master Business Continuity Professional
MCNE	Master Certified Novell Engineer
MCP	Microsoft Certified Professional
MCSD	Microsoft Certified Solutions Developer
MCSE	Microsoft Certified Systems Engineer
MOE	Microsoft Office Expert

Appendix B: Certification Programs

TABLE B.1 CERTIFICATION PROGRAM ABBREVIATIONS *(CONTINUED)*

Abbreviation	Certification Program
MOES	Microsoft Office Expert Specialist
MOPS	Microsoft Office Proficient Specialist
N+	Network+ Certification
NCIP	Novell Certified Internet Professional
PSE	Professional Server Expert
PSS	Professional Server Specialist
RCDD	Registered Communications Distribution Designer

TABLE B.2 COMPUTER AND NETWORKING CERTIFICATION PROGRAMS

Company	Certification Program Name
Ascend	Ascend Certified Technical Expert
Baan	Baan Basic Certification Baan Advanced Certification in Enterprise Logistics Baan Advanced Certification in Enterprise Finance Baan Advanced Certification in Enterprise Tools Baan Advanced Certification in Enterprise Modeler
Banyan	Certified Banyan Specialist Certified Banyan Specialist: Windows NT Certified Banyan Engineer
Bay Networks	Bay Networks Certified Specialist: Hub Technology Bay Networks Certified Expert: Hub Technology Bay Networks Certified Specialist: Router Technology

TABLE B.2 COMPUTER AND NETWORKING CERTIFICATION PROGRAMS (CONTINUED)

Company	Certification Program Name
	Bay Networks Certified Expert: Router Technology Bay Networks Certified Specialist: Network Management Technology Bay Networks Certified Expert: Network Management Technology Bay Networks Certified Specialist: Switching Technology Bay Networks Certified Expert: Remote Access Technology
BICSI	Registered Communications Distribution Designer Registered Communications Distribution Designer: LAN Specialty
Cisco	Cisco Certified Internetworking Expert: WAN Switching Expert Cisco Certified Internetworking Expert: ISP Dial Expert Cisco Certified Internetworking Expert: Routing and Switching Expert
Compaq	Associate Accredited Systems Engineer Specializing in Novell IntranetWare Associate Accredited Systems Engineer Specializing in Microsoft Windows NT Accredited Systems Engineer Specializing in Novell IntranetWare Accredited Systems Engineer Specializing in Microsoft Windows NT
Computer Associates	Certified Unicenter Engineer
CompTIA	A+ Certification Certified Document Image Architect Network+ Certification
CNX Consortium	Certified Network Expert
Disaster Recovery Institute International	Associate Business Continuity Professional Certified Business Continuity Professional Master Business Continuity Professional

Appendix B: Certification Programs

TABLE B.2 COMPUTER AND NETWORKING CERTIFICATION PROGRAMS *(CONTINUED)*

Company	Certification Program Name
Hewlett-Packard	HP OpenView Certified Consultant: Unix
	HP OpenView Certified Consultant: Windows NT
	HP OpenView Certified Consultant: Unix and Windows NT
IBM	Certified Solutions Expert: Net Commerce
	Certified Solutions Expert: Firewall Resources
	IBM Certified AIX User
	IBM Certified Specialist: AIX System Administration
	IBM Certified Specialist: AIX Support
	IBM Certified Specialist: AS/400 Associate System Operator
	IBM Certified Specialist: AS/400 Professional System Operator
	IBM Certified Specialist: AS/400 Associate System Administrator
	IBM Certified Specialist: AS/400 Professional System Administrator
	IBM Certified Specialist: OS/2 Warp Server Administration
	IBM Certified Expert: OS/2 Warp Server
	IBM Certified Specialist: OS/2 LAN Server Administration
	IBM Certified Expert: OS/2 LAN Server
	IBM Certified Systems Expert: OS/2 Warp
	IBM Certified Advanced Technical Expert: RS/6000 AIX
	Professional Server Expert
	Professional Server Expert: Novell NetWare
	Professional Server Expert: OS/2 Warp Server
	Professional Server Expert: Windows NT Server
Informix	Database Specialist: Informix Dynamic Server
	System Administration: Informix Dynamic Server
	Informix-4GL Certified Professional

TABLE B.2 COMPUTER AND NETWORKING CERTIFICATION PROGRAMS (CONTINUED)

Company	Certification Program Name
Institute for Certification of Computing Professionals	Certified Computing Professional
International Information Systems Security Certification Consortium	Certified Information Systems Security Professional
Information Systems Audit and Control Association	Certified Information Systems Auditor
Learning Tree International	Internet/Intranet Certified Professional Internetwork Certified Professional Local Area Networks Certified Professional PC Service and Support Certified Professional Unix Systems Certified Professional Wide Area Networks Certified Professional
Lotus	Certified Lotus Specialist Certified Lotus Professional: Application Developer Certified Lotus Professional: Principal Application Developer Certified Lotus Professional: System Administrator Certified Lotus Professional: Principal System Administrator Certified Lotus Professional: cc Mail System Administrator
Microsoft	Microsoft Certified Professional +Internet Microsoft Certified Systems Engineer +Internet

Appendix B: Certification Programs

TABLE B.2 COMPUTER AND NETWORKING CERTIFICATION PROGRAMS *(CONTINUED)*

Company	Certification Program Name
	Microsoft Certified Systems Engineer
	Microsoft Certified Solutions Developer
	Microsoft Office Proficient Specialist
	Microsoft Office Expert Specialist
	Microsoft Office Expert
Network Professional Association	Certified Network Professional
Novell	Certified Novell Administrator
	Certified Novell Engineer
	Certified Novell Instructor
	Novell Certified Internet Professional
	Novell Internet Architect
	Novell Internet Business Strategist
	Novell Intranet Manager
	Master Certified Novell Engineer
	Novell Web Designer
Oracle	Certified Database Administrator
	Certified Application Developer
SCO	Advanced Certified Engineer: Server Track
	Advanced Certified Engineer: OpenServer Track
	Advanced Certified Engineer: UnixWare Track
Silicon Graphics	Certified IRIX System Administrator
	Certified IRIX Network Administrator

TABLE B.2 COMPUTER AND NETWORKING CERTIFICATION PROGRAMS *(CONTINUED)*

Company	Certification Program Name
Sun	Certified Java Programmer
Microsystems	Certified Java Developer
	Certified Solaris Administrator
	Certified Solaris Network Administrator
Sybase	Certified PowerBuilder Developer Associate
	Certified PowerBuilder Developer Professional
	Certified Database Administrator
	Certified Performing and Tuning Specialist
Xylan	Xylan Certified Switch Specialist
	Xylan Certified Switch Expert

Appendix C: ASCII Character Set

Table C.1 shows the first 32 characters (0–31) from the American Standard Code for Information Interchange (ASCII), also known as the control characters.

Table C.2 shows the 7-bit standard ASCII character set (comprising characters 32–127), which is implemented on all computers that use ASCII.

Table C.3 shows characters 128–255 of the 8-bit IBM extended ASCII character set.

TABLE C.1 ASCII CONTROL CHARACTERS

DECIMAL	CHARACTER	CONTROL COMBINATION
0	NUL (Null)	Ctrl+@
1	SOH (Start of heading)	Ctrl+A
2	STX (Start of text)	Ctrl+B
3	ETX (End of text)	Ctrl+C
4	EOT (End of transmission)	Ctrl+D
5	ENQ (Enquire)	Ctrl+E
6	ACK (Acknowledge)	Ctrl+F
7	BEL (Bell)	Ctrl+G
8	BS (Backspace)	Ctrl+H
9	HT (Horizontal tab)	Ctrl+I
10	LF (Line feed)	Ctrl+J
11	VT (Vertical tab)	Ctrl+K
12	FF (Form feed)	Ctrl+L
13	CR (Carriage return)	Ctrl+M

TABLE C.1 ASCII CONTROL CHARACTERS (CONTINUED)

DECIMAL	CHARACTER	CONTROL COMBINATION
14	SO (Shift out)	Ctrl+N
15	SI (Shift in)	Ctrl+O
16	DLE (Data link escape)	Ctrl+P
17	DC1 (Device control 1)	Ctrl+Q
18	DC2 (Device control 2)	Ctrl+R
19	DC3 (Device control 3)	Ctrl+S
20	DC4 (Device control 4)	Ctrl+T
21	NAK (Negative acknowledgement)	Ctrl+U
22	SYN (Synchronous idle)	Ctrl+V
23	ETB (End transmission block)	Ctrl+W
24	CAN (Cancel)	Ctrl+X
25	EM (End of medium)	Ctrl+Y
26	SUB (Substitute)	Ctrl+Z
27	ESC (Escape)	Ctrl+[
28	FS (File separator)	Ctrl+/ Ctrl+_
29	GS (Group separator)	Ctrl+]
30	RS (Record separator)	Ctrl+^
31	US (Unit separator)	Ctrl+_

Appendix C: ASCII Character Set

TABLE C.2 STANDARD 7-BIT ASCII CHARACTER SET

DECIMAL	CHARACTER	DECIMAL	CHARACTER
32	space	51	3
33	!	52	4
34	"	53	5
35	#	54	6
36	\$	55	7
37	%	56	8
38	&	57	9
39	'	58	:
40	(	59	;
41	)	60	<
42	*	61	=
43	+	62	>
44	,	63	?
45	-	64	@
46	.	65	A
47	/	66	B
48	0	67	C
49	1	68	D
50	2	69	E

TABLE C.2 STANDARD 7-BIT ASCII CHARACTER SET *(CONTINUED)*

DECIMAL	CHARACTER	DECIMAL	CHARACTER
70	F	89	Y
71	G	90	Z
72	H	91	[
73	I	92	\
74	J	93	]
75	K	94	^
76	L	95	_
77	M	96	`
78	N	97	a
79	O	98	b
80	P	99	c
81	Q	100	d
82	R	101	e
83	S	102	f
84	T	103	g
85	U	104	h
86	V	105	i
87	W	106	j
88	X	107	k

Appendix C: ASCII Character Set

TABLE C.2 STANDARD 7-BIT ASCII CHARACTER SET (CONTINUED)

DECIMAL	CHARACTER	DECIMAL	CHARACTER
108	l	118	v
109	m	119	w
110	n	120	x
111	o	121	y
112	p	122	z
113	q	123	{
114	r	124	
115	s	125	}
116	t	126	~
117	u	127	DEL

TABLE C.3 IBM EXTENDED ASCII CHARACTER SET

DECIMAL	CHARACTER	DECIMAL	CHARACTER
128	Ç	135	ç
129	ü	136	ê
130	é	137	ë
131	â	138	è
132	ä	139	ï
133	à	140	î

TABLE C.3 IBM EXTENDED ASCII CHARACTER SET (CONTINUED)

DECIMAL	CHARACTER	DECIMAL	CHARACTER
134	å	141	ì
142	Ä	161	í
143	Å	162	ó
144	É	163	ú
145	æ	164	ñ
146	Æ	165	Ñ
147	ô	166	à
148	ö	167	ó
149	ò	168	¿
150	û	169	²
151	ù	170	³
152	ÿ	171	¹ / ₂
153	Ö	172	¹ / ₄
154	Ü	173	¡
155	¢	174	«
156	£	175	»
157	¥	176	⋮
158	℞	177	⋮
159	f	178	⋮

Appendix C: ASCII Character Set

TABLE C.3 IBM EXTENDED ASCII CHARACTER SET (CONTINUED)

DECIMAL	CHARACTER	DECIMAL	CHARACTER
160	á	179	
180	┘	199	
181	┘	200	└
182	┘	201	└
183	┘	202	└
184	┘	203	└
185	┘	204	└
186		205	=
187	┘	206	┘
188	┘	207	┘
189	┘	208	┘
190	┘	209	┘
191	┘	210	┘
192	┘	211	┘
193	?	212	┘
194	>	213	┘
195	£	214	┘
196	-	215	┘
197	1	216	┘

TABLE C.3 IBM EXTENDED ASCII CHARACTER SET (CONTINUED)

DECIMAL	CHARACTER	DECIMAL	CHARACTER
198	⌋	217	⌋
218	⌋	237	f
219	■	238	'
220	■	239	\
221	l	240	[
222	■	241	6
223	■	242	j
224	~	243	π
225	i	244	(
226	G	245	)
227	p	246	4
228	Σ	247	<
229	s	248	8
230	m	249	8
231	t	250	-
232	F	251	=
233	Q	252	n

Appendix C: ASCII Character Set

TABLE C.3 IBM EXTENDED ASCII CHARACTER SET *(CONTINUED)*

DECIMAL	CHARACTER	DECIMAL	CHARACTER
234	V	253	²
235	W	254	■
236	l	255	

Appendix D: EBCDIC Character Set

Table D.1 shows all 256 characters that make up the Extended Binary Coded Decimal Interchange Code (EBCDIC) character set.

TABLE D.1 EBCDIC CHARACTER SET

Decimal	Character	Decimal	Character
0	NUL (null)	14	SO (shift out)
1	SOH (start of heading)	15	SI (shift in)
2	STX (start of text)	16	DLE (data length escape)
3	ETX (end of text)	17	DC1 (device control 1)
4	SEL (select)	18	DC2 (device control 2)
5	HT (horizontal tab)	19	DC3 (device control 3)
6	RNL (required new line)	20	RES/ENP (restore/enable presentation)
7	DEL (delete)	21	NL (new line)
8	GE (graphic escape)	22	BS (backspace)
9	SPS (superscript)	23	POC (program-operator communication)
10	RPT (repeat)	24	CAN (cancel)
11	VT (vertical tab)	25	EM (end of medium)
12	FF (form feed)	26	UBS (unit backspace)
13	CR (carriage return)	27	CU1 (customer use 1)

Appendix D: EBCDIC Character Set

TABLE D.1 EBCDIC CHARACTER SET (CONTINUED)

Decimal	Character	Decimal	Character
28	IFS (interchange file separator)	42	SM/SW (set mode/switch)
29	IGS (interchange group separator)	43	CSP (control sequence prefix)
30	IRS (interchange record separator)	44	MFA (modify field attribute)
31	IUS/ITB (interchange unit separator/intermediate transmission block)	45	ENQ (enquiry)
32	DS (digit select)	46	ACK (acknowledge)
33	SOS (start of significance)	47	BEL (bell)
34	FS (field separator)	48-49	not assigned
35	WUS (word underscore)	50	STN (synchronous idle)
36	BYP/INP (bypass/inhibit presentation)	51	IR (index return)
37	LF (line feed)	52	PP (presentation position)
38	ETB (end of transmission block)	53	TRN (transport)
39	ESC (escape)	54	NBS (numeric backspace)
40	SA (set attribute)	55	EOT (end of transmission)
41	SFE (start field extended)	56	SBS (subscript)

TABLE D.1 EBCDIC CHARACTER SET (CONTINUED)

Decimal	Character	Decimal	Character
		57	IT (indent tab)
58	RFF (required form feed)	90	!
59	CU3 (customer use 3)	91	\$
60	DC4 (device control 4)	92	*
61	NAK (negative acknowledge)	93	)
62	not assigned	94	;
63	SUB (substitute)	95	¬ (Logical NOT)
64	SP (space)	96	-
65	RSP (required space)	97	/
66-73	not assigned	98-105	not assigned
74	¢	106	(broken pipe)
75	.	107	,
76	<	108	%
77	(	109	-
78	+	110	>
79	(Logical OR)	111	?
80-89	not assigned	112-120	not assigned

Appendix D: EBCDIC Character Set

TABLE D.1 EBCDIC CHARACTER SET (CONTINUED)

Decimal	Character	Decimal	Character
121	`	138-144	not assigned
122	:	145	j
123	#	146	k
124	@	147	l
125	!	148	m
126	=	149	n
127	"	150	o
128	not assigned	151	p
129	a	152	q
130	b	153	r
131	c	154-160	not assigned
132	d	161	~
133	e	162	s
134	f	163	t
135	g	164	u
136	h	165	v
137	i	166	w
167	x	210	K

TABLE D.1 EBCDIC CHARACTER SET (CONTINUED)

Decimal	Character	Decimal	Character
168	y	211	L
169	z	212	M
170-191	not assigned	213	N
192	{	214	O
193	A	215	P
194	B	216	Q
195	C	217	R
196	D	218-223	not assigned
197	E	224	\
198	F	225	NSP (numeric space)
199	G		
200	H	226	S
201	I	227	T
202	SHY (syllable hyphen)	228	U
203-207	not assigned	229	V
208	}	230	W
209	J	231	X
232	Y	245	5

Appendix D: EBCDIC Character Set

TABLE D.1 EBCDIC CHARACTER SET (CONTINUED)

Decimal	Character	Decimal	Character
233	Z	246	6
234-239	not assigned	247	7
240	0	248	8
241	1	249	9
242	2	250-254	not assigned
243	3	255	EO (eight ones)
244	4		